Northern Nut Growers Association Report of the Proceedings at the 43rd Annual Meeting

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———————————+ |DISCLAIMER | | | |The articles published in the Annual Reports of the Northern Nut Growers| |Association are the findings and thoughts solely of the authors and are | |not to be construed as an endorsement by the Northern Nut Growers | |Association, its board of directors, or its members. No endorsement is | |intended for products mentioned, nor is criticism meant for products not| |mentioned. The laws and recommendations for pesticide application may | |have changed since the articles were written. It is always the pesticide| |applicator's responsibility, by law, to read and follow all current | |label directions for the specific pesticide being used. The discussion | |of specific nut tree cultivars and of specific techniques to grow nut | |trees that might have been successful in one area and at a particular | |time is not a guarantee that similar results will occur elsewhere. | +

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43rd Annual Report

OF THE

Northern Nut Growers Association

Incorporated

AFFILIATED WITH THE AMERICAN HORTICULTURAL SOCIETY

Annual Meeting at

ROCKPORT, INDIANA

August 25, 26 and 27, 1952

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Officers for 1952-53

EXECUTIVE APPOINTMENTS 1952-53

Program Committee:

Dr. J. W. McKay, Royal Oakes, Gordon Porter, Gilbert Becker, A. A. Bungart, W. D. Armstrong.

Local Arrangements:

George Salzer, Victor Brook.

Place of Meeting Committee:

R. P. Allaman, Dr. Lloyd L. Dowell, Edwin W. Lemke, Alfred L. Barlow.

Publication Committee:

Professor George L. Slate, Professor Lewis E. Theiss, Dr. L. H. MacDaniels.

Varieties and Contests Committee:

Dr. L. H. MacDaniels, J. C. McDaniel, Sylvester M. Shessler, H. F. Stoke, Royal Oakes.

Standards and Judging Committee:

Dr. L. H. MacDaniels, Dr. H. L. Crane, Louis Gerardi, Spencer Chase, Professor Paul E. Machovina.

Survey and Research Committee:

H. F. Stoke (With all the state and foreign vice-presidents).

Exhibits Committee:

Sylvester M. Shessler, Dr. L. H. MacDaniels, H. F. Stoke, Royal Oakes, A. A. Bungart, J. F. Wilkinson.

Root Stocks Committee:

Professor F. L. O'Rourke, J. C. McDaniel, Albert F. Ferguson, Dr. Aubrey Richards, Louis Gerardi, Dr. Arthur S. Colby, Max Hardy, Gilbert Smith.

Auditing Committee:

Raymond E. Silvis, Sterling A. Smith, Edward W. Pape.

Legal Advisor:

Sargent H. Wellman.

Finance Committee:

Sterling A. Smith, Ford Wallick, Edward W. Pape.

Necrology:

Mrs. Herbert Negus, Mrs. C. A. Reed, Mrs. G. A. Zimmerman.

Nominating Committee:

(Elected at Rockport, Indiana), Max Hardy, Gilbert Becker, Dr. William Rohrbacher, Professor George L. Slate, J. Ford Wilkinson.

Membership Committee:

George Salzer (With all the state and foreign vice-presidents).

State and Foreign Vice-Presidents

Alabama Edward L. Hiles, Loxley

Alberta A. L. Young, Brooks

Arkansas W. D. Wylie, Univ. of Ark., Fayetteville

Belgium R. Vanderwaeren, Bierbeekstraat, 310, Korbeek-Lo

British Columbia, Canada J. U. Gellatly, Box 19, Westbank

California Thos. R. Haig, M.D., 3021 Highland Ave., Carlesbad

Colorado J. E. Forbes, Julesburg

Connecticut A. M. Huntington, Stanerigg Farms, Bethel

Delaware Lewis Wilkins, Route 1, Newark

Denmark Count F. M. Knuth, Knuthenborg, Bandholm

District of Columbia Ed. L. Ford, 3634 Austin St.,
S. E. Washington 20

Florida C. A. Avant, 960 N. W. 10th Ave., Miami

Georgia William J. Wilson, North Anderson Ave., Fort Valley

Hawaii John F. Cross, P. O. Box 1720, Hilo

Hong Kong P. W. Wang, 6 Des Voeux Rd., Central

Idaho Lynn Dryden, Peck

Illinois Royal Oakes, Bluffs (Scott County)

Indiana Edw. W. Pape, Rt. 2, Marion

Iowa Ira M. Kyle, Box 236, Sabula

Kansas Dr. Clyde Gray, 1045 Central Ave., Horton

Kentucky Dr. C. A. Moss, Williamsburg

Louisiana Dr. Harald E. Hammar, 608 Court House, Shreveport

Maryland Blaine McCollum, White Hall

Massachusetts S. Lathrop Davenport, 24 Creeper Hill Rd.,
North Grafton

Michigan Gilbert Becker, Climax

Minnesota R. E. Hodgeson, Southeastern Exp. Station, Waseca

Mississippi James R. Meyer, Delta Branch Exp. Station, Stoneville

Missouri Ralph Richterkessing, Route 1, Saint Charles

Montana Russel H. Ford, Dixon

Nebraska Harvey W. Hess, Box 209, Hebron

New Hampshire Matthew Lahti, Locust Lane Farm, Wolfeboro

New Jersey Mrs. Alan R. Buckwalter, Route 1, Flemington

New Mexico Rev. Titus Gehring, P. O. Box 177, Lumberton

New York Stephen Bernath, Route No. 3, Poughkeepsie

North Carolina Dr. R. T. Dunstan, Greensboro College, Greensboro

North Dakota Homer L. Bradley, Long Lake Refuge, Moffit

Ohio Christ Pataky Jr., 592 Hickory Lane, Route 4, Mansfield

Oklahoma A. G. Hirschi, 414 North Robinson, Oklahoma City

Ontario, Canada Elton E. Papple, Cainsville

Oregon Harry L. Pearcy, Route 2, Box 190, Salem

Pennsylvania R. P. Allaman, Route 86, Harrisburg

Prince Edward Is. Canada Robert Snazelle, Forest Nursery, Route 5, Charlottetown

Rhode Island Philip Allen, 178 Dorance St., Providence

South Carolina John T. Bregger, P.O. Box 1018, Clemson

South Dakota Herman Richter, Madison

Tennessee W. Jobe Robinson, Route 7, Jackson

Texas Kaufman Florida, Box 154, Rotan

Utah Harlan D. Petterson, 2076 Jefferson Ave., Ogden

Vermont A. W. Aldrich, R. F. D. 2, Box 266, Springfield

Virginia H. R. Gibbs, Linden

Washington H. Lynn Tuttle, Clarkston

West Virginia Wilbert M. Frye, Pleasant Dale

Wisconsin C. F. Ladwig, 2221 St. Lawrence, Beloit

CONSTITUTION

of the

NORTHERN NUT GROWERS ASSOCIATION, INCORPORATED

(As adopted September 13, 1948)

NAME

ARTICLE I. This Society shall be known as the Northern Nut Growers Association, Incorporated. It is strictly a non-profit organization.

PURPOSES

ARTICLE II. The purposes of this Association shall be to promote interest in the nut bearing plants; scientific research in their breeding and culture; standardization of varietal names; the dissemination of information concerning the above and such other purposes as may advance the culture of nut bearing plants, particularly in the North Temperate Zone.

MEMBERS

ARTICLE III. Membership in this Association shall be open to all persons interested in supporting the purposes of the Association. Classes of members are as follows: Annual members, Contributing members, Life members, Honorary members, and Perpetual members. Applications for membership in the Association shall be presented to the secretary or the treasurer in writing, accompanied by the required dues.

OFFICERS

ARTICLE IV. The elected officers of this Association shall consist of a President, a Vice-President, a Secretary and a Treasurer or a combined Secretary-treasurer as the Association may designate.

BOARD OF DIRECTORS

ARTICLE V. The Board of Directors shall consist of six members of the Association who shall be the officers of the Association and the two preceding elected presidents. If the offices of Secretary and Treasurer are combined, the three past presidents shall serve on the Board of Directors.

There shall be a State Vice-president for each state, dependency, or country represented in the membership of the Association, who shall be appointed by the President.

AMENDMENTS TO THE CONSTITUTION

ARTICLE VI. This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or copy of the proposed amendments having been mailed by the Secretary, or by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

(Revised and adopted at Norris, Tennessee, September 13, 1948)

SECTION I.—MEMBERSHIP

Classes of membership are defined as follows:

ARTICLE I. ANNUAL MEMBERS. Persons who are interested in the purposes of the Association who pay annual dues of Three Dollars ($3.00).

ARTICLE II. CONTRIBUTING MEMBERS. Persons who are interested in the purposes of the Association who pay annual dues of Ten Dollars ($10.00) or more.

ARTICLE III. LIFE MEMBERS. Persons who are interested in the purposes of the Association who contribute Seventy Five Dollars ($75.00) to its support and who shall, after such contribution, pay no annual dues.

ARTICLE IV. HONORARY MEMBERS. Those whom the Association has elected as honorary members in recognition of their achievements in the special fields of the Association and who shall pay no dues.

ARTICLE V. PERPETUAL MEMBERS. "Perpetual" membership is eligible to any one who leaves at least five hundred dollars to the Association and such membership on payment of said sum to the Association shall entitle the name of the deceased to be forever enrolled in the list of members as "Perpetual" with the words "In Memoriam" added thereto. Funds received therefor shall be invested by the Treasurer in interest bearing securities legal for trust funds in the District of Columbia. Only the interest shall be expended by the Association. When such funds are in the treasury the Treasurer shall be bonded. Provided: that in the event the Association becomes defunct or dissolves, then, in that event, the Treasurer shall turn over any funds held in his hands for this purpose for such uses, individuals or companies that the donor may designate at the time he makes the bequest of the donation.

SECTION II.-DUTIES OF OFFICERS

ARTICLE I. The President shall preside at all meetings of the Association and Board of Directors, and may call meetings of the Board of Directors when he believes it to be the best interests of the Association. He shall appoint the State Vice-presidents; the standing committees, except the Nominating Committee, and such special committees as the Association may authorize.

ARTICLE II. Vice-president. In the absence of the President, the Vice-president shall perform the duties of the President.

ARTICLE III. Secretary. The Secretary shall be the active executive officer of the Association. He shall conduct the correspondence relating to the Association's interests, assist in obtaining memberships and otherwise actively forward the interests of the Association, and report to the Annual Meeting and from time to time to meetings of the Board of Directors as they may request.

ARTICLE IV. Treasurer. The Treasurer shall receive and record memberships, receive and account for all moneys of the Association and shall pay all bills approved by the President or the Secretary. He shall give such security as the Board of Directors may require or may legally be required, shall invest life memberships or other funds as the Board of Directors may direct, subject to legal restrictions and in accordance with the law, and shall submit a verified account of receipts and disbursements to the Annual meeting and such current accounts as the Board of Directors may from time to time require. Before the final business session of the Annual Meeting of the Association, the accounts of the Treasurer shall be submitted for examination to the Auditing Committee appointed by the President at the opening session of the Annual Meeting.

ARTICLE V. The Board of Directors shall manage the affairs of the association between meetings. Four members, including at least two elected officers, shall be considered a quorum.

SECTION III.—ELECTIONS

ARTICLE I. The Officers shall be elected at the Annual Meeting and hold office for one year beginning immediately following the close of the Annual Meeting.

ARTICLE II. The Nominating Committee shall present a slate of officers on the first day of the Annual Meeting and the election shall take place at the closing session. Nominations for any office may be presented from the floor at the time the slate is presented or immediately preceding the election.

ARTICLE III. For the purpose of nominating officers for the year 1949 and thereafter, a committee of five members shall be elected annually at the preceding Annual Meeting.

ARTICLE IV. A quorum at a regularly called Annual Meeting shall be fifteen (15) members and must include at least two of the elected officers.

ARTICLE V. All classes of members whose dues are paid shall be eligible to vote and hold office.

SECTION IV.—FINANCIAL MATTERS

ARTICLE I. The fiscal year of the Association shall extend from October 1st through the following September 30th. All annual memberships shall begin October 1st.

ARTICLE II. The names of all members whose dues have not been paid by
January 1st shall be dropped from the rolls of the Society. Notices of non-payment of dues shall be mailed to delinquent members on or about December 1st.

ARTICLE III. The Annual Report shall be sent to only those members who have paid their dues for the current year. Members whose dues have not been paid by January 1st shall be considered delinquent. They will not be entitled to receive the publication or other benefits of the Association until dues are paid.

SECTION V.—MEETINGS

ARTICLE I. The place and time of the Annual Meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the Board of Directors shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the President and Board of Directors.

SECTION VI.—PUBLICATIONS

ARTICLE I. The Association shall publish a report each fiscal year and such other publications as may be authorized by the Association.

ARTICLE II. The publishing of the report shall be the responsibility of the Committee on Publications.

SECTION VII.—AWARDS

ARTICLE I. The Association may provide suitable awards for outstanding contributions to the cultivation of nut bearing plants and suitable recognition for meritorious exhibits as may be appropriate.

SECTION VIII.—STANDING COMMITTEES

As soon as practical after the Annual Meeting of the Association, the President shall appoint the following standing committees:

1. Membership 2. Auditing 3. Publications 4. Survey 5. Program 6. Research 7. Exhibit 8. Varieties and Contests

SECTION IX.—REGIONAL GROUPS AND AFFILIATED SOCIETIES

ARTICLE I. The Association shall encourage the formation of regional groups of its members, who may elect their own officers and organize their own local field days and other programs. They may publish their proceedings and selected papers in the yearbooks of the parent society subject to review of the Association's Committee on Publications.

ARTICLE II. Any independent regional association of nut growers may affiliate with the Northern Nut Growers Association provided one-fourth of its members are also members of the Northern Nut Growers Association. Such affiliated societies shall pay an annual affiliation fee of $3.00 to the Northern Nut Growers Association. Papers presented at the meetings of the regional society may be published in the proceedings of the parent society subject to review of the Association's Committee on Publications.

SECTION X.—AMENDMENTS TO BY-LAWS

ARTICLE I. These by-laws may be amended at any Annual Meeting by a two-thirds vote of the members present provided such amendments shall have been submitted to the membership in writing at least thirty days prior to that meeting.

Forty-Third Annual Meeting

Northern Nut Growers Association

August 25, 26, 27, 1952

Spencer County Court House, Rockport, Ind.

The opening session of the Forty-third Annual Meeting of the Northern Nut Growers Association convened at 9:20 o'clock, a.m., at the Spencer County Court House, President L. H. MacDaniels presiding.

PRESIDENT MacDANIELS: The gavel with which we open this forty-third annual meeting of the Northern Nut Growers Association has some historical significance. It was made from a pecan tree which grew in the orchard of Mr. Thomas Littlepage in Maryland, near the city of Washington, and it has been the custom of the Association to open its meetings with that gavel.

The forty-third meeting of the Northern Nut Growers Association will be in order. To open the session we will have the presentation of the colors. You will all stand, please, and remain standing through the invocation. (Colors presented by Boy Scouts and the invocation given by the Reverend William Ellis of Rockport.)

PRESIDENT MacDANIELS: At this time we will call on Mr. Hilbert Bennett to bring us greetings from the people of Rockport. Mr. Bennett of Rockport.

Address of Welcome

HILBERT BENNETT, *Rockport, Ind.*

Some are here that were here in 1935 and 1939. I was on the Citizen's Committee in each of those years. It was the purpose of the Citizen's Committee to take notice of your coming and to try to make you appreciate our interest in you and in your coming.

Why was I on that Committee in 1935?

Why was I on that Committee in 1939?

Why am I on that Committee in 1952?

I will tell you.

When I was a boy two other young men, somewhat older than I, were young men in the same township and somewhat closely located. I knew those boys and I knew them well. You came to know them and know them well. One of those boys was the late Thomas P. Littlepage, a charter member of this Association. It was my good pleasure to teach school with him. We attended College together. At college we roomed together. We attended conventions together and were close personal friends. I think I was in position to know him and know him well. The other boy was R. L. McCoy. We too, were close personal friends. We too, taught school in the same territory and contemporary with T. P. Littlepage. Prior to any organization of the N.N.G.A. I went with these two boys (men by that time) on trips of investigation and inspection of certain nut trees about which they had heard and which they wanted to examine.

If the trees examined met the proper standards, they wanted to use them in propagation. If not they would pass them up.

Another boy somewhat younger than myself and the two above mentioned boys, joined most heartily into the nut discussions and investigations and explorations of promising clues. With them he helped to run down clues when they would hear of a promising prospect. The jungles were never too dense, the distance too far, the road too muddy or rough, for those three characters to run down in those horse and buggy days, any prospect in which they were interested. This boy also became a member of your most valued organization. I have a special interest in this boy. I was, especially closely associated with him and his family. He went to school to me. My signature appears on his Common School Diploma. Their home was my home whenever I sought to make it so. I was free to come and go. I came a lot. Ford Wilkinson, the third character, and I have been close friends ever since.

Another one of your fine members became a good friend of mine. He came into our county and planted a farm to nut trees and nut production. It is now the largest nut orchard in the county. I am informed that at that time it was the largest nut farm of hardy northern varieties in the world. I got acquainted with him early and became endeared to him. It was none other than the late Harry Weber.

When it became known that you were to meet here in 1935, it was a natural sequence that Ford Wilkinson, knowing that I would gladly help in any way I could and knowing I was his genuine friend saw fit to place me on the Citizen's Committee. If he had not, I positively would have climbed aboard anyway. You couldn't have driven me out with a peeled hickory club. I was just going to be in on it whether or no.

Whether I performed well in 1935 or whether he couldn't find any one else to serve in my place, I never knew; but he again placed me on the Committee in 1939.

Now here I am in 1952 an old broken down fossil, broken in health, but not in spirit, of little consequence to anybody or anything, I am still on the Committee.

That answers the question of some of you of why that old man Bennett is always on the local committee and that you have wondered if there is no other person in this whole community that will serve but him. No, friends, we have many who would gladly serve and I doubt not that would serve much more efficiently.

I have prepared a short "skit" that I wish to present.

* * * * *

1st. *Introducing Joan Flick, of Washington, D. C.*

I am a pecan plucked from a small orchard planted by a retired business man. He had some surplus ground near his premises that was too rough for easy cultivation. He thought that he would plant it to pecans so that his family and his children's families would have nuts for their own use and pleasure. He took good care of the trees. He fertilized them every year and sometimes oftener. In the course of a few years he not only had more pecans than all of the families could use, but he sold hundreds of pounds of nuts from these trees. He developed a commercial orchard unconsciously.

2nd. *Palma Smith of Cincinnati, Ohio.*

I am the hican, I have no commercial value of consequence. I demonstrate the ability, the interest, the development and the possibilities of improvement by the determined efforts of the members of your association. Knowing your ability and determination to make improvements in nut culture, I have every feeling that in the not too distant future you will develop me into a profitable commercial product.

3rd. *Sandra Wright of Rockport, Indiana.*

I am the walnut, a most valuable tree for fine fruit and fine timber for many uses. I have been noted for my fine grain and my ability to take a fine polish. Our forefathers immediately found the walnut to be the choice timber out of which to build fine furniture, gun stocks, home furnishings and many other things that required high grade material. We have never lost sight of its significance.

Thin shelled nuts, easily cracked, and hulled out in halves have been developed. Walnuts will grow almost any where. Originally it was a common forest tree and would continue to be if it had the opportunity. There is little danger of the walnut becoming extinct. It is too valuable. I suggest that you plant liberally to high grade walnut trees.

4th. *Jo Ann Hall of Rockport, Indiana.*

I am the once popular beech under whose folds thousands of picnickers have gathered and enjoyed life's most savory and pleasant moments. I have built thousands of American homes and farm barns. I have built thousands of miles of old farm plank fences. I have built car load after car load of beautiful, useful and valuable furniture. In the early period of this country I furnished mast for thousands of swine that fed many families. I have filled many minor places of usefulness. As sad as it is to do and as much as I hate to do so, I am now bidding you a last farewell.

Self interest, the slowness of my growth and the impracticability of propagation of this once valuable tree leaves but one course, that I pass to my reward with the firm hope that the other trees now being developed, and grown will fill all of the purposes for which I have been so useful, and fill them with increased usefulness. With this sad but necessary adieu, I bid you one and all goodbye.

5th. *Pattie Jones of Rockport, Ind.*

I am the oak, the sturdy oak, the king of the forests. I am stout. They make beams, spars, sills, fulcrums and what not from me that require strength. I grow fairly fast. I came into usefulness as the world came into need of heavy timbers.

I am dainty and refined as well as strong. I am used in making fine flooring, fine furniture and many other useful things. Please do not discard me from production. Please do not let me pass into oblivion. I am very very valuable. I deserve to be perpetuated.

6th. *Marcia Smith of Cincinnati, Ohio.*

I am a pecan plucked from the tree of a man who in the early years of his married life planted pecan trees in unused spots on his farm that were unsuitable for cultivation. As the trees grew into nut bearing trees his family of children grew. In the October days, with great gaiety, glee and happiness, the children would gather the fruit of those trees. The children grew to maturity and went to the city to work; but when those October days came they returned home and with similar happiness as of their youth they gathered the nuts from those trees. With pleasure I say I am one of those trees.

7th. *Jean Morris, Joyce Morris and Sandra Wright, all of Rockport, Indiana.*

We are a group of clusters, the filbert, the pecan and the walnut. We came from a nut farm within the bounds of Spencer County. This farm was planted and developed by a former enthusiastic member of your wonderful organization. He spent much time and energy in behalf of your organization. He developed the largest nut orchard in the county. I refer to Harry Weber, who came from a neighboring state and endeared himself to this community by his superb manhood, his genial disposition and his intense interest in his subject matter. We commend his efforts to others.

8th. *Virginia Mae Daming of Rockport, Ind.* She was carrying the former Reports of the N.N.G.A.

This cluster is plucked from a "Tree" of great magnitude and significance. Today it has its roots firmly set in Rockport, Indiana. Its branches reach from the Atlantic to the Pacific, from Canada to Mexico. Its influence is felt throughout the world.

Its inception was in Spencer County, Indiana, not specifically detailed, but in the main, by boys that were reared among the native nut trees of this community of which there were many. It was born in the great City of New York under the care of the late Thomas P. Littlepage, Dr. Wm. C. Deming, Dr. Robert T. Morris and Prof. John Craig. It was nurtured throughout the land of the detailed history you know much more than I.

It has had an enormous growth. It is a most meritorious organization. Language will not express the extent of its benefits to humanity and to civilization. It adds to the comfort of untold thousands of happy homes. It furnishes employment for thousands of people. It furnishes food of vital importance to many families. It is the main stay in the manufacture of all kinds and grades of furniture. It furnishes food for thought. It keeps the scientific and investigating minds busy in the constant development and improvement of its processes and benefits. Its possibilities are boundless.

That this "Tree" may continue to grow and develop in the future as it has in the past in the interest of humanity and help us to realize its importance and help us to continue its forces in accord with nature and nature's God is my earnest prayer. May God bless you one and all.

PRESIDENT MacDANIELS: Thank you very much, Mr. Bennett. You have made us feel most welcome in Rockport, as you have before on two other occasions. I don't believe that there is any other man who has welcomed this organization three times in the same locality.

We also thank you for bringing in the trees and the children to greet us on this occasion. It isn't very often that the trees themselves come into the assembly room to greet us, and we appreciate your effort in doing this for us.

We will now proceed with the business of the Association.

There appears to be no record of the members elected to serve on the nominating committee for this session. As near as we can determine this committee is as follows: Mr. Silvis, Mr. Allen, Mr. Wilkinson, Mr. McKay and Mr. Gerardi.

Is there a motion to approve these names?

The committee was approved by vote.

PRESIDENT MacDANIELS: This Committee will bring in a slate of officers of the Association for the next year at our final business session.

I will now call for the reports of standing committees. There are eight of these. The Program Committee. Royal Oakes is the chairman. The fact that we are having a meeting indicates the functioning of the Program Committee.

MR. OAKES: I believe I have nothing to report at this moment. I would like to say the other members did a good part of the committee work.

PRESIDENT MACDANIELS: We appreciate the part that all of you have played in arranging these meetings.

The Publications Committee, Editorial Section. Dr. Theiss, I believe, is not here. Dr. Theiss received the manuscripts and either had them read or read them himself.

The Printing Section of the Publications Committee, Mr. Slate.

MR. SLATE: Our proceedings are on the press and probably will be finished and in the mail this week.

PRESIDENT MacDANIELS: The Place of Meeting Committee. Mr. Allaman is the chairman. In the absence of Mr. Allaman, I present the invitation secured by Mr. Salzer, to meet in Rochester, New York in 1953. Their convention bureau offers very attractive facilities and the invitation is seconded by the Mayor, Joseph J. Naylor, the president of the Rochester Convention and Publicity Bureau, the President of the Rochester Hotel Association, the President of the Junior Chamber of Commerce of

Rochester, and the Deputy Commissioner of the Rochester Parks, which just about covers the board.

It doesn't seem to me worthwhile to read all of this material. What it boils down to is that Rochester would be a very good place to meet. The Rochester parks are very interesing places to go, and as I understand it, there are facilities which would not be expensive to the Association. Is that true, Mr. Salzer?

MR. SALZER: Yes, there would be no charge for exhibit rooms if they are held in the hotel, because we are classed as a scientific organization. And we would have the facilities of the Bausch Memorial Museum. There would be facilities for showing moving pictures or slides, and for an exhibit.

PRESIDENT MacDANIELS: It would be in order at the present time to take definite action on this Rochester invitation, if you care to do so. A motion would be in order to accept.

It has been moved, seconded, and carried that we have our 1953 convention in the City of Rochester, the dates will be determined by the Board of Directors.

The general thinking of the Board of Directors is that we will go to Lancaster, Pa. again in 1954, and in 1955 come back into the Middle West. Mr. Allaman has been working on the Lancaster proposal and I think there has been some spade work done in Michigan already. Have you anything to say about that, Mr. O'Rourke?

MR. O'ROURKE: We will be very glad to have you at Michigan State College at any time. Unfortunately, however, we do not have any nut plantings there. The nut plantings are either in the eastern part of the state or the western part. It's quite a drive either way.

PRESIDENT MacDANIELS: I don't think we have to make a commitment at this time, but it is something to be brought to the attention of the Place of Meeting Committee.

I think we might have a little further explanation from Mr. Best about his bacon breakfast.

MR. BEST: We said in our membership drive that anyone who would go out and work would bring home the bacon, and we further fortified the deal that we were going to furnish the bacon here at Rockport at this session. So in the morning over at Cotton's restaurant we will have bacon, all you want to eat, and the only requirement is that you either got a member last year in the membership drive we have been working on, or that you tried to get a member. That's all that's necessary.

MR. GRAVATT: You have spoken about the meeting in 1954. As you know, I have represented this country at the International Chestnut Meeting for two years. There has been some talk about the possibility of the N. N. G. A. inviting the International Chestnut Meeting to meet in this country in 1954 or '55. At the last meeting the delegates from Japan recommended that they meet in the United States in 1954. The matter is not decided, and I think if you will put off decision about Lancaster until later, it would be a little better.

PRESIDENT MacDANIELS: The committee on Standards and Judging, Mr.
Spencer Chase.

MR. SPENCER CHASE: Mr. President, we contemplated having a report on hickory standards for this meeting, but because of circumstances beyond our control, we didn't get the project under way.

PRESIDENT MacDANIELS: I will call on our secretary at this time for the report of the meeting of the directors.

MR. McDANIEL: There were several things brought up last night at the meeting of the Board of Directors of the Northern Nut Growers Association. One matter was the subscription to the American Fruit Grower magazine which we give our membership.

The American Fruit Grower had been selling subscriptions to the Association for its members at 30 cents a year. Since the first of July this year their rate is 50 cents. The opinion of the directors and committee members present last night was that we should drop that subscription to the American Fruit Grower for our members. It will be sent to all members who join for this year and up to the beginning of the next fiscal year. After October 1st, no subscriptions to the American Fruit Grower through the Association. Do we have any discussion on this proposal? (Considerable discussion followed.)

PRESIDENT MacDANIELS: I suggest that we hear the report of the Board of Directors and then act on the various items one by one in executive session.

MR. McDANIEL: You have heard something about the membership drive, and we will have more on that later. The directors suggested that we encourage more memberships, contributing memberships and sustaining memberships in the Association at $5.00 and $10.00 per year. Some of us feel we can't pay any more than $3.00 for our membership; others will be able to support the organization financially by taking memberships at the

$5.00 or $10.00 rate, and we are still offering our life membership at $75.00.

Another matter discussed was offering the set of 34 volumes of back reports in The Nutshell at the price of $20.00 for the 34 volumes now available.

We suggest also that the Association authorize the appointment of a Publicity Committee to work with the Membership Committee in attracting new members.

That is about all I have as the report of the directors' meeting last night, Mr. President.

PRESIDENT MacDANIELS: This matter of the Board of Directors reporting to the business session is a pattern which I think is a good one. The proposition has been placed before you as to whether or not you wish to continue our affiliation with the American Fruit Grower magazine. As you will recall, the reason the question comes up at the present time is that they have raised their rate from 30 cents a member to 50 cents a member, which is 50 cents of our $3.00, which with the 50 cents secretarial expenses leaves but $2.00 to run the society. As the Treasurer will explain to you later, we are in somewhat of a financial difficulty.

It has been moved and seconded that the Association subscription to the American Fruit Grower be discontinued.

This matter is up for discussion.

MR. MCDANIEL: We have much more space available in The Nutshell than in the American Fruit Grower, and there is the possibility of more frequent publication.

MR. DOWELL: If we could actually get it bi-monthly or quarterly, in place of the Fruit Grower, I think most all of us would be better informed and actually have more information. And The Nutshell is a very excellent means of showing somebody what the organization is about. You give them a copy of the American Fruit Grower, and if he is interested in nuts, most copies aren't going to convince him of much.

PRESIDENT MacDANIELS: I think this question is related to the appointment of a Publicity Committee which will explore what can be done to secure more publicity and give more information about nuts to our members than has been possible in the Fruit Grower.

The members of the Board of Directors felt that $300-plus is a high price to pay for what we got out of The American Fruit Grower.

(The question was called for.)

The motion is passed without dissent.

The question of authorizing the appointment of a Publicity Committee is introduced mainly as a matter for your information, also because it's much better if the society as such were to authorize such a committee. Do I hear such a motion?

Moved by Mr. Salzer, seconded by Colby and passed that the appointment of a Publicity Committee be approved.

I will ask for the report of the Treasurer, Mr. Prell.

Treasurer's Report

MR. PRELL: Mr. Chairman, ladies and gentlemen, Mr. Best has asked that I help in connection with his report. That certainly is not because I can make his report better than he can, but probably because a new member is not a new member until his check has arrived and has been recorded, and I happen to have those figures. I will be happy to do that, but perhaps we should start first with the report that the President has asked for, the Treasurer's report.

I imagine that you are uninterested in an itemized, detailed report of receipts and expenditures; I imagine you are interested in the question: How are we doing? We are not doing too well. The annual report for this year indicates that our financial condition is not satisfactory. For the second successive year we have spent more money than we have taken in, and that would be the third successive year, if it hadn't been for the fact that due to the lateness of the publication in 1950—that it, the annual report—we did not pay for an annual report that year. That means there are three years in a row that we have gone downhill.

The picture is not entirely black, however. There are some bright spots. For instance, all our bills are paid. Second, we have money in the bank. Third, our $3,000 investment in Government bonds is still intact, and fourth, our deficit this year was less than it was last year, which may indicate that we have already touched bottom and are starting up.

The cause of our deficit is easy to put your finger on. We are operating on budgets that are ten years old, and costs have gone way, way beyond. Dues were increased several years ago, but even at that time they were not increased adequately, and since then costs have skyrocketed.

The membership situation is not too bad, though the cost situation is bad. The two don't jibe at all. The reason we have a lesser deficit this year than last is Mr. Best's work and the work of his vice-presidents in increasing the membership, and the results of that work; I think, have only begun to show.

Specifically, we came within $417 of collecting enough money this year to pay our expenses. It was over $500 last year, making a total of a thousand dollars that we have spent above our receipts. While we have some money in the bank, there will be a bill due in about 30 days on the publication of the annual report, that will be mailed within the next few days. And that will take all the money that is in the bank, plus what we are able to collect in dues immediately, and I hope that many of them are paid at once. But that still leaves us without money to operate through the year, and by January, unless conditions change, we will be borrowing money.

The Board of Directors has discussed this. They have some thoughts on the subject which will be presented to you by Dr. MacDaniels. I think that one of the obvious things that you all think of and I may mention is the matter of increased membership. That's an obvious solution, and as I said a minute ago, it's a very possible solution.

The work that was started by Mr. Best last February is only now beginning to bear fruit. New memberships, even as late as this for this year, in August, are coming in very, very well. I personally see no reason why the membership cannot be increased to a thousand members next year, providing all of us bring in a member or two.

I asked a friend of mine on The Country Gentleman for some data on state population compared to farm population. I forget just exactly now how it runs on various states, but I do recall Indiana. We have a population here of four million people. There are about 700,000 of these people on 166,000 farms. The farms in this state produce a wealth of $75,000,000 a year. With 700,000 farmers in this state and population of 4,000,000 with a wealth of $75,000,000 a year, it would seem to me that the State of Indiana should have more than only 39 members. Out of that group we should certainly increase that ten times. We should have 400 members, and if the same proportion is carried throughout the nation, why, this organization can easily obtain a roll of 7500 to 10,000 members. A thousand members next year should be a pushover. So much for the financial report.

Mr. Best's campaign started last February. His vice-presidents were given material and the inspiration to work for new members, and they responded. For Mr. Best I compiled the list of the new members who have been brought in, with the people who have brought in the greatest number, but that thing went galley-west in the last few days by the strong finishers. Mr. Best himself came in yesterday with a pocket full of 11 new members, and he already had a couple on the list. Up to that time—and I am not giving credit to the Secretary, because several of the members that show his sponsorship have come naturally through his office. So disregarding the sponsored members of the Secretary, Spencer Chase was top man, up until Mr. Best upset him yesterday, followed by Dr. Rohrbacher, who was a late finisher with members who were not recorded in this report. All through the year it was a battle between Pennsylvania and Illinois as to who would have the greater number of members.

Illinois, with 36 members, hopped up to 60, and Mr. Best's 11 make 71. And just this morning they got two others from Illinois, making 73. So I think Illinois has the second place position firmly nailed down.

Last year we had 563 members all together. This year now we have 170 new members. We can't add that to 563, because in every organization there is a loss of membership every year, and it's to be expected that our membership should have a 10 per cent turnover through circumstances of people leaving their places where they have their nut tree plantings, deaths and other circumstances. So there was a net gain of 86 members to date.

TREASURER'S REPORT

August 25, 1951 to August 18, 1952

RECEIPTS

Membership Dues $1,907.00
Sales of Annual Reports 190.00
Interest on U. S. Bonds 37.50
Donations 48.95
U.S.P.O. Unused Balance, Permit 3.05
Petty Cash 1.97

TOTAL $2,188.47

DISBURSEMENTS

41st Annual Report (Pleasant Valley) $1,375.86
Plates and printing, 900 copies $1,271.16
Envelopes, 2500 31.65
Mailing 73.05
The Nutshell 86.55
Printing & mailing Vol. 4, No. 3 28.64
Printing & mailing Vol. 5, No. 1 57.91
American Fruit Grower 191.60
582 Subscriptions at 30c 174.60
34 Subscriptions at 50c 17.00
Urbana Meeting 163.68
General Expenses 20.28
Reporting & Transcribing 143.40
Secretarial Help, 50c per member 317.00
Stationery and Supplies 179.81
Association Promotion 114.91

Application Folder, 5000 90.02
Supplemental Folder, 650 17.69
Things-of-Science 7.20
Secretary's Expense 77.23
Treasurer's Expense 94.04
Dues, American Horticultural Society 5.00

TOTAL $2,605.68

Cash on deposit, First Bank, South Bend $1,313.78
Disbursements 2,605.68

$3,919.46

— — — —

On hand August 26, 1951 $1,730.99
Receipts 2,188.47

$3,919.46

— — — —

U. S. Bonds in Safety Deposit Box $3,000.00

I know that Mr. Best has still some more material that he will supply to any of you who are anxious to go out and help in getting the new members. It's only a matter of every person getting a couple, or like Spencer Chase getting 10. That would put us well toward our goal of a thousand members, on which the Association probably can operate without deficit. I thank you. (Applause.)

PRESIDENT MacDANIELS: Thank you very much, Mr. Prell. We are very much indebted to you for your business-like handling of the affairs of the society. It is sometimes bitter to know the facts, but the only way that we are ever going to get anywhere is by knowing the facts and facing them.

Either fortunately or unfortunately we are not like the federal government, which can go on piling up deficits. We have to do as each one of us as individuals has to do: If our operating-expense exceeds income, we either have to get more income or cease out-go. That is the situation under which we are confronted at the present time.

A little later we can take up some of the things we have in mind. Did you have a further report, Mr. Secretary?

I think probably the Treasurer stole some of the thunder that you might otherwise have.

MR. MCDANIEL: He did that, and the Membership Committee also. You know something of the activities of the secretary's office during the current year, a matter of getting out three issues of The Nutshell and assisting with the editing of the annual report, which I hope you will receive about the time you get home.

One other activity in which the Secretary participated, in addition to the usual task of answering letters to beginning nut growers, was this project "Things of Science". Perhaps Dr. McKay could tell us more about that. Is Dr. McKay in the room? Will you come up now?

DR. MCKAY: We being near Washington, were, of course, the logical people to come in contact with this suggestion early when it was made. As a matter of fact, the very beginning of this movement goes back to Harry Dengler. Some of you may know of him. He is Extension Forester at the University of Maryland and is also Secretary of the American Holly Association.

Harry Dengler was very much interested in this "Things of Science" program and happened to mention to the Science Service paper, of which

Watson Davis is editor, that it would be a desirable thing to work up a test on nuts.

For the benefit of those of you who do not know what "Things of Science" is, it is a movement sponsored by Science Service, located in Washington, D. C, whereby 12,000 subscribers to "Things of Science" receive every month a little kit through the mails dealing with all kinds of subjects in science. It is usually a little box, as in the case of the one on nuts, or it may be simply an envelope with some things in it to taste. The idea is to give people all over the country who are interested enough to pay $5.00 a year one kit a month, each one dealing with a different phase of science.

Many groups subscribe to this service; for instance a boy scout troop, libraries and industrial plants. So it goes to literally many thousands more people than the 12,000 actual subscribers that it has.

So when Science Service came to us and said, "Would you be interested in helping us work up a kit on nuts", naturally, we wanted to do what we could towards helping these people, and our first thought was this organization as an official sponsor for it. So we contacted the directors, the officers, Dr. MacDaniels and J. C. McDaniel, and as a result, the Northern Nut Growers, through its board of directors, because we had no other means to authorize it, went ahead and sponsored this move.

To do it, we approached the California Walnut Growers Association, the California Almond Growers Association, the Northwest Nut Growers Association, and the Southeastern Pecan Growers Association, with the idea of having their names mentioned in the kit, and in return they would furnish samples to distribute. The Northern Nut Growers Association furnished the hickory nut samples. The kit was composed of, as I recall, six different kinds of nuts—Persian walnuts and almonds from California, filberts from the Northwest, Pecans from the Southeast, hickory nuts from the Northern Nut Growers Association, and pistachio nuts furnished through the Department of Agriculture by Captain Whitehouse at Beltsville. He secured the pistachio nuts from the trees in California. The kit was composed of a little box about four inches long, an inch and a half deep and three inches wide, containing two or more nuts of the various kinds, together with a brochure that we helped the science people work up. Dr. MacDaniels and the various cooperating groups worked up this brochure of information. The kits include a set of directions for the subscriber to follow in using the material. There are several different possibilities, all along the lines of scientific experimentation.

The idea is to get these youngsters and young people to become familiar with different kinds of nuts.

I think that's all I should say, Mr. President. That covers pretty well the effort that was made and those who made the effort. (Applause.)

PRESIDENT MacDANIELS: Thank you very much, Dr. McKay. This project is one in which there were deadlines as to time, and we had to work rather fast. Air mail, special delivery, the long distance telephone and telegraph played quite a part in it. The Science Service was paying the cost of assembling and mailing. The only cost to the Association was for the hickory nuts.

MR. MCDANIEL: We were late on that and unable to get the quality nuts we would like, but we did get enough to fill the kits, not all of which were worthy.

PRESIDENT MacDANIELS: We would like to have secured Carpathian walnuts, but the nuts from known sources of supply were so discolored with husk maggot that we were ashamed to send them out. We were not able to locate and to furnish any considerable amount of any kind of northern nuts. Twelve thousand of these kits went out, and each one of them is in a position where it probably contacted a dozen or more on the average, so that I am sure as a result of the effort a great many people not only became more familiar with nuts and their various sources and uses, but also learned that the contest was sponsored by the Northern Nut Growers Association. Mr. Prell, who knows something about advertising, thought it was a very worthwhile project.

That completes the reports of the officers and of the committees. We will now take ten minutes recess.

PRESIDENT MacDANIELS: The session will be in order.

As your treasurer said, there are several other things which we discussed in the directors' meeting. We discussed this matter of how, the situation being such as it is, the Association could improve its position through gaining more members and through either making more money or cutting down expenditures.

The Publicity Committee was one of those suggestions, who were to explore this matter of getting better publicity for less money. That is, whatever publicity we got from the American Fruit Grower cost us about $300, and we think we can do a lot better in some other way.

Another matter was to place the financial situation of the society squarely before the membership and ask that as many as could and felt so inclined take out a contributing or a sustaining membership. We felt quite strongly that raising the dues was not the answer, because there are a lot of people sort of on the fringe who don't work too actively for the society but who do take out regular memberships but who, if we raised the dues even another 50 cents, would probably fail to renew their memberships. So that at least for the present we are not going to go ahead on that basis, unless you want that to come up for further discussion.

Another point which we, I think, should explore was the matter of advertising in the proceedings. Some other associations, the pecan association, particularly, as Dr. McKay pointed out, make a substantial part of their revenue from advertising in the proceedings. We have tried that before, but times have changed, and I think it should be considered again.

Then the matter of speeding up sales of sets of the proceedings to libraries, that is, further publicity in The Nutshell about sets that are for sale and, perhaps, circularizing the library lists to sell complete sets, or as complete as we have.

Another matter that might be explored is having some kind of a "give-away program", some inducement for those who take out memberships for the first time. Other societies do it in one way or another. Unfortunately, our material does not lend itself to that sort of thing as well as some others, but we might be able to give nuts of Carpathian strains that could be used as seed nuts, or perhaps the hybrid hazels.

MR. MCDANIEL: One suggestion made in a letter from Dr. Crane was to distribute hybrid walnuts to grow to fruiting size. That might be explored if there is a source of enough seedlings or seed nuts of Juglans Regia crossed with Juglans Nigra.

PRESIDENT MacDANIELS: We would welcome any further suggestions which you may have, either as to saving money or making money, or increasing our membership, which amounts to making money, of course.

Another thing that might be done to present the possibilities of nut growing to your communities is to sponsor exhibits at your own county or state fairs.

Mr. Slate wanted to make a comment along these lines.

MR. SLATE: That matter of urging sustaining and contributing memberships has been mentioned by you. I think it would be one of the best things we could do to send a statement of our financial condition to the members of the Association pointing out the need for additional funds and suggesting that all who can possibly afford it take out sustaining and contributing memberships. It seems to me that this is just about the only alternative to increasing the dues. I am not sure whether an increase in the dues would result in the loss of many members or not. Perhaps they are getting rather used to the higher price level, and it might be well to have an expression of opinion from some of those here as to whether they thought there would be serious objections to an increase in the dues. Surely, there are many who can afford to carry sustaining or contributing memberships.

PRESIDENT MacDANIELS: That is the opinion of the Board of Directors. Mr. Slate has raised a question as to the validity of the conclusion of the Directors regarding the advisability of raising the dues. Our thinking was that to raise the dues beyond the present level would result in sufficient loss of membership to offset any gain in revenue. The last time we raised the dues what was the effect?

MR. MCDANIEL: When we raised the dues to $3.00 we had a membership of 650. It dropped to about 580; a loss of 60 or 70.

MR. PRELL: We in effect raised dues 50 cents this morning. It won't affect new members, but it may cause some of the older ones who are members to drop. They know that at present 50 cents of their dues are going to the Fruit Grower; now they aren't getting the Fruit Grower.

MR. MACHOVINA: They were getting for $2.50 what they will now get for $3.00.

PRESIDENT MACDANIELS: Any other discussion?

MR. KINTZEL: I have given this problem of increasing the membership quite a bit of thought, and have an idea which might be used. Let's see by a show of hands how many live in the city but own farms outside of the city.

PRESIDENT MacDANIELS: The question is how many live in the city but have farms outside. Sixteen or 17, probably about 20.

MR. KINTZEL: You might call me a city farmer. Like many other city people, I own a small farm near the city in which I live, which is Cincinnati, Ohio. I am intensely interested in the work of the N.N.G.A. There must be many others who, too, are owners of land but who use the land for experimental farming and to get a little diversion from the daily grind in the busy, noisy city. These people would consider it a favor to have their attention called to the interesting work of our organization.

A practical plan for getting in touch with this reservoir of future members is to secure the names and addresses of such land owners from the records at the various county court houses fringing the cities. A personal letter should be written to these future members. A friendly invitation to join the N.N.G.A. should be extended, and a printed brochure describing and explaining its work and objects should be included.

I believe that by working systematically on the city dweller, who also owns acreage outside the city limits, we could give our membership list a big boost.

PRESIDENT MacDANIELS: That is a good suggestion for the Membership Committee.

Is there anything further?

MR. CALDWELL: This is not a suggestion, but a comment following up the idea of the previous speaker. In Syracuse there was a woman with an estimated 160 acres of land, who about 15 or 16 years ago became interested in planting hybrid chestnuts. Unfortunately, the land was not suitable for raising chestnuts and the two or three hundred trees she planted failed to grow. I don't think there are two alive there now. So you will have to be a little bit careful in encouraging city people to plant nut trees. She spent a lot of money and right now if you mention that, she will just practically tear you apart. She wasted money and time, so be careful in getting people going too strong unless you are sure the trees are going to grow for them.

MR. SNYDER: According to the chart outside, cutting off the Fruit Grower will leave us just a few cents per member in the red.

PRESIDENT MacDANIELS: Right.

MR. SNYDER: Well, don't we have $3,000 in bonds? What are they for, if it isn't to tide us over a hard period like this?

PRESIDENT MacDANIELS: That is a suggestion for the Board of Directors.

MR. SNYDER: If inflation keeps up, the bonds will be worth nothing. We might as well use them up. I would suggest we use every method to balance the budget without them, but if necessary, use some of them up. If it is necessary, use the bonds to balance the budget.

PRESIDENT MacDANIELS: The question of whether or not we use the bonds, I think, would have to be considered very carefully. I think one of the Ohio men has a suggestion.

MR. DOWELL: This discussion would follow along with that on membership. The active members of the Ohio section were organized back in 1946, and in 1948 the national body put in its by-laws a provision that there could be state sections formed. That is Article 1 and also Article 2, that you could have affiliated bodies. Now, as far as I know, there is no other state section.

MR. MCDANIEL: Michigan has one, now.

MR. DOWELL: Michigan has not actually affiliated yet, and when it does come in it will be an affiliated society. According to the by-laws it will not be necessary for all its members to be members of the N. N. G. A.

Now, we feel that some strong state section is the main support in membership interest and a lot of other lines, and I think that if you check the rolls you will find where you have had a state organization, whether it's affiliated or otherwise, particularly Ohio and Michigan, that our membership has not really dropped down in total numbers. Of course, there is a turnover every year. If it has dropped down, it's been slight in comparison with the overall drop down.

MR. MCDANIEL: Ohio is only holding its own now. You have one more member than you had a year ago.

MR. DOWELL: That's right, we are holding our own, and previous to this last run, the total number in the Association was down a hundred. That has not dropped in Ohio, which has the state section. Neither has it recently in Michigan, which has recently organized the Michigan Nut Growers.

The Executive Committee of the Ohio section wishes to present the following resolution for the consideration of this body:

RESOLUTION

> "WHEREAS we feel that membership in a state section has been a definite advantage in maintaining and increasing membership in the National Organization, as has been demonstrated in the Ohio Section of the N. N. G. A.;
>
> WHEREAS a National Organization becomes strong because of its strong local sections which help maintain interest;
>
> THEREFORE the National Organization should encourage and foster the formation of local sections.
>
> We therefore submit the following motion: That the N. N. G. A. amend its constitution to provide for the organization of local sections. These amendments should include the following provisions:
>
> 1. Membership in the N. N. G. A. shall be a requirement for full membership in the local section; however this shall not exclude local sections from accepting associate members.
>
> 2. That each member of the N. N. G. A. shall automatically become a member of a local section when he resides in a location where a recognized local section exists.

3. Wherever a local section has become established, the local chairman shall serve as vice president of the N. N. G. A. for that area.

4. The N. N. G. A. shall refund to the treasurer of each local section ten percent (10%) of the N. N. G. A. dues paid annually by members of that section."

PRESIDENT MacDANIELS: I conclude that you are presenting this for the consideration of the Association. It would be an amendment to the by-laws, I take it, rather than the constitution. Such an amendment would have to come up for consideration at the next meeting after consideration by the Board of Directors; either that, or else vote on it by mail.

MR. DOWELL: It is purely a motion now, if passed or rejected. But if it is passed, then previous to the Rochester meeting, the proposal would have to be in a suitable form to be either passed or rejected for the by-laws.

PRESIDENT MacDANIELS: We have this resolution in printed form. That will be transmitted to the Board of Directors for consideration at the next meeting.

MR. DOWELL: We make it as a motion that the mass accept or reject it here.

PRESIDENT MacDANIELS: The motion is, then, to accept the resolution and present it to the Board of Directors. Is that right? Is there a second?

MR. KINTZEL: I second it.

PRESIDENT MacDANIELS: Are there further remarks? If not, all in favor, signify by saying "Aye." (Chorus of "ayes"). Opposed? (None.) It is carried.

MR. O'ROURKE: I am very sorry I was not recognized before the vote was taken.

PRESIDENT MacDANIELS: I am sorry.

MR. O'ROURKE: I am speaking, I think, for the Michigan Nut Growers, of which we have quite a group here today, and we are quite anxious to maintain an independent state organization. We feel that it is perfectly all right for this motion to have been adopted as it has been, if there will be no attempt made to delete that section which now refers to affiliation.

PRESIDENT MacDANIELS: I think there would be no attempt to do that.

MR. O'ROURKE: Is that clearly understood that there will be no attempt made to delete the section on affiliation?

MR. DOWELL: That is the understanding. Now, there are two ways in the present by-laws. Now, this would either be a third or replace the first. It would have nothing to do with affiliating groups.

PRESIDENT MacDANIELS: I think that is right, and I think the thing to do, Mr. Dowell, would be to be sure that the new president is apprized of the Michigan point of view in that regard. He will be the chairman of the new Board of Directors, and this is simply a motion to consider it. It doesn't go any further than that.

Is there any further business to come before this group at this time? If not, the other item on the agenda, as it is stated, I believe, is a presidential address.

The Forward Look

Presidential Address, by L. H. MacDaniels

As the retiring president of our Association, it is a time honored custom and a privilege to give what is often referred to as the presidential address. I do not have in mind giving an address but rather to consider with you informally the present situation of the Northern Nut Growers Association and to give my ideas as to what we might do to improve our position and forward the purposes for which the Association was organized in 1910.

Time does not permit recounting the history of the development of the Association. This has been done on several previous occasions. I will, however, go back to the 1945 report in which under the title "Where Do We Go from Here" I tried to pick up various aspects of the condition of the Association immediately following the war and point out areas to which special attention should be given at that time.

Considering our situation in 1952, it appears that many of our problems are about the same as they were in 1945 although in some areas definite progress has been made. A quick look at our problems then and now is perhaps pertinent to the present discussion. One of these is variety evaluation. This still remains one of the important areas where we need much more information particularly as to the success or failure of different named clones of nut trees in various regions. Perhaps it is time for us to carefully summarize whatever data we have accumulated as to the adaptation of varieties or at least make plans for extending a program of evaluation. Since 1945 our survey committees have been active and have secured information that will certainly be helpful.

The problem of judging standards has been clarified somewhat. It is my personal opinion that the judging schedule for varieties of black walnuts worked out with the assistance of Dr. S. S. Atwood is on a sound basis and might well receive much wider use. Following along somewhat the same pattern, suggested schedules have been proposed for the hickories and butternuts. These should receive further consideration and adoption, if approved at least on a tentative basis. A schedule for Persian walnuts is very much needed as indicated by the recent contest in which confusion occurred related to there being no recognized standards of evaluation. With the Persian walnut such matters as the method of cracking and the importance of such characters as sealing of nuts, recovery of whole halves and others should be agreed upon.

Our procedure in naming varieties is still somewhat chaotic. Possibly we should adopt the general pattern of the American Pomological Society. Their example of setting up an approved list of varieties for planting on a regional basis is worthy of consideration. Even though such a list were tentative and incomplete, a start which would embody the best information we have would be valuable.

Securing new varieties of, hardy nut trees through breeding has made some progress. Most encouraging is the work of the Federal Experiment Station at Beltsville where Doctor Crane and Doctor McKay and their associates are using modern techniques in securing new varieties of hardy nut trees. Some progress in hybridization, of course, has been made, particularly with the filberts, the hybrids developed by J. F. Jones, G. L. Slate, S. H. Graham, Heben Corsan and some others, showing great improvement over previous European varieties in their adaptability to the northern United States. At the present time there are filbert varieties of hybrid origin better than those in the nursery trade which should be propagated and made available. Work with the Chinese chestnuts has also been valuable.

It is my opinion, which I believe is shared by most of those who are familiar with progress in securing new varieties, that we are not likely to find in the wild, varieties or clones which show any marked improvement over those already found and named. There is, of course, always the possibility of the "perfect nut" arising as a chance variation. The recent walnut and hickory contests, however, have been somewhat disappointing for they have not discovered any variety of black walnut better than the Thomas for instance, or a hickory much better than some of those located years ago. This does not mean that members of the Association should not keep a sharp lookout for new varieties occurring spontaneously which will be better than existing sorts. It does mean, however, that if real "breaks" are to be secured, it will be necessary to apply some of the more effective techniques which are known in the plant breeding field. Any such program is a long time project and can only be effectively attempted by experiment stations, or by some of the young men, who begin now to make crosses under the direction or at least with the advice of those who are familiar with plant breeding techniques.

Progress has been made in the Association organization. The constitution has been thoroughly overhauled and amended, particularly to provide for regional groups. Certainly such groups are to be encouraged and have done and will do much to strengthen the national organization in the various states. It is my personal opinion that these regional groups can be of particular value in working with the experiment stations and legislatures to promote the interests of the Association. The state associations should be on the alert to build on the interests of conservation departments as related to wildlife preserves and sportsmen's clubs and other agencies which put the growing of nut trees in proper perspective. I am not at all in favor of securing either federal or state support for every minor project which comes along. However, the Northern Nut Growers Association need make no apologies for its program, particularly as it is related to the conservation of our natural resources; to the promotion of better living on the farm and those values which are real and great, even though they do not show up large in dollar value of crops produced.

Unfortunately, projects in nut growing have been started in various states, particularly Ohio and Michigan only to be eliminated before they really got under way because of lack of support. Experiment station directors, if they are confronted with a shortage of funds, are likely to run the blue pencil through items which cannot be backed up with economic considerations. The approach of the Northern Nut Growers Association it seems to me should not be to seek support on an economic basis but rather on the basis of better living on the farm, improvement of gardens and farmsteads and the advantages of growing nut trees as compared with any other horticultural activity. There has been a real increase in the importance which is given to this approach in recent times and an active state association, which can keep in touch with local conditions and call on the national association for additional support, will certainly be of great assistance in the future.

I personally am not in favor of any sort of a set up by which the national association gives a kick back of national dues to a regional association. The dues are inadequate for the national association at the present time. Looking at the whole situation with some perspective, it would seem that the regional associations might contribute to the national association rather than the reverse. If the constitution and by-laws of the Association are not such as to make affiliation with the national association and the formation of regional associations easy, they can readily be changed to secure the very best pattern that can be devised.

Perhaps one of the most acute problems with which the Association is faced is the struggle to keep financially solvent. We are all aware of our changing economy, particularly the increased costs of printing and in fact of everything that our organization uses or needs, even postage. In my thinking, the finances of the Association are much the same as those of an individual, who is confronted with expenditures that exceed his income. The things that have to be done are obvious and the same in both cases. One is to spend less and the other is to secure more funds. In the judgment of your directors and executive committee, expenditures have been reduced as low as is safe in order to keep a going organization. Members join the Association for the value which they get out of it and a large part of this value is in the form of reports, newsletters, information made available and the organization of annual meetings. If these services were discontinued or curtailed, membership falls off. This has been the experience of other plant societies, of which there are many.

In my judgment retrenchment is not the answer in the present situation. Securing additional funds is the best forward-looking policy. The question comes up as to how this may be done. Experience in our Association and I believe other associations as well, has shown that $3.00 is about as far as dues can be raised. There comes a point with every society when, if the dues are increased, there is a falling off of membership, which more than offsets the gain. Other obvious procedures are: (1) increasing the number of members; (2) providing different types of memberships to encourage larger contributions; (3) gifts; and (4) special fund raising projects. Of these various ways and means, certainly increasing the number of members is by far the more promising. The overhead of the association is not increased with additional memberships anywhere near in proportion to the contributions of those members. This is particularly true for additional copies of the report and general office expense. The drive for new members under President Best's leadership has produced gratifying results and I believe if this is continued effectively through the next few years, a membership increase can be secured that will assure the Association's balancing its budget. Somewhere in the neighborhood of a thousand paid memberships would solve most of our financial difficulties. Provision is already made for different types of memberships and it is to be hoped that many who can do so will join the contributing member class at least until we are out of our present financial woods.

Other societies raise considerable revenue through special projects such as the sale of publications of one kind or another, seed distribution or slide rental. The type of material with which the Northern Nut Growers Association deals is not comparable to some of these other organizations but certainly the possibilities of revenue through special projects need to be explored.

Research with northern nut trees is exceedingly important from the standpoint of accomplishing the objectives of the Association. The matter of breeding new varieties has already been touched on. Other types of research are such that a large part must be carried on by experiment stations which have a continuing program. Much has been done in securing observational information by Association members themselves but some problems are such that they must be continued over a long period of time and set up with adequate checks and provision for securing significant data. Otherwise the results are of no real value. Granted we need all the sound observational experience that all the members can bring to our problems, there are still aspects of culture of northern nut trees that need continuing program of scientific research.

Fortunately, much of the cultural information secured with nut crops of economic value is directly applicable to northern nut trees. This is true of the work with northwestern filberts, western walnuts, southern pecans and even the tung industry. There comes a point, however, when information thus gained needs to be checked under the specific conditions where the crops are grown and very little research has been done in the northern states where the hardy nuts are important.

Of special importance to the northern nut growers is the control of diseases and insects. At the present time the bunch disease of walnuts is becoming increasingly more troublesome and very little is known as to how this is spread or how it may be controlled. In my own filbert planting, the hazel bud mite during past years has made the crop practically a failure. Little apparently is known as to the life history of this insect or when miticides might be applied. Examples such as the bunch disease and mite damage are multiplied many times with other diseases of local or regional importance. In my thinking our best hope for getting something done is to encourage the Departments of Entomology and Plant Pathology in the experiment stations to take up these disease and insect problems, which might be attacked by graduate students as thesis subjects, even though the economic importance is not great.

As I see the situation of the Association, there is need for its members to produce more nuts of better quality. Nothing intrigues the interest of potential members as much as actually seeing and tasting locally grown samples of nuts of superior varieties. On several occasions I have tried to assemble collections of nuts for exhibit or to buy them for one purpose or another and found great difficulty in finding sources of supply. This was particularly true in the fall of 1951 when we were trying to assemble nuts for "The Things of Science" project. We wanted very much to secure Carpathian walnuts that could be sent out and used for seed purposes. There was no source to which we could turn. In several possible sources of supply, husk maggots had so infested the crop that the nuts were discolored and unattractive. It might have been possible to secure enough black walnuts to include in the kit but the problem of state quarantines against the bunch disease could not be easily adjusted.

Finally I believe the Northern Nut Growers Association is doing a very significant work. Our emphasis at the present time at least might very well be on nut growing as a hobby and for conservation, for better shade trees and for better living on the farms and homesteads rather than to emphasize the commercial angles. This will come in time if it can really be demonstrated that growing northern nut trees is a profitable venture. In these days of job specialization everyone needs a hobby and an outlet for special interests. I know of few other fields of endeavor for those who like growing things than the rewards that are to be found in the growing of hardy nut trees.

MONDAY AFTERNOON SESSION

The Monday afternoon session was convened at one o'clock p.m.

PRESIDENT MacDANIELS: The afternoon session will please be in order.

The first paper this afternoon will be, "The Future of Your Nut Planting," Mr. W. F. Sonnemann, Vandalia, Illinois.

The Future of Your Nut Planting

W. F. SONNEMANN, *Vandalia, Ill.*

Ladies and gentlemen, it is a great pleasure to appear before the Northern Nut Growers Association. I am just a sprout as far as nut growing is concerned, when we consider the age of some of our old hickory nut trees.

About 25 years ago, I became interested in nut growing and, in particular, the river-bottom hickory nut tree. Then we had so many nut trees growing in the bottom that we never thought of trying to plant a tree or look after one. People could gather all the nuts they wanted and often the trees were cut just to get the nuts. They'd lay a stick of dynamite at the base of the tree to shake the nuts off.

After a few years of that, I thought we might do something to save the nut trees for the future generations. That's when I first started to plant some nuts. Incidentally, I made a big mistake, by not joining the Northern Nut Growers Association.

Naturally, I wanted the largest pecan I could find. I went to the St. Louis market and bought and planted nice Papershell pecans—very nice pecans, but the trees do not mature their crop. Mr. McDaniel and I tried to top-work them, but that's a big job. Had I joined the Northern Nut Growers Association, I could have avoided a lot of those mistakes.

There are some things that I found out in practicing law that can very well apply to nut growing. If you will pardon the reference to personal experience, I can bring forth to you about four situations. One, a good, close friend of mine had a vacant lot close to his home. He had been planting nut trees and papaw trees and persimmon trees for years. On this vacant lot he had a 25-year-old Busch walnut growing back on the alley, on the lawn was a beautiful Japanese flowering cherry, and there were two pecan trees in the yard proper. He sold the lot to a neighbor whose wife was just crazy about flowers, little dreaming that those trees would ever be cut down. I don't believe the ink of the recorder had been cooled or dried before that English walnut was cut down, the Japanese cherry grubbed out of the front lawn, and one of the pecan trees was cut. It just about broke the old owner's heart, and all he could say was, "I am just disappointed in my neighbors." And now there is a house being erected there, and the pecan tree that was 12 inches in diameter was cut. That could have been prevented, had this man given thought to the future.

Another man, named Hagen, who was instrumental in getting me interested in nut growing, had a nice group of river-bottom shellbark trees growing in his field. One of these has been propagated and named the Hagen, and although it isn't a good cracking quality, it's a very large nut.

A pipe line was laid close to that field, and this man had the fore-*sight to put a clause in this pipe-line right of way which gave him the protection of collecting adequate damages for the destruction of the trees. Didn't even need a lawyer, which is something bad for the law business. It is a suggestion, that when a pipe line, or telephone company is buying a right of way, it is possible to protect your interests in valuable trees.

Another instance of protecting nut trees was when the new U. S. Highway 40 was built across Illinois. I had the job of condemning the right of way and when the engineer and I were out walking over it we noticed a fine group of hickory nut trees on the hillside. I remarked what a nice group of trees it was. He said, "Yes, that's going to be a borrow pit up there." I said, "You mean they are going to destroy those trees?" He said, "Yes, dirt from this borrow pit will make the fill across this bottom."

I said, "Why can't we get the dirt somewhere else? Dirt is dirt."

And the engineer said, "Well, that's the plans." We had a little contrariness there, and I had to threaten to drop the case as far as that tract of land was concerned. If you fight long enough and hard enough in such cases you may find some other person who is interested in nut trees. We did; we found an engineer higher up, and that group of hickory trees is now a picnic area. They used a borrow pit somewhere else, and it gives me a great pleasure to drive past that group of hickory trees and see them still standing there. In the fall of the year you'd be surprised at the number of people at that picnic area, and they keep those hickory nuts picked up clean as fast as they fall.

In our county hospital just started they happened to select a piece of ground I own an interest in for a county hospital. On that are some good hickory nut trees. I told them they'd never get the land until they made some arrangements in regard to those nut trees. The engineer that designed that hospital must have had some sense, because they are building a canopy around one of the trees adjacent to that hospital, and have arranged to cut only one scrub oak. The other trees will be mentioned in the deed with restrictive covenants to protect them.

If you sign anything a company gives you, you are liable to have anything cut on your land. Remember the saying that "the big print gives it to you and the fine print takes it away." And it's the fine print you want to watch in all your right of ways or in your condemnation proceedings.

I know a man who had almost 160 acres of river-bottom hickories. During his lifetime he was very careful about those trees. He would cut the brush around the trees and harvest those hickory nuts as if it was a crop of corn or beans. Upon his death his children were scattered over the various states. They didn't care anything for this hickory grove. It's been cut. Now there is a bulldozer in there trying to clean out those hickory stumps. They are not making much progress. All you now have in that farm is 160 acres of old tree stumps, wild honey-suckle vines, poison ivy and poison oak, and even a coon hunter gripes when he has to take his dogs through there on a coon hunt. Those heirs care nothing about it.

In selling land it doesn't make any difference whether it's a sale to a neighbor, or to a friend or a stranger, you should protect any trees that you have growing upon that land by what we term a covenant running with the land, and that means if a deed is made it will provide that certain trees shall not be cut within a certain period of time. In one case where I am forced to sell some land I am protecting the trees for 10 years.

Each of these situations requires research under your own state laws. I had hoped to be able to tell you something definite and precise as to each situation, but when I considered the membership in the Northern Nut Growers, the many states it covers and the great difference in the state laws, it's just impossible to lay your hand upon one set of facts that governs. You should consult your attorney who is dealing with your transactions and tell him specifically what you have in mind and what you want to protect. He will know whether your state recognizes covenants running with your land and what provision can be made to protect trees that you want to save or secure damages.

Remember, in any transaction, if it is not in the written instrument that you sign, it's just an oral agreement that you make on the side, and it doesn't mean a thing. It has to be in the paper that you sign.

As I mentioned briefly, in what they call "eminent domain", the state has a right to take property for public use. The only thing you can do there is just get your head square and fight, and if you are stubborn enough, you may find someone in the organization that you are dealing with who has some interest in trees. They may not be members of the Northern Nut Growers Association or any tree association, but there are some people who appreciate trees and who do realize how long it takes to have a nice pecan tree or nice hickory nut tree growing.

If they call you contrary, that you won't give in to anything, let them call you contrary, let them call you nuts, but you can protect your trees and make sure that their future is secure.

What will happen to your trees after you are dead? Each individual's situation has to be considered separately. In many states you can provide by will to whom you want your nut planting to go, or you can, by making a trust, give the trees to trustees with certain powers and duties to care for and manage them for a period of time or perpetually, depending on the laws of your state. Usually it is limited to the life of some person or 21 years. In that length of time if your heirs or the person you desire these trees to go to have not educated themselves to the value of the tree, then the planting will be lost anyway.

In all of these cases and all the transactions that you make, if you value your trees—and you surely do when you will carry water for them and plant them and dig that large hole for those roots—it is worth while to look after them during the trees' lifetime, not your own lifetime. And if you will consult with your attorney, particularly mention those trees to him and just exactly what your ideas are, I think you will be assured that you will have a future for nut trees.

PRESIDENT MacDANIELS: Thank you, Mr. Sonnemann.

Are there any questions you wish to ask on this subject. Here is a chance to get free legal advice on the spot. That's unusual.

DR. GRAVATT: There is one point I'd like to bring out, backing up what the gentleman just said. You know we introduced back in 1928 to 1936 very large numbers of Chinese and Japanese chestnuts. Most of them went out to state forestry departments and such; somewhere around a half million trees. We have had some very valuable cooperative orchard plantings, which have been lost because something happened to the man, he moved away, sold his property, or died. With these gentlemen who have passed away, experimental orchard plantings and other trees were part of their lives, but their children, or whoever inherited the property, had no interest in continuing the work.

We have had the same experience with some agricultural experiment stations where one of the horticulturists is interested in the plantings, but has moved away, and we have lost our plantings.

PRESIDENT MacDANIELS: Thank you, Dr. Gravatt. Mr. Becker, do you wish to say something about the Reed Memorial?

MR. BECKER: This is just a word of appreciation to a number of the Northern Nut Growers members who have helped out with the C. A. Reed Memorial.

When we organized the Michigan Nut Growers Association last January it was Professor O'Rourke's idea to have a memorial at Mr. Reed's home town, which is Howell, Michigan. With Mrs. Reed's approval we planned as our first project, planting a nut tree with a suitable plaque in memory of the late Dr. Reed.

As a followup, we issued a little bulletin asking for contributions toward the memorial. We sent these out to people who knew Mr. Reed, many of whom are among this group.

Response has been gratifying and we now have approximately $95 toward the tablet. On Arbor Day a Michigan variety of shagbark hickory called the Abscoda was planted at Howell on the library grounds. The services were conducted with the cooperation of the Michigan State College and the Livingston County garden group. This is a word of appreciation and also to explain what we have done. Thank you. (Applause.)

PRESIDENT MacDANIELS: We will go on to the next paper, "The Value of a
Tree," Ferdinand Bolten, Linton, Indiana. Mr. Bolton. (Applause.)

MR. BOLTEN: Members of the Northern Nut Growers Association and ladies and gentlemen: I am just a farmer. I am not a speech-maker, like the lawyer here who makes his living talking. I make my living farming, and I have some ideas, views that I'd like to bring before you.

The Value of a Tree

FERD BOLTEN, *Linton, Ind.*

Members of the Northern Nut Growers Association, ladies, and gentlemen. It may be a little unusual for a fruit grower and farmer to be on this program; however, I have lived a lifetime working with trees on the same farm I was born on sixty-six years ago last May. We have one hundred acres of orchard, several varieties of nut trees, including English walnut, pecans, hybrid pecans or hicans, hickories, filberts, hazelnuts, heart nuts, butternuts, black walnuts; also, persimmons, pawpaws, hybrid oaks and many of the native forest trees. In operating a farm this size, you naturally get a lot of experience and headaches. A very good friend of mine told me a joke that I think fits in with my farm very well. He said a fruit grower delivered a load of apples to the insane asylum. One of the inmates was helping unload the apples. The inmate kept talking about apples, so the grower asked him if he was ever on a fruit farm. The inmate replied that he was before he came to the asylum and, in return, asked the grower if he had ever been in the asylum. The grower replied that he had not. Then the inmate said, "Mr., I have been both places, and I can tell you something. It is a lot nicer here than it is on a fruit farm".

My subject is,

THE VALUE OF A TREE

A tree out of its natural habitat sometimes becomes worthless. As an extreme example, the orange tree in Indiana has no commercial value and the apple tree in Florida has no commercial value. Therefore, it seems that we should, in Indiana, endeavor to develop better trees in the trees which are at home here. This includes the native hickory and the black walnut, hazels, filberts and the pecans in Southern Indiana. Personally, I am spending quite a bit of time with the Crath Carpathian English or Persian Walnut. Last winter, I lost seven out of fifty trees from some cause, after they had gone through the winter of 1950 and 1951, at a temperature of nineteen below zero without injury. It may have been they were caught last fall by a hard freeze in full foliage, early before the apples were all picked; and, again, it may be blight. I hope not. But this I do know, the hickory and black walnut in their natural habitat were not injured.

I wonder why hickories are so erratic in their bearing habits. Could it be the winter rest period? For example, the peach has to have from seven hundred hours, in some varieties, to twelve hundred hours, in others, of below forty-five degrees temperature, or they will not set a good crop of fruit. The value of a variety of peach in Georgia sometimes is determined by the number of hours of rest period below forty-five degrees that the variety has to have. It has happened that the same variety of peach has produced a good crop in Northern Georgia and a poor crop in Southern Georgia. Where the winter was not as cold in Indiana we never lose crops from the lack of enough cold weather; we lose them from sub-zero temperatures. So you see, the value of a variety in Georgia is different to Indiana.

The value of a tree may be in the wood or in the food its produces, or its beauty in winter. Many a picture is taken of evergreens covered with snow. Its value may be its beauty in summer, or the coloring of its leaves in the fall. There is also a sentimental value; a limb that is just right for a child's swing, the Constitutional Elm at Corydon, or the Harrison Oak at North Bend, Ohio. They have a historical value and are visited by many people.

A man said to me some time ago, "I wonder why God made the hicans the cross between the pecans and the hickory?" There may be a valuable nut tree show up in the second or third generation of the hybrid trees when certain characteristics begin to revert to the parent trees. I have on my farm some hybrid oaks grafted, and am very anxious to see them produce acorns so I can plant them and watch the results. This hybrid originated in the Greene and Sullivan County Forest in Indiana, and is called the Carpenter Oak after Mr. Carpenter, the district forester. It is, apparently, a cross between the shingle oak and the pin oak because it is comparable with both of them.

The value of a tree is not always the one that wins first prize in the show. The best plate of nuts in the show may not be from the most valuable tree, because it may be biennial in bearing habits, it may be a shy bearer, it may be an early bloomer and subject to frost. My most productive Crath Carpathian tree is not the best walnut and would not get anywhere in the show, but it is hardy, blooms late, and is productive; so its value is in these traits. The number of chromosomes in the Crath Carpathian walnut may be different. There is quite a difference in the size of nuts produced on individual trees. This indicates that there may be a difference in chromosome count. If this is true, it will be a great help in improving the size of the nuts produced. It may be of value in pollination. The triploid apple needs to be pollinated by the diploid variety. By setting them close together, you get a much better set of fruit.

Sometimes I think trees are as temperamental as people. Some trees, especially the apple, lose their value because they are subject to certain diseases. Some are susceptible to scab, blight, codling moth, rots, blotch, and other diseases, to a point where they become worthless as commercial varieties. The honey locust has been considered one of the trees on farms to be destroyed, because it was thought to be worthless. Now, its value is being found in the correcting of sugar deficiency in dairy cattle. The pods of the honey locust are one of the best foods to correct sugar deficiency and cattle like them and eat them freely. I have on my farm a thornless honey locust that produced ten bushels of pods one year. The honey locust is also a legume and produces nitrogen which, in turn, is used by the pasture grasses and makes more pasture for the cattle.

The mulberry tree that ripens when cherries are ripe has a value in the fact that every mulberry eaten by a bird saves a cherry and the birds are valuable because they destroy insects that cause the worms in cherries.

After observing trees for years, I am convinced that there are certain strains or families of trees in the forest that have outstanding traits. Those traits in growth might be dwarfs or they may be giants; they may have short lives or long lives, like different varieties of apples. The fruit or seeds may be large or small. I believe as reforestation progresses there will be certain trees located which have value as seed trees and which will improve the forest equal to the improvement in livestock on the farms today. The razor back hog that roamed the forest is gone and has been replaced by animals much improved; yet, the forest in which it roamed is the same. Now we are turning to man made forests and a chance to improve them by selecting the more valuable trees for our source of seed. In the native hickory and black walnut, there is a great need for more interest in searching for and preserving the most valuable trees for their cracking quality, flavor, and productivity. There have been and are now, nut trees on farms that were valuable trees, but were known only to the owner and the small boys of the community. These trees should have been preserved for posterity, but many of them are lost forever.

In forestry, a tree's value may be in its ability to re-seed itself. In the kinds of pine, the Virginia pine is one of the best, and also, one of the youngest to produce seed cones. I have counted twenty-five cones on a five year old Virginia Pine tree. In forestry, the red cedar is good to re-seed itself in the area in which it grows. The maple ash, cotton wood, and poplar also grow freely from nature's seeding.

Every tree that grows has a value. The leaves help purify the air; the persimmon and the tree with a wild grapevine are food for wild life. The old hollow tree is a refuge for the coon and o'possum and other wild life. I have a hollow white oak on my farm I let stand because a family of squirrels is raised in it every year. I also have a bee tree and the bees help pollinate my fruit trees so they produce better. A world without trees would be a desolate place. The value of a park is in its trees.

I have spoken of the value of trees for the preservation of wild life, but how do trees affect the life of man and how does man affect tree life? Man is the builder or destroyer of tree life; although the tree is the oldest living thing in plant or animal life, man is master over trees. A man came into my farm office one day and said, "Everything in this room either grew from the earth or was mined from the earth." How about everything in this room? The furniture, the clothing you wear, the ring on your finger, the glass in the windows, etc.? Let us think for a minute, what are the things of the greatest value in this room? We have an organization, The Northern Nut Growers Association. It did not grow from the earth, there is knowledge of science here, there are doctors' degrees (I wish I had one), there is ambition, honesty, love, pride, and patriotism. Man's knowledge is the key. What he leaves alone or what he destroys. So the greatest value is man's knowledge. After all, the greatest values are the things that come from the minds and the hearts of men. By man's efforts, we find or develop these valuable trees.

The value of a home is increased by trees. The love of trees and the pride in owning a home is hard to separate. The privilege in America to own a home and plant a tree on your own ground is of great value. It has been said that he, who plants a tree, is truly a servant of God. I sometimes wonder if this great value of the privilege of owning a piece of ground and building a home and planting a tree is in danger of being lost under the present creeping grip of socialism and communism. This privilege of planting and owning a tree is of greater value than any tree, and we must not lose this valuable inheritance in America.

PRESIDENT MacDANIELS: Mr. Magill, are you all set with your program?

MR. MAGILL: Yes, sir. This is to be a discussion of "Methods of Getting Better Annual Crops on Black Walnut—A symposium led by W. W. Magill (Kentucky)—Discussion by a panel made up of W. G. Tatum, Spencer Chase,
W. B. Ward and Mr. Schlagenbusch." Will those men come here? We will get
started.

My business in life is Extension peddler down in Kentucky, working on fruits and nuts and berries, and naturally that takes me into a good many counties. We have 120, and I have been in all of them. Some places didn't have anything, so no reason to go back. But I pick up a lot of conversation, people give you ideas and things to think about.

We were talking about the conditions of the world—everybody's got a good job and plenty of money and biggest incomes that the country has ever known. That's true, but if you take down in the hills and hollows into some places that I go and you take the financial status of certain of those families, it's not measured in thousands of dollars, some cases not hardly measured in hundreds of dollars. It's measured in terms of gratuities and things to eat and not measured by greenbacks, and the families don't pay income tax.

Last fall I was out on a farm in the foothills some 70 miles from Lexington, in a place that most of you folks wouldn't want to live in and call home, a little farm, probably 16 acres, with a widow lady probably 65 years old, living there with her daughter. And among other things, she said, "Mr. Magill, I understand that you are supposed to know something about nuts. See that tree standing right out there?" She says, "I will give you a $20 bill if you will tell me how to make that nut tree bear annual crops."

Well, I was a little bit surprised. I listened, and I got to asking her questions. Some member of the family had gone to Chicago years ago, and she knew about all the black walnut packing firms in Kentucky. This relative had worked in the market, and had indicated she could get a dollar a pound for all the nut meats she would pick out and send to this relative in Chicago. And that nut tree meant about 30 to 35 dollars a year when it had a crop but only bore every other year.

Well, that drove home just a little more to me than ever before the question of why certain nut trees bore and others didn't bear. To that lady there it meant $30 the year it bore and no income from that tree on the year it didn't bear. And she stood there beside the home and pointed out other trees that bore regularly. And she said, "Why do they bear regular crops and this good tree that makes so many fine, big kernels bears every other year?" That's a challenge I am throwing out to this audience today to all the members on this panel.

I am hoping that Pappy Ward or Friend Chase will answer that question completely. The thing I have in mind, is that in a group like we have here today, as many nuts as we have got here, if we think about this question and talk to the folks back home, I believe in a year or two we can have worked out and have printed in the records of the report some pretty reasonable answers as to why nut trees don't bear, or why they bear heavy crops on certain years and are off certain years.

Mr. Ward, I know you have observed this over a period of years. What, in your opinion, is the one factor that is more responsible for this alternate bearing of black walnuts?

Why Black Walnuts Fail to Bear Satisfactory Crops

W. B. WARD, *Department of Horticulture, Purdue University, Lafayette, Ind.*

When man or nature, and sometimes both, change the natural habits of a tree, most anything can happen. There are years when the black walnut sets very few fruits either on the seedling trees or trees of named varieties. Some few trees have alternate years of production, while other trees bear annually and some not at all. Good results and good crops may be expected only when several factors are normal and conditions favorable. After twenty years of keeping records and observations on nut trees and through correspondence with other growers, I consider the main reason for crop failure or light production to be climatic conditions and the weather for an entire year.

The black walnut produces a pistillate flower at the end of the present season's growth. The staminate flowers, or catkins, come from last year's wood. Good growing conditions are desirable for wood growth and fruit bud formation and any retarding of growth the previous season means little or no production. Winter injury to wood and bud, diseases or insects attacking the foliage, soil moisture, and summer temperatures will lower tree vitality. There are times when strong vigorous trees fail to fruit which could be due to a high or low carbohydrate-nitrogen balance. Soil type, plant food, age of tree, and location will have some influence on annual or even biennial production but yet are not the all important reasons for light crops.

The pollen of the black walnut is mostly wind borne as few insects ever visit the flowers and pollination is dependent on wind borne pollen. Trees planted in groups and close together are generally more productive than trees planted in orchard rows even as close as 40' by 40'. When the weather is cold and rainy during bloom, one should not expect much of a crop.

The staminate flowers opened early in Indiana the years of 1950, 1951, and 1952. The weather was more or less ideal during the time the catkins had elongated and about ready to shed pollen. This warm spell was followed by a fairly cool weather and considerable rain, which delayed the opening of the pistillate flowers, consequently the pollen dried and was lost before the pistil was receptive.

The few walnut trees in the University plantation have always had the best of care. The trees have been mulched, fertilized (both through root and leaf feedings), sprayed, cultivated and seeded to grass with the grass clipped. The trees are some distance away from other seedling walnuts and a bit off the beaten path of the right direction of the spring winds. The varieties are Ohio, Stambaugh, Stabler, Rohwer, and Thomas. When the spring weather is balmy at flowering time, the trees bear a respectable crop but let the weather change to cool and moist and then that is the time one begins to think about calling up the sawmill to see if there is any need for some good walnut logs.

MR. MAGILL: That's a mighty good discussion. I see Mr. Ward has been observing walnut trees closer than I assumed he had.

Mr. Chase, I know you have seen a lot of things in Tennessee that you are not going to tell us about, but I suggest that you discuss some of the things you have observed about walnut trees bearing anywhere.

MR. CHASE: Alternate bearing has been a problem with fruits and nuts since time immemorial. I know a tremendous amount of work has been done with the apple, which has a definite biennial bearing habit. There have been all sorts of things tried to make it bear annual crops, and as far as I know, there has not been anything effective developed along that line. Of course, there are varieties of apples that tend to bear annual crops.

As Mr. Ward brought out—he took all my thunder, so I don't have much to say—a tree may set a heavy crop of nuts one year because frost or poor pollination the year before destroyed the crop so that a large amount of carbohydrates were built up in the tree. Now, the tree in producing a heavy crop of well filled nuts utilizes every bit of carbohydrates it has stored and can manufacture. While it is doing this the terminal bud is being formed for next year's crop, and if there isn't a sufficient amount of carbohydrates in the tree at that critical period, there is not likely to be a flower bud formed.

This is not limited just to walnuts, but occurs with nearly all fruits and nuts, with the possible exception of the chestnut.

We made a study which was reported in the 1946 report by Mr. Zarger in which he reported the bearing habits of some 135 trees over a 10-year period, and there were definite bearing cycles, or bearing habits. It was not always an on year followed by an off year, but possibly two years in a row, then nothing. There were some trees that went three years without a crop, then a crop. Very few, however, had annual crops, and the annual crops were heavy or moderately heavy, followed by what we consider a light crop.

These trees were scattered through seven states and, of course, conditions were not the same. They were all seedling trees, but careful records were kept on the bearing habits. There was a group of trees that could not be classified into any definite bearing habit. In those instances we suspected unfavorable weather at pollination time, but as a general rule, in our section I don't believe we are concerned with that factor.

The Thomas, which we can watch carefully in a nearby orchard, is definitely on one year and off the next. Quite a few are on one year and off two years. We haven't found any way to make that an annual crop, because when it sets a crop, it sets a bumper crop, and there is simply not enough food in the tree to set a sufficient number of fruit buds for the following year's crop. I am sure that a lot of you folks have observed this, and I think, Mr. Magill, that you might sound out some of them.

MR. MAGILL: Going back to an observation I made as a kid, money didn't grow in bushes around our place, and back in those days you could go out and kill ten rabbits and sell them for 8 cents apiece, and if you only used 4 cents apiece for ammunition, you have made 40 cents off of the deal and had $20 worth of fun, and that was a good day's work. You remember those days, Pappy? Back in those same times, I used to get money out of hauling black walnuts to an old corn sheller and having people who didn't have an interest in the corn sheller sell them for 50 cents a bushel. That was also pin money. Come in mighty useful.

We had a certain group of trees on the farm I was raised on that bore every other year, and I can think of two fields where we rearranged the fences in such a way as to make pasture fields out of them, and two of those trees were where 15 or 20 cattle pastured. These were the only shade trees, and naturally they manured those trees. And I recall for a few years I was getting annual crops from them. Apparently they got something supplied by cattle that they didn't have otherwise. Others in the foothills of Kentucky, have come to the same conclusion.

I know a man who has pecan holdings in Alabama. He told me up to the time he got the farm the trees had a few blooms but wouldn't set pecans. He applied 15 mineral elements and claims to have got results from it. I have talked to at least three people in my travelling around who tried the same treatment on pecans, one in Georgia, one in Alabama and one in Mississippi. They reported that they had improved yield on pecans by using complete mineral fertilizer. That's in addition to nitrogen, phosphorus and potassium.

I am foolish enough to think that that nice, young orchard of Mrs. Weber's would make an excellent place to try it. I understand that the trees are not behaving as well as they should. I'd like for Ford Wilkinson to be made chairman of a committee to see that they are fertilized according to some kind of a schedule that could be worked out and do some observing. That is one of the few places I know of in the several states that would be as adequately laid out. I'd like to see a complete fertilizer including nine or ten mineral elements used.

I don't mean spend a lot of money, but you can do a lot of observing for relatively few dollars. I just throw that out as a hint.

I would like to open up this discussion. Mr. Bolten talked a while ago about things he was growing out of the ground, or out of minerals. Everything comes from the ground, and I reckon you'd say this Northern Nut Growers Association is a little like Topsy, it just developed, as the fellow about the weeds. He said they weren't created, they just come all at once. Now I believe that out of this Northern Nut Growers assembly here that we have got some keen observers that might have something on their minds they want to tell us about. Who wants to speak first?

MR. CALDWELL: This is just an observation I am throwing out for the benefit of those who are here. I spent some time in China, and I was interested in the fact that their walnuts there produced yearly crops. In trying to find out why they produced yearly crops, I also discovered that their persimmons, their plums and their peaches did the same thing. The reason for that apparently goes back to their mythology. They believe in signs and doing certain things according to certain seasons of the year, and one of the things that they did was to gather together in the dark of the moon on one particular night at a certain time and beat the living daylights out of these trees with big bamboo clubs. I wouldn't suggest that people here do that, but it's been known to foresters quite a while that by transplanting or severely pruning or girdling trees that you could produce fruits on these trees the following year. Apparently the Chinese so injured the cambium during the severe beating that they have caused that wound stimulus to induce the formation of flower buds for the following year. By so doing in their English or Persian walnuts they did have yearly crops. I have seen this myself, and I checked back to see why. Perhaps they could explain it. The only explanation we made was not fertilizing, but in the wounding of the cambium. Now, perhaps there could be something done of that nature for walnuts, but I wouldn't suggest getting around and beating the trees up.

MR. MAGILL: In that connection, one man in Kentucky got the same answer. He said about five years ago a cyclone came through there and blew the chimney off the house and uprooted a number of apple trees and leaned over three walnut trees, and he said they have borne five crops in succession. Now, this is the same story that you have got there.

MR. STOKE: I'd just like to remark that I think that's a sort of negative approach. I noticed a boy who had an apple tree that was about to die. He girdled it and got a tremendous crop of blossom. You probably have secured the same results. That is one of Nature's ways to perpetuate itself. But I think there a constructive angle in those trees that respond to nitrogenous fertilizer or manure. I believe the secret, if there is a secret, is that a tree in bearing a crop exhausts itself more or less. It recuperates the following year and then is ready to bear another crop. And the way to meet that situation is to fertilize heavily, especially with nitrogen, the season of the heavy crop so that you will have not only enough leaf growth to produce that crop, but to build up nutrients the following year. I believe that will help break the cycle and establish more regularity.

Some trees do that themselves; that is, they will bear a moderate crop every year. I have the Land walnut at home. It bears every year. Certain chestnuts will bear every year, not excessive crops, but Hobson bears a pretty good crop every year. I believe the secret of breaking that on-and-off cycle is to fertilize heavily the year of production not the year of non-production. If you apply nitrogen on the off year you produce perhaps an excess of wood growth that year and overbearing the following year.

MR. MAGILL: Referring to apples, any of you apple growers well know that the Golden Delicious and York Imperial grow crops in alternate years. Now, you come along with hormone sprays and take half or two-thirds of the young fruits off soon after the trees blossom and throw them into regular production. That's the same thing that you are talking about, Mr. Stoke. I never heard of anybody thinning walnuts. I don't know whether they do or not. A lot of things I don't know, but I don't know of anybody ever thinning walnuts, except squirrels.

MR. WARD: Last year a lady from Kokomo, Indiana, wrote me that she had a very fine walnut tree growing near Mr. Bolten's place in Greene County, and as far as she could remember that tree had borne an annual crop for the past 70 years. I wrote to Mr. Bolten asking him to investigate. If I remember correctly, these trees were grown in the poorest possible place. Is that right, Mr. Bolten?

MR. BOLTEN: Yes.

MR. WARD: There were two or three trees right close together that had a nice crop and the ground was covered with a lot of nice nuts which Mr. Bolten thought worth propagating, and he has a tree already started.

We have other varieties that we call the Saul, the Goose Creek and the Alley, which are all seedlings and which have produced almost every year with about the same size of crop.

In our own planting, at the University, we have tried a lot of things without telling anybody about it. Every once in a while the boys mow the orchard, and have bruised and barked a lot of these trees with no effect whatever on bearing. We have time and time again taken the Stambaugh, Ohio, Thomas, Stabler, and Aurora and have given them a good shot of fertilizer in the spring after a rain, and have produced wonderful growth in all of those years but still had only a light crop.

A few years ago some of the boys were spraying the apple orchard with Nu-Green and Urea at the rate of 5 pounds to 100 gallons of water, and had a little extra. They said, "Well, we don't like Ward's nut trees over there, we will put this stuff on them, and if it kills them, that's all right, and if they live, that's all right, too." They gave them some feeding throughout the summer and we haven't found any different results.

MR. STOKE: May I say just one more thing to clarify my suggestion? I was assuming that potash and phosphate were present in sufficient quantity. What I wanted was leaf growth to store up energy and nutrients for the following year and to apply that on the year of heavy crop, so besides maturing the crop, it will provide that leaf growth, and not in the year of no crop.

MR. WARD: We have tried that both ways, and going back, Mr. Stoke, again to the lack of pollination, it seems like both the pistillate and staminate flowers are there, but they just don't set a crop of fruit.

MR. STOKE: One thing more I wanted to say, and it slipped my mind. We know any tree that grows too rapidly will not produce seed nor fruit, and excess nitrogen on apple or walnut or anything else will not cause the formation of fruit buds, but the normal amount is necessary for the formation of buds.

MR. MCDANIEL: We have even got alternate bearing on persimmons in Urbana now. Trees that bore extremely heavily didn't bloom this year.

MR. MAGILL: We hill-billies have been taking a pass at that. I wonder if Dr. Slate couldn't give us some scientific facts about this. How about it, Slate?

DR. SLATE: Mr. Caldwell's remarks about the beating of the walnut trees in China reminds me of an ancient saying that, "A dog, a woman, a walnut tree, the more you beat them the better they be."

MR. DAVIDSON: One of my seedlings began to bear seven years ago, and has borne steadily every year exceptionally large crops. It never failed until this year, and the only explanation that I can give is that just as the bloom was incepted we had continuous rains. There was no pollination of that tree, whereas other trees that were receptive at other times are pretty well filled.

Out of two or three thousand trees you will find some exceptional ones. I have some that bear fairly good crops but do not fill. Walnut trees are just as different from each other as are apple trees. There are some things you can't do anything about at all, and weather is one of the things. One shouldn't be too much mystified by an occasional failure, because it may be due to continuous rains during the period of pollination and when they are receptive.

PRESIDENT MacDANIELS: This matter of alternate bearing is one that has plagued the pomologist for a great many years, and one in which we made little progress, with apples for example, until with hormone sprays the trees could be thinned very early in the year. Any thinning done after the fruit was the size of your thumb was too late. However, now that the fruit can be thinned when it is very young, real progress is being made in securing annual bearing on varieties that previously were a serious problem in alternate bearing.

The failure to fruit is due to many different factors. Some of these are external such as frost and rain at pollen shedding. There is nothing you can do about these. Other factors are internal and determine the formation of fruit buds. If the tree is carrying an exceptionally heavy crop, the chances are it will not have enough of the material which determines the setting of buds to form buds for the following year. With the apples we can do something about this by thinning the crop at the time it blooms. With walnuts, I don't see how we are going to do it. Fertilization is another approach.

Certainly we should make conditions just as favorable as possible for growth and for the development of the buds and by all means control insects and diseases. If you do not have a good leaf surface good crops will not be set the next year. It's a complex problem, but I don't think it is insoluble.

DR. MCKAY: Mr. Chairman, in connection with this matter of annual bearing of black walnut trees we believe that in doing all sorts of things you will not influence the yielding of most of our black walnut varieties. The black walnut, *Juglans nigra* is probably—some of us think, at least—constituted genetically in such a way that the varieties we have do not yield annual crops simply because they are not constituted that way. I know some of you may disagree with me, but one of the greatest arguments for this idea is the fact that in some of our other nut species we do have varieties that are genetically heavy producers. For instance, we have a selection of Chinese chestnuts right now that will bear annual crops on the poorest soil under any conditions imaginable. You can graft scions of that tree on other stocks and plant them anywhere you choose under differing conditions and it will have a heavy set of burs. It may not fill the nuts, it may not attain the size, but genetically speaking, inherently it is a heavy bearer. Perhaps our black walnut species are inherently not annual producers. This is hard to prove, I admit, because the breeding of the species takes so long that we cannot actually demonstrate it.

We have felt also that the black walnut species as a whole does not have the characteristics of thin shells and good cracking qualities that we want. For this reason we have begun a program of crossing the black walnut with the English or Persian walnut, in order to get the thin shell that we want from the other species. Perhaps the same thing is true in the question of yield and the species as a whole does not have the characteristic of yielding heavy annual crops.

MR. MAGILL: I think we can readily see that we haven't settled this problem but it is time to close the discussion.

PRESIDENT MacDANIELS: The next paper that we have is by H. F. Stoke of Roanoke, Virginia, "Survey on Hickory Varieties." Mr. Stoke is the chairman of our Survey Committee. Last year he brought us very valuable information about walnuts, and this year he is going to talk about the hickories. Mr. Stoke.

MR. STOKE: They delegated the job to the Survey Committee to make a hickory survey for this year, using the different state and provincial and national vice-presidents to collect the data. I am going to read this.

The 1952 Hickory Survey

By the Survey Committee

H. F. STOKE, *Chairman*

In compiling this report the pecan has been omitted from the list. As it is the most important member of the hickory group it was felt that the national and state pecan associations are far more competent to compile complete and reliable data on the species than is this organization.

The response by our vice-presidents to the questionnaire sent out has been rather disappointing, replies having been received from slightly less than half their number. It is apparent that interest in the hickory is considerably less than in the black walnut, which was surveyed in 1951.

Perhaps the most beloved and widely distributed of the hickories is the shagbark, *Carya ovata*. It is reported from Massachusetts on the east to southeastern Minnesota, southward to Texas and eastward to the Carolinas where it mingles with and is sometimes confused with the scalybark. In the opinion of many the superb distinctive flavor of its nuts is not equaled by those of any species.

The domain of the Shellbark or Kingnut *C. laciniosa* lies within the same area but is slightly less extensive. Like the pecan, it is partial to the rich alluvial bottom lands along streams and is seldom found elsewhere. It occurs rarely in Virginia and North Carolina, and there only in the Appalachian area.

The Scalybark or southern Shagbark, *C. Carolina septentrionalis*, is reported only by Virginia, West Virginia and the Carolinas.

The White Hickory or Mockernut, *C. alba*, covers the South and is reported as far north as Pennsylvania, Ohio and Indiana and, rarely, in Michigan. It is found from the Atlantic coast to east Texas.

The widely distributed Bitternut, *C. cordiformis*, covers virtually the same territory as the shagbark.

The Sweet Pignut, *C. glabra*, is reported from New Hampshire to Wisconsin and southward to North Carolina. Its south-westward occurrence has not been defined in reports received.

In addition to these better-known species, the Water Hickory, *C. aquatica*, is reported from Louisiana, and the Black Hickory, *C. buckleyi*, from Indiana and Texas.

In an unusually full report Indiana lists all of sixteen hickory species and sub-species as appearing in The Flora of Indiana, a book by Mr. Charles Deam, former State Forester. The list follows.

1. *C. pecan*

2. *C. cordiformis*

3. *C. ovata*

3a. *C. ovata*, var, *fraxinifolia*

3b. *C. ovata*, var. *nuttali*

4. *C. laciniosa*

5. *C. tomentosa (alba)*

5a. *C. tomentosa* var. *subcoriacea*

6. *C. glabra*

6a. *C. glabra* var. *megacarpa*

7. *C. ovalis*

7a, b, c. *C. ovalis* var. *odorata*

7d. *C. ovalis* var. *obovalis*

7e. *C. ovalis* var. *obcordata*

8. *C. ovalis* var. *pallida*

9. *C. ovalis* var. *buckleyi*

Doubtless many sub-species and variants are actually hybrids of obscure ancestry. Virginia has many such.

There is no reason to doubt that the hickories will grow anywhere ecological conditions approximate those of their native habitat. This is true in the Pacific coast states. Mr. Julio Grandjean, of Hillerod, Denmark, reports that there are several white hickories, *C. alba* or *C. tomentosa*, growing in the Horsholm Royal Park that were planted about 1790. There is no reason to believe that such northern species as the shellbark and shagbark would not also succeed. He reports winter-killing of pecans from southern sources. Inasmuch as extreme winter temperatures in Denmark are less than in some places where the pecan is grown here, it would appear that the more northern strains should succeed there, though lack of summer heat would prevent the maturing of nuts.

There appears to be much less interest in planting hickories on home grounds than the value of the species justifies. Only five states, Indiana, Ohio, Michigan, Pennsylvania and West Virginia, indicated any local interest. In each case the shagbark was the preferred species. Apparently we must still depend on the much-abused squirrel for the future of the hickory.

R. E. Hodgson of the Southeast Experiment Station, Waseca, Minn., reports 15 named varieties of hickory under test, but no evaluation of their worth can be made as yet.

Dr. R. T. Dunstan of Greensboro, North Carolina, has also a considerable number of hickory varieties under more advanced test. Results have been highly variable. He finds that Schinnerling has filled poorly; Whitney and Shaul are "Excellent growers and highly satisfactory bearers." Whitney, however, with a kernel of superb quality, cracks poorly and the husk is thick and heavy. Shaul is reported as having a rather thin kernel and cracking poorly, also.

Romig, that has been late in coming into bearing, is described as producing a large, handsome nut of good quality that cracks unusually well. Grainger, good in other respects, has borne light crops as also have Glover and Weschcke. Fox is described as superb in every respect except cracking quality.

Among the hicans, Burton is declared to be outstanding in vigor and health of tree, and production of good regular crops of delicious nuts that crack well.

It is interesting to note that in his extensive hickory experiments Dr. Dunstan is using pecan stocks. He uses the bark-slot method of grafting and hot wax compounded of 10 parts resin, 2 parts beeswax and one part Kieselguhr. Both method and wax he finds highly successful.

Dr. Dunstan also reports a Mahan pecan grafted on a white or mockernut hickory stock that produces heavy crops of well-filled nuts. This is an exceptional performance for this variety.

Mr. Fayette Etter, of Pennsylvania, supports Dr. Dunstan in the use of pecan stocks for hickories. He states that the young trees grow more rapidly in the nursery, transplant better, and grow faster thereafter than when on hickory stocks.

Mr. A. G. Hirschi, of Oklahoma reports that in the hilly "blackjack" country of southeastern Oklahoma the scrub has been cleared away and a 40-acre project of grafting the native hickory (probably white or mockernut) with pecan has been established. The land has been terraced and is cropped with cotton. The results have been so satisfactory that this plot in one year carried off more prizes on pecans than any other entry within the state.

Mr. Harald E. Hammar reports from Louisiana that there has been some grafting of pecan on hickory, species not specified. The older trees show a decided overgrowth of the hickory stock by the more vigorous pecan, in some cases the diameter being almost double above the graft of that below.

In virtually all cases of topworking hickory on pecan, or vice versa, the bark slot graft has been used.

In point of preference of named varieties, Michigan suggests Abscoda, Ohio suggests Stafford, while Pennsylvania recommends Glover, Goheen, Whitney and Weschcke, in that order.

In naming the insects and diseases that attack the hickories, Pennsylvania offers the following rather appalling list:

Nut curculio
Hickory shuckworm
Galls
Spider mites
Twig girdlers
Fall web worm
Pecan phylloxera
Black pecan aphids
Flathead apple borer
Other unnamed borers

Those that know Mr. Etter will understand that this formidable list is due to his excellent powers of observation and his integrity rather than to the likelihood that the state of Pennsylvania is worse plagued with insects than others. Dr. Dunstan lists leaf-spot along with some of those listed above, but adds that none are generally serious. This is corroborated by other reporters.

Wild nuts are generally harvested for home use. Commercial marketing, reported by Pennsylvania, Indiana, West Virginia, Virginia and North Carolina, is in all cases local. Usually the nuts are marketed whole, but occasionally home-picked kernels are sold.

Good stands of second-growth shagbark hickory are reported in Pennsylvania. Kansas reports limited shellbark and bitternut stands. West Virginia reports considerable stands of young shagbark and pignut, while North Carolina reports small stands of mockernut.

The industrial use of hickory reached its height in the horse and buggy days. Nothing equalled its strong, tough wood for the wheels and running gears of horse-drawn vehicles. Old-timers will recall "hoop poles", tall slender young saplings of shagbark hickory that were split and fashioned with the "drawshave" into barrel hoops.

The market for hickory still remains, however. It is universally used for hand tool handles, if obtainable. In the mountains of the South hickory "splints" are still woven into imperishable baskets and chair seats. Louisiana insists it is still the only fuel for roasting barbecue and there is, indeed, no finer wood fuel of any species.

Those propagating hickory trees for sale and distribution should be given every encouragement. They are contributing a real patriotic service. No tree is more characteristically American. Except for a related species in China, it is found nowhere else in the world. In beauty, utility and durability no tree has greater appeal. Who plants a hickory plants for generations unborn.

MR. STOKE: If there are any misstatements, I'd be glad to have them publicly corrected.

PRESIDENT MacDANIELS: Thank you, Mr. Stoke. The comment that you made that there wasn't as much enthusiasm about the hickory as about the black walnut, although true, is not the way I personally feel about it. I have at Ithaca a number of trees of various kinds of nuts, and I think that the enjoyment I get out of the hickories, which we grow, is as great or greater than that from the black walnuts. The Davis hickory is one of the best that matures, the Wilcox—that's an Ohio nut—probably has a bushel and a half of nuts in the shuck this year, and the Kentucky will give a pretty good record. Of about 20 varieties, those are the only ones which amount to anything, and we have a fairly good selection.

There was a good deal said about stocks in Mr. Stoke's discussion. We have a short paper here by Gilbert Smith on his experience with stocks, and I have asked Mr. Chase to read it. Mr. Smith began topworking seedling trees on a side hill many years ago and has trees of good size at the present time.

MR. CHASE: This is a short discussion of several species of hickory which Mr. Smith has used as stocks to graft named varieties.

A Discussion of Hickory Stocks

Gilbert L. Smith, *Rt. 2, Millerton, N. Y.*

This is a discussion of several species of hickory as stocks on which to graft the named varieties of shagbark, shellbark and hybrid hickories. We have never had any experience grafting pecan as we are too far north for it. This paper is limited to the species with which we have had experience.

SWEET PIGNUT, *Carya ovalis*

This species will be discussed first because it is the poorest stock of any of the hickory species which we have used. This is probably because it is a tetraploid while the shagbark, shellbark and hybrids are diploids.

We have grafted many of the named varieties of hickory onto pignut stocks, using several thousand scions. We have found only one variety (the Davis shagbark) that will grow on pignut stock. We have heard of one or two others but have never tried them.

Nearly all varieties grow well the first season but fail to leaf out the following spring. They appear to winterkill. Davis has continued to grow on it for over fifteen years but growth is slower than on shagbark or bitternut stocks.

PIGNUT, *Carya glabra*

I have never been able to positively identify this species of pignut. Pignuts growing here vary considerably in roughness of the bark, some being smooth while others are as rough as the shagbark. In other respects they are essentially the same, all having seven leaflets per leaf. However, I have observed a very few pignut trees having smooth bark and five leaflets per leaf. The leaves are finer and smaller than on the seven leaflet trees.

These may be the *glabra* species, but if so, grafting results have been no better on these than on the seven leaflet trees.

As nursery stock the pignuts are worthless. However if one has some nice young pignut trees growing where he wants them, it is feasible to graft them to Davis or some other variety which has proven its ability to grow on pignut stocks. It is not advisable to graft hickory trees growing in dense woods.

MOCKERNUT, *Carya alba*

While the mockernut is also a tetraploid, it is a somewhat better stock than the pignuts, in that more of the named varieties will grow on it and as the mockernut is faster growing than the pignut, such grafts will usually grow faster.

It is of little value as a nursery stock, but if one has young mockernut trees growing where hickory trees are wanted, they would be somewhat better to graft than would pignut trees. One would at least have a larger selection of varieties and the grafts would grow faster.

PECAN, *Carya illinoiensis*

While we have read many favorable reports on the use of the pecan as a stock on which to graft shagbark, shellbark and hybrid hickories, our own experiences with it have not been very favorable. This may be due to the fact that we have used only two varieties of shagbark on pecan-stocks and may have happened to use two varieties that are not well adapted to pecan.

Pecan seedlings are much faster growing than are shagbark seedlings and for this reason would be valuable as a nursery stock if satisfactory in other respects.

BITTERNUT, *Carya cordiformis*

All of our experiences with bitternut as a stock, both in the nursery and as young trees growing in permanent locations, have been very favorable.

We have heard reports of grafts failing on bitternut stocks after a few years growth. All such reports have come from regions considerably farther south than our location. It may be that the bitternut does not thrive as well in the South as it does here.

Bitternut seedlings are much faster growing than are shagbark seedlings. This is of considerable value in the nursery.

SHAGBARK, *Carya ovata*

The shagbark makes the best stock on which to graft the named varieties of shagbark, shellbark and hybrid hickories. However it has one very serious drawback in that young shagbark seedlings are so very slow growing. It usually takes five or more years to grow a shagbark stock from seed to a size large enough to graft in the nursery row.

However, when shagbark stocks are large enough to be grafted, all of the named varieties we have grafted onto it have grown well.

SHELLBARK, *Carya laciniosa*

We have never had any experience with shellbark seedlings as stocks, but as it is so similar to the shagbark, I expect that it would make a good stock.

The production of grafted hickory trees is a serious problem in the nursery, taking many years to grow the stocks and the grafted trees are difficult to transplant, resulting in a high rate of mortality.

However, the grafting of young hickory trees growing in a permanent location is not difficult, and such grafts will grow much faster and bear younger than will grafted hickory trees from a nursery.

PRESIDENT MacDANIELS: My experience with bitternut stock with only two varieties, the Strever #1 and the Champigne, has not been good. The grafts have been stunted, the stocks have tended to sprout and make vigorous growth, and the fruiting has been sparse. Neither have I had success with the pecan stock with only three varieties. The trees have been very slow coming into bearing and have made rather stubby growth.

MR. MCDANIEL: I was about to remark that we have had similar experience at Urbana with bitternut stock with pecan and shagbark varieties. It warps the shagbark and very likely those trees won't live long. We have already lost the Weschke hickory grafted on bitternut.

MR. CRAIG: Have you tried hickory on pecan? The pecan is O. K. there.

PRESIDENT MacDANIELS: Tomorrow we are to have a round table on hickory propagation and suggest that further discussion of stocks might be left until then. Has anyone any comments on hickory varieties?

MR. KEPLINGER: (North Central Michigan) I was born and raised in Saginaw County where the Saginaw River is fed by five or six different runs and you have prairie farms. More hickories grow there than any place in the United States—enormous size. We think we have better hickories than anyone.

PRESIDENT MacDANIELS: Why couldn't you send some in for testing? Mr. Becker would be glad to take them. Any other discussion on hickory varieties? How many are growing the Wilcox? (5 hands). How many find it a good variety? (Two). How many have Davis? (Three). The shucks are fairly thin, compared with the Wilcox.

Who else has a variety that is doing very well? We ought to have a hickory show here sometime and see who has the best hickory.

DR. MCKAY: I'd like to ask if anyone has the variety Lingenfelter.

PRESIDENT MacDANIELS: We have it at Ithaca; doesn't mature.

DR. McKAY: We have two varieties at Beltsville that are outstanding as far as bearing is concerned. One is Lingenfelter, which has been a consistent bearer for us for a number of years, and the variety Shaul, that was mentioned in Mr. Stokes' report and has been mentioned here before, is a very good producer.

MR. MCDANIEL: What species is the Shaul, is it *ovata* or *laciniosa*?

DR. MCKAY: It's *ovata*. It's a shagbark, as also is Lingenfelter. The one characteristic that is outstanding with these two varieties with us is the fact that they bear while they are young trees; from the time our trees were as tall as one's head, they have been full of nuts.

MR. MCDANIEL: Have you fruited the Weschke at Beltsville?

DR. MCKAY: No.

PRESIDENT MacDANIELS: How about the Barnes?

MR. STOKE: I have been growing it on mockernut or white hickory. It produces moderate crops and is the one that came into bearing about first on mockernut. In fact, I have several varieties on mockernut that haven't borne yet. It's been on there about 12 years.

PRESIDENT MacDANIELS: The Barnes, with us, has yielded more at a younger age than any other variety, but it never filled. It began early and bore heavy crops, but the season is not long enough or hot enough.

MR. STOKE: In Virginia they fill well, but they are not easily extracted. The shell is rather thin and fills well.

PRESIDENT MacDANIELS: I don't want to prolong this discussion longer than seems profitable.

DR. MCKAY: Did I understand you to mention the variety Schinnerling?

MR. GERARDI: I have got that at home. That's one that's bearing, but if it's that variety I have there, I wouldn't give it yard room.

DR. MCKAY: It is also one of our best. We have three, the Shaul, the Lingenfelter that I mentioned, and the third one is Schinnerling, all three of which are extremely heavy bearers and the three hickory varieties that we are interested in.

MR. GERARDI: How big is that Schinnerling?

DR. MCKAY: It's an average-sized nut.

MR, GERARDI: Big as your thumb?

DR. MCKAY: Oh, yes, about an inch long, I'd say.

MR. BECKER: I was wondering about the Stratford. That's not supposed to be a pure shagbark, but it's the only one we've got, I think, that bears.

PRESIDENT MacDANIELS: I have the Stratford. It grows very well, but it doesn't quite fill. What does it do with you?

MR. SNYDER: It's not been doing well the last year or two. Of course, none of them have for that matter. Used to bear tremendous crops and filled well. I wouldn't say it's the best quality of any tree, but it's easy to graft and bears young.

PRESIDENT MacDANIELS: That's been my experience, that it was a young bearer and bears fairly consistently.

If there is no other discussion, on the hickories, we will close that discussion. We stand adjourned until this evening at 7:20.

Adjournment at 4:30 o'clock, p.m.

MONDAY EVENING SESSION

Called to order at 7:20 p.m.

PRESIDENT MacDANIELS: We will call on Dr. McKay as chairman of the
Nominating Committee to present the slate of officers for the next year.
Dr. McKay.

DR. MCKAY: Mr. Chairman, the Nominating Committee, as you know, is charged with the responsibility of selecting a slate of officers that will be presented to the meeting.

The committee, composed of myself as chairman, Mr. Allaman, Mr. Silvis, Mr. Ford Wilkinson and Mr. Gerardi, have the following slate of officers for next year: For president, Mr. R. B. Best; for vice-president, Mr. George Salzer of Rochester; for secretary, Mr. Spencer Chase; for treasurer, Mr. Carl Prell.

PRESIDENT MacDANIELS: You have heard the report of the Nominating Committee. At this time we will entertain further nominations from the floor, if any.

The only action to be taken now is to accept the report of the Nominating Committee. Do I hear such a motion?

The motion to accept the report was moved, seconded and carried.

Going on with the program of the evening, are you ready to show the film?

MR. MCDANIEL: The film comes to us from the Northwest Nut Growers now located in Portland, Oregon. They are an organization for marketing filberts, and you will see, "The Filbert Valleys", the title. I haven't seen it myself and don't know exactly what the contents are. We will look at it now and judge for ourselves.

The film, "The Filbert Valleys", was shown.

PRESIDENT MacDANIELS: We appreciate very much your running it.

The next item will be our discussion of filbert varieties and their culture. Mildred Jones, who was to be here, could not come. She telephoned the last minute that she was ill and could not be with us. I have asked George Slate to be the moderator in the discussion, with his panel, D. C. Snyder, Raymond Silvis, A. M. Whitford, Louis Gerardi and H. F. Stoke.

MR. SLATE: I just learned when I arrived here that I was to be on this discussion group, and I learned a few minutes ago that I was to lead it, so I can assure you that this is wholly unrehearsed, and I may have to flounder around a bit before we get things running smoothly.

I thought I might review the variety situation rather briefly. We have done quite a lot of variety testing of filberts at Geneva; in fact, about the only nut cultural work we have done at Geneva has been the filbert project. We started out with about 25 or 30 varieties that we secured from American nurseries, many of them from a firm in Rochester which imported them from Germany. Later we added varieties from England, France and Germany. I picked up nearly all the varieties that I could locate until we had about 120 varieties growing there at Geneva. These were there for some years, and it became evident that many of them were not of great value. Then we had a hard winter in 1933 and 1934, and although it did not kill the trees, most of them were blackhearted and began going back soon after that. However, I felt at that time that I knew enough about the varieties to discard most of them. Many of them were discarded because they had poor nuts, many of them were unproductive, and many of them lacked hardiness of catkins. I laid a great deal of emphasis on the hardiness of catkins in testing the varieties.

Out of that variety test were three varieties which we considered to be most satisfactory of the lot. These were Cosford, an English variety, rather a small nut but very thin-shelled. The catkins were hardy and one of the heavier croppers of the lot. Medium Long, a nut which I believe originated as a seedling in Rochester, was another one, and Italian Red, which later proved to be Gustav's Zellernuss, a German variety, was another.

As a result of that variety test it became evident that varieties from Germany, many of which originated in the colder portions of Germany and Northern Germany, were distinctly more hardy than the varieties that we got from French sources and English sources. In some of the proceedings of the Association published during the '30's I have reported on the different varieties and their hardiness and those varieties that I thought were most valuable. I don't recall the names of many of those German varieties. These three varieties which we consider the best of the lot were turned over to the New York Fruit Testing Association to propagate and distribute, because they were not available from American nurseries. I am not sure how many of them were available from other sources, but they are still available from the Fruit Testing Association.

Then out of that variety test a grading project developed. We got our start from about 500 seedlings that Clarence Reed sent us in the early '30's. We made crosses there at Geneva, using the Rush variety of *Corylus americana* as the seed parent in many cases. We also made some crosses between *Corylus avellana* varieties, and with these seedlings from Mr. Reed and seedlings of our own crossing, we have grown about 2,000 filbert seedlings there at Geneva. These have all been evaluated and discarded, except possibly 30 or 35 selections still on hand, some of them being propagated for a second test planting. Stock of one or two has been turned over to the Fruit Testing Association for increase and eventual naming and introduction.

The work of the United States Department of Agriculture was along similar lines. Mr. Reed did not send us all of his seedlings. A number of them were fruited at Beltsville, and from that work at Beltsville I believe two varieties have been named, Reed and Potomac. I am not sure whether they are available yet from commercial sources.

MR. MCDANIEL: Two of them are.

MR. SLATE: Mr. Graham of Ithaca, a long-time member of this Association and very much interested in filberts, had also made some crosses and raised several hundred seedlings. He used the Winkler variety as a seed parent. I believe he raised some seedlings of the Jones hybrids, which would make that material second generation stock from the original cross between Rush and the *avellana* varieties.

Mr. Graham's planting was in rather a cold area; he had considerable winter killing. Eventually filbert blight got into his planting, and it really cleaned house. There were a very few seedlings in his planting which remained free of filbert blight. I think it is a fairly safe guess to say that they were probably very resistant to blight. So far these have not been propagated to any extent.

There are a few cultural problems. The ones that we have encountered at Geneva have been winter injury, particularly of the catkins, and also some of them have not been as hardy in wood as we would like. We have had no trouble with filbert blight, presumably because we are isolated from the wild hazel, which harbors this blight. Dr. MacDaniels has had trouble with his planting at Ithaca with filbert mite.

With this introduction, which is mostly varieties and breeding, because that seems to be my interest, I'd like to call on some members of the panel to get their experiences. Mr. Snyder raises nut trees in Iowa where winter injury is probably much more serious than we have at Geneva. At Geneva we have a fairly respectable climate and can get a crop of peaches about nine years out of ten. In Iowa they have a lot more sunshine, and I think probably sharper drops of temperature than we have at Geneva. I'd like to have Mr. Snyder tell us what his experiences have been with filbert varieties.

MR. SNYDER: I really didn't know that I was to be on this panel until I got here. I thought I was on the hickory panel. As Mr. Slate says, our climate is more severe that that at Geneva. We can get the very hardiest peaches to bear about two years out of three, and the trees are severely injured in between. So that will give you a little idea as to the climate in that respect.

We made quite a planting at one time, maybe 30 of the Jones hybrids, and they did quite well for several years, and then between the winter-killing and the blight most of them are dead now. The Winkler, of course, is an

Iowa nut and was introduced by our people and did very well for a number of years but has backed out on us the last several years, too, I believe due to this same mite trouble that Dr. MacDaniels reports in New York. They just don't bear. The bushes are quite healthy, and we get plenty of catkins, but we don't get any nuts to amount to anything.

We have a little bush of the Mandchurian hazel. It isn't worth mentioning as a nut producer, but it does have very attractive foliage and seems to be entirely healthy, produces perhaps three to five nuts a year on a bush as high as your head. You may be familiar with it. The foliage is very distinct from anything I know. The leaves are truncate at the end, cut off quite square, with just a little point in the middle.

MR. SLATE: I don't have that.

MR. SNYDER: That is standing our conditions all right, and several years ago Mr. Reed sent us what he said at the time were Chinese tree hazels, but later he retracted and said that they were not Chinese tree hazels but they were hybrids of the Chinese tree hazel. There were four of those plants; one of them was a tremendous grower. It would grow six feet or more a year and commence bearing in a year or two. But the blight hit it and cleaned it out. There is only one left now, one of the slower-growing ones, and while it promises to become a tree, it is a very irregular-growing one. I think it had half a dozen nuts on this year.

The Turkish tree hazel, of which I have two trees, were very badly damaged by a very severe hailstorm 12 or 15 years ago, which completely peeled off the bark on one side. That was in early July, and we were afraid to cut them off and let them grow up new for fear it would kill them. They have finally developed into quite beautiful upright trees. Also they have more than one stem from the bottom. One of them produces a great abundance of catkins, but neither of them has produced any nuts yet, and they are 14 feet high or more, good-sized trees and very attractive. The foliage is very beautiful, and it remains healthy. I don't know that there are any other varieties that I can name.

MR. SLATE: We have had several of the Turkish tree hazels, *Corylus colurna*, growing at Geneva for two or three years. They came from the Rochester State Park. We have one tree which Mr. Bixby imported from China, as *Corylus chinensis*, but recently I had it checked by Dr. Lawrence of the Bailey Hortorium and he assured me that it was *Corylus colurna*. I think these make a very handsome tree. I like that rough, corky bark they have as they get older. The trees in Highland Park at Rochester are the largest, perhaps, in the country, certainly the largest that I know anything about. They are at least as large as a very large apple tree. They have been fruiting for some years. The trees at Geneva have not fruited very much. I don't think you can expect much in the way of nuts until the tree is about 15 years old. This year one of our trees has a number of nuts on it. The nuts are too small and too thick-shelled to be of any great value for nuts.

Now, Mr. Whitford, you have had some experience with the filbert varieties. Which one would you recommend?

MR. WHITFORD: I haven't had a whole lot of experience with the filberts, but we had some of the old varieties, like Barcelona and DuChilly, and they didn't bear many nuts, and eventually they went out with blight. And we have some of the Potomac and Reed, about five years old, and they don't bear well as yet. I don't know what the outcome is going to be on the Potomac and Reed. They make a nice ornamental bush, anyway, and that's about the sum and substance of my experience with filberts.

MR. SLATE: The Barcelona and DuChilly at Geneva have not been very satisfactory. During the first two years Barcelona outyielded the other varieties, but as the trees became older they experienced winter injury.

DuChilly or Kentish Cob makes a small tree, but the nut is about the best of the nuts. There is a German variety not in circulation in this country, Langsdorfer, which is much like DuChilly, but it seems to make a much better tree. I think if they were put into circulation it might be a good substitute here in the East for DuChilly variety.

Let's hear from you, Mr. Gerardi. I know you are testing filbert varieties now.

MR. GERARDI: Yes, I have DuChilly and Kentish Cob. So far, at our place we have no blight or mite damage to speak of. The original plantings were the Bixby and Buchanan. We have them yet, and they are still as healthy as the day we put them out. They show no damage; even the Winkler hazel has had no damage or disease. It may be the soil, although we have them on high ground and low ground both. Among the newer ones this year the Reed has the most on. The Potomac, though it is the strongest grower of the two, has less nuts. Although it appeared to me that the catkins were all killed in February of this year, still we have some nuts. The Jones hybrids, when the catkins are killed, have very few, if any nuts. Some years we have a crop, if some of the catkins are held back and bloom late. Winter killing in February before they have had a chance to pollinate, has been our main trouble. If we could get a variety that this wouldn't bother, we'd have what we are looking for.

MR. MCDANIEL: The Winkler will bloom for you almost every year. Doesn't the Winkler hold its catkins most years?

MR. GERARDI: Yes, sir, I'd say at our place the Winkler has never failed entirely. Even though the catkins are killed, they still bear quite regularly.

MR. MCDANIEL: I can say that for it at Urbana.

MR. WHITFORD: The catkins might have been killed, but you might have had some cross-pollination from other sources.

MR. GERARDI: There is a chance of that, of course. There is a wild hazel within a quarter of a mile, but apparently the wild hazel bloomed first. They were on a south slope and naturally came out first. I tried to keep them on the north slope, or on the cool side of any particular planting, because if you can hold them back more, you have got a better chance. If you plant them on the south side, you rarely get anything.

MR. SLATE: The hybrids bloom later than the *avellana* varieties, and they mature nuts later. Is that your experience?

MR. GERARDI: That's true, I will admit your hybrids are a little later blooming, because your American hazel nuts around our place bloom very early, sometimes in January in full bloom.

MR. SLATE: *C. avellana* starts blooming in March and blooms for about a month. Some years when you have had considerable open weather, they have bloomed as early as the middle of February. They will, of course, stand considerable freezing when they are in bloom.

As regards the pollination, I believe about all the information we have is the work that was done at the Oregon Experiment Station a number of years ago. All of the varieties tried were self-unfruitful or self-incompatible. The term, "self-sterile" is often used, but I think it is a little more exact to say self-unfruitful or self-incompatible. They are not sterile, because the pistillate flowers are normal and so is the pollen produced by the staminate flowers. It's just a question of inability of the pollen to fertilize the pistillate flowers on the same variety.

We know nothing about the pollination requirements of any of these *Corylus avellana* or *Corylus americana* hybrids. We do know that when the cross is made that the *Corylus americana* variety must be the seed-parent. The cross doesn't work the other way around. That's about all we know about the cross-pollination of these filberts.

MR. SAWYER: We have had them to bloom in April or the first week in May.

MR. SLATE: The seedlings?

MR. SAWYER: The seedlings.

MR. SLATE: What is the origin of the seedlings?

MR. SAWYER: They are the natives.

MR. SLATE: The native *C. rostrata*, or *C. cornuta* to some botanists, it seems to me has nothing that we want in the way of a nut, if we can possibly grow these other varieties, the *americana* selections or the hybrids. It's a miserable little nut with that long, prickly husk. It's very difficult to get the nut out of it. For that reason, I have never been very much interested in it.

MR. SAWYER: How is the Ryan?

MR. MCDANIEL: Mr. Gellatly out in British Columbia has named several hybrids between *avellana* and the *Corylus cornuta*. Have you seen it?

MR. SLATE: No, I haven't seen it.

MR. MCDANIEL: They described them in their catalog.

MR. COLBY: I have preference for the Winkler hazel, as you know. I bought and put them in the greenhouse several years ago and shook the pollen on the pistils and got a full set. So I felt that was self-fruitful.

MR. SLATE: That was pretty good evidence, then, that it was self-fruitful.

Now, Mr. Silvis, you raise nut trees, and the climate is somewhat like that in Western New York, perhaps a little milder in the winter. What have you to say about the filbert varieties?

MR. SILVIS: It's Warmer, and in spite of all the statistics of previous gentlemen, I find that *avellana* types which I had growing in my back yard three years ago produced pollen on January the 25th. It was unseasonably warm. It was 70 degrees, and most of the pollen was dispersed. And this year I found the wild hazel pollen much later than the early types, due to the different situation. The wild ones which I had seen were growing in semi shade under tall trees, and my bushes and plants are growing in the back yard south of our house. And I think I have the largest planting in the State of Ohio, about two dozen plants, and I am in production.

Besides numerous seedlings, I have the following varieties: Italian Red, Cosford, Medium Long, DuChilly. They are in bearing. Italian Red and DuChilly planted together, I believe, are good for one another for the production of nice filbert nuts. I have, from scion wood you sent me several years ago, Cosford, and now on their own roots Neue Riesenuss, and what I thought the tag said, not "Langsdorfer," but Langsberger.

MR. SLATE: There is a Langsdorfer, and I think there is another variety which Langsberg is part of the name. I am not sure, I will have to look that up.

MR. SILVIS: Well, I have it as Langsberger. I have shown last evening the picture of Harry L. Pierce's orchard at Willamette in Oregon, or in Salem, Oregon. I have one of his trees with staminate blooms only, no pistillate blooms. But I also have what Fayette Etter in Pennsylvania calls his Royal, and I just cannot get two fellows together with paper and pencil to determine whether those two Royals are the same, but I am hoping to find out whether the two Royals are identical. I had Fayette Etter find me scion wood, and now I have it growing as a graft and layered on its own roots.

I think you people do yourselves an injustice by not learning to graft and learning to work with the filbert. You only have to have three compatible plants. If you have more, you will have more nuts. I see no reason why anyone who owns a city lot cannot grow filberts. They are much easier to take care of, and you are not going to prejudice the plant by having it associate with its wild cousin, and I think you will find a lot of enjoyment in the filbert bush.

MR. SLATE: What variety do you think is best? What two or three would you plant?

MR. SILVIS: For eating I like DuChilly, and the catkin is hardy with me, and I am between the 40th and 41st parallel. I'd say anyone who lives from Iowa to the East Coast within one hundred miles north or south of the 40th parallel should have the same luck that I have. And as to a group planting, I would suggest, as you recommended to me when we first started out the Medium Long, Cosford and Italian Red. If you want only two bushes, Italian Red and DuChilly will work well together.

MR. MCDANIEL: Do you have Medium Long?

MR. SILVIS: Yes, I do.

MR. MCDANIEL: Is that doing well?

MR. SILVIS: I don't think it fruits as well as Cosford or DuChilly. That's been my experience. My DuChilly was plastered with nuts last year and this year, and I believe it's due to the Italian Red which New York Fruit Testing Laboratory sold me.

MR. SLATE: Thank you.

MR. WHITFORD: Do you fertilize those bushes?

MR. SILVIS: Due to the fact I have started to mulch with sawdust I have been using nitrate and rock phosphate, so my teeth don't fall out when I chew them.

MR. SLATE: I crack mine with a hand cracker, I don't crack them with my teeth.

DR. COLBY: Mr. Chairman, we can grow filberts. How does the chairman keep the squirrels from eating them?

MR. STOKE: I will tell you that.

MR. SLATE: Mr. Stoke raises his nut trees in the Sunny South, and he has problems down there that we don't have up north. I think he has to worry a lot more about winter killing than we do way up north where we are in Central New York. What's been your experience with some of the varieties and what are your principal cultural problems with the filberts?

MR. STOKE: I wish to answer Dr. Colby's query about squirrels. I find that squirrels are very highly allergic to these BB caps or the CP caps used in a 22 rifle. It works. In my back yard there is a Brixnut filbert, which originated in Oregon. I guess it's been there 15 years. There are four trunks to it, the largest about 16 inches in diameter. One of those I grafted to Giant, as a pollinizer for Brixnut. It's similar in shape, somewhat smaller in spite of its name, but it's pretty effective. Then about ten years ago there was an old gentleman from Halsey, Oregon. I don't know whether any of you have corresponded with him or not. He bought the Breslau Persian walnut—I pretty nearly said the English walnut, and I'd have been disgraced—and furnished me scions and I got a start of it from him. Russ sent me some scions from a filbert he called Jumbo. You will see it out on the table there. It's rather a long nut, little larger than DuChilly and not quite so flat, that I grafted in there. It absolutely is hopeless as a pollinizer for anything, because it loses its staminate blossoms by Christmas. But the Hall's Giant pollinizes them, and it's the best filbert I have, all things considered. This year off that one scion—of course, it's four inches in diameter—I got about 7 quarts of nuts, and they began ripening at least three weeks ago, and the crop is all off now. And the foliage is unusually heavy, almost in clusters, and it drops cleanly and freely from the husks, and I think it is a very nice filbert. Whether it's a recognized variety in the West I have no idea, and I haven't corresponded with the old gentleman for some years, and he probably has passed on by this time, because he was an elderly man and not in good health at the time I had my correspondence with him. I consider that an excellent filbert, and I think anyone wishing to plant filberts should investigate with the Oregon nurseries or Washington nurseries and see if that is a recognized variety. I tried to find out once and failed so far. I do not have it on its own roots. I hope that I will have it rooted in another year.

In my back yard also I have one that I bought in Oregon. That's as tall as up to that beam, maybe almost to the ceiling, very vigorous growth, larger nut than Longfellow, thicker nuts and also longer. But I think the thing he sold me was a graft and the graft died and this came from the root. It bears very sparingly, but it's a very large nut, and I wondered why it was always so spare, and I caught it blooming in December, staminate blossoms in December this year. So that's that.

Ten miles east of my home, east of the Blue Ridge Mountains in the granitic, very heavy clay soil of what we call the Piedmont down there, I have a planting that was made 15 years ago of filberts, some on their own roots and some that I grew on the Turkish tree hazel stocks. Those grew well, and the main advantage was they put up no suckers. You had a nice clean trunk, and you didn't have that problem of getting rid of the sprouts all the time. And it looked very good for a while.

I find where you graft that way, the stocks get old and do not renew themselves, and eventually the life will be shorter than if you had a shrub that might last for a century, when you are renewing your stalks when they reach maturity and cease to grow enough to be productive.

Two years ago I had most of the standard varieties you mentioned here in that planting, about three-quarters or perhaps an acre planted in between chestnut trees. Planted the chestnut trees 40 feet apart and then interplanted with the filberts at 20 feet. Two years ago we had an unusually wet season, and the blight, of which I had had some before, hit hard and virtually ruined the whole planting. And in addition to that, we have leaf miner. It's an insect that lays a tiny egg in the leaf and develops a little larva or worm that eats out the chlorophyll between the two membranes of the leaf, just hollows it out and makes unsightly spots in there and, of course, kills that portion of the leaf. But the blight, known as the eastern filbert blight, according to Mr. Gravatt, has just ruined that planting. Some of the trees have been killed outright, and most of the tops are either dead or dying. This year the blight wasn't apparently active on the living part, because it was very dry up until the first of August, and since then it's been very wet. That's what happened to my filberts there.

Now, in that same location I have some younger, second-generation or third-generation plantings that I grew from scions from the Jones hybrids and so far those have not been attacked by the blight and not much by the leaf miner. I used them to replace some of the others that had died several years ago, so they are right in there together. About the best I have of those are also on exhibit out there and marked as the Jones Hybrid.

At the same time I put out some seedling Colurna or the Turkish tree hazel in that same plot. They were attacked somewhat but not badly by the blight. Today you'd never know they had any blight. They look healthy, and as has already been said, they make a beautiful tree. And if you want an avenue of trees on a drive that don't spread too wide and run up like Lombardi poplar, they'll beat Lombardi poplar all to pieces. And if you crowd them a little, they will grow up like a spire and retain their branches, so you really have a tree.

There was one in the J. F. Jones yard at Lancaster that I think was at least 14 inches in trunk diameter 20 years ago when I saw it. Do you know whether that is still there at the Jones place, that Turkish tree hazel, Mrs. Weber?

MRS. WEBER: Where is it located?

MR. STOKE: It's right near the house, it seems to me between the house and the side near the barn.

DR. MCKAY: Mr. Stoke, that tree is gone. We were there last fall.

MR. STOKE: But it was a very nice tree, and for shade it's very nice. The Manchurian hazel has been spoken of, and I might mention that, because I have dabbled in everything, I guess. I got seed from the University of Nanking along with some other things, and those seedlings were quite variable. The nuts compared rather favorably with the American hazel. Some were thick-shelled, but they will average almost as good as the American hazel, and they bore quite freely for me until I let the bushes get right thick. They will send out suckers and make a very spreading growth. If you dig them out and leave a piece of root in the ground, it will come up just like sassafras or persimmon will on that piece of root. But it is an attractive bush, and mine has a reddish-brown little spot in the middle of the leaf in most cases. It seems to be characteristic of that strain that I have. The nuts were quite variable and, as I say, they bore right well until I let them get too thick. I believe that's all.

MR. SLATE: I neglected to answer your question, Dr. Colby, but the squirrels have not been much of a problem with our filberts at Geneva, strange as it may seem. They have never taken a very high percentage of the crop. We have a Lancaster heartnut, and they clean up every nut on that tree every year before the end of August.

I'd like to comment on this matter of the name of Halle's Giant, I think you called it. I think the name is Halle, the German town where the variety originated. I prefer the name Halle, because calling it Hall's Giant is more or less a sign its origin is a man named Hall.

MR. STOKE: In some catalogs it is one way and some the other.

MR. SLATE: We have other items on the program tonight, and as the Latin student said, "Tempus is fugiting very fast," so I think we had better turn the meeting back to Dr. MacDaniels.

PRESIDENT MACDANIELS: The next two talks have slides to be shown, and it is suggested that you take about ten minutes, take a stretch and then come back when the slide projector is set up.

My Experiences With Chinese Chestnuts

W. J. WILSON, *Fort Valley, Ga.*

When I was asked to appear on this program to tell my experiences as a grower of chestnuts, I felt like a child, appearing before a group of grown-ups to tell them how to make marriage succeed. When I see the sages of chestnut knowledge seated before me I realize that I can only relate my experiences and ask your advice.

My father was a pioneer peach and pecan grower; he loved trees and has told me time after time that if I ever made more than just a living, farming, it would have to come from trees, not row crops. He was what I would call a self-educated man. He had small chance of formal education, being the sickly son, one of eight sons and three daughters, of a couple who eked out an existence on the poor, unproductive, sandy, soils of Crawford County, Georgia, growing the one and only cash crop of those days, cotton. The combined wages of these boys often amounted to more cash money than their own cotton crop returned because the supplier got most of the money from their own crop. They helped neighbors pick out their cotton crops after finishing their own. Grandfather must have liked to experiment in his limited way. Each spring as Grandfather would plant his small patch of Spanish peanuts and yellow corn, Grandmother would tongue-lash him, saying, 'so long as you fool away your time with Spanish peanuts and yellow corn you will remain a poor man. Time has proven Grandfather right and Grandmother wrong. Spanish peanuts is a huge industry; most of our hybrid corns, which have added millions of bushels to our yields are yellow.

My father wasted his time back at the turn of the century planting a peach orchard on his best cotton land. He planted pecans each winter, beginning about 1912, often to the ribbing of friends who still worshipped at the feet of King Cotton. One told him that he had a pecan tree or two about his home and the damn flying squirrels ate all of the nuts. Another told him that if he wanted a load of stove wood he would just as soon cut down a pecan tree as any other kind. At his death in 1942, my father had planted six hundred acres of pecan orchards, each acre having been interplanted with peaches, to produce income while the pecans were reaching bearing age.

I give you this background so that you may better understand my attitude toward chestnut growing. The scale on which I have set out on chestnut growing I know to some of you will seem rather bold or foolhardy.

About ten years ago I found that the U. S. D. A. Pecan Experiment Station at Albany, Georgia had a small chestnut orchard. Max Hardy, was doing the chestnut work and was so much interested in them that I caught fire and have been burning ever since. When I found that the harvest came between the peach harvest and the pecan harvest it fitted right into my kind of farming. The fact, that it was a possible tree crop made chestnut growing still more attractive to me. Max suggested that I join the N. N. G. A. when I complained that I couldn't find much information on chestnuts. I attended my first convention at Norris. I have tried to make most of them since that time. Of all the discussions at the Norris meeting, the one that stuck in my mind was whether nurseries should recommend seedlings or grafted trees. I thought then, and still think, that for commercial production one must have varieties, because seedlings are so variable. I believe, that when, chestnut growing comes of age, the major part of the production will go through processing plants. It will be a great advantage to have nuts of uniform quality and size, which is and will be impossible with seedlings.

Of the fifteen trees that I planted in 1946, only one fruited in 1951. It bore only 3-1/4 pounds of nuts. The other fourteen did not fruit. This year there are a few scattering burs at seven years of age, on those that I did not graft this spring. I am now too old to wait seven or eight years for a chestnut tree to begin bearing. These trees came from a Virginia nursery. The trees I planted in 1947, I started grafting in 1950, to Nanking, Meiling, and Kuling, and finished this spring, except for a few replants. I also grafted ten trees in 1950 to Abundance. These tops bore the second year, several bearing good burs the same year the scions were set. These grafted trees are anxious to go to work, because they bloom in the spring and again in late July and early August. I have used the in-lay bark, modified cleft, the cleft, and what I call a saddle graft, bevelling two sides of the stock and splitting the scion, thus slipping the split scion down over the prepared stock. I have had equally good take on all types of grafts used. In 1948 I planted two hundred seedlings bought from Max Hardy, grown from seed from the Experiment Station orchard. I believe the production record of this orchard has been given to this convention at previous meetings. You will recall that the off-type trees were rogued, leaving the parent trees of Nanking, Kuling and Meiling and others of good bearing habits. In 1951 four trees out of this lot, were outstanding in precocity. The earliest started dropping nuts the fifteenth of August and bore 7-1/4 pounds. The next matured September 5th and produced 8-1/2 pounds. The third tree is unusual. I noticed it the 4th of October. The ground was covered with nuts, but only an occasional bur. All of the burs were wide open and still on the tree. The crop weighed 6-1/2 pounds. The fourth tree I found on the 5th of October with all of its nuts on the ground, the tree retaining the burs. The yield of this tree was 4-1/2 pounds. Mind you, this was the fourth summer after planting. These trees have repeated this year with another good heavy crop. The other trees in this block bore from none to one or two pounds of nuts in 1951. This year less than ten trees in the block are not bearing. Next spring these ten will be growing new tops, because their present tops are not satisfactory. I noticed that one tree in this block bloomed long after the rest this spring, several weeks in fact. It might have possibilities in northern areas because of its late blooming.

Of the eleven hundred trees planted in 1950, one bore nuts in 1951. I didn't know it until this spring, when I was pruning the trees in this block, and found nuts on the ground under this tree. It is bearing a good crop this year for its size and age. There are a number of these trees bearing this year. Dr. Crane in a hurried inspection of these trees this summer thought those trees bearing were offspring of a certain tree in the Philema orchard.

I do not give my chestnut trees special care. They are fertilized and cultivated the same as young peach orchards. We try to bring in a peach orchard the third summer, with enough fruit to make it worth spraying. I see no reason to wait seven or eight years to get a chestnut orchard into bearing. If you will keep down competition from weeds, cultivate frequently, and give the tree plenty of nitrogen you will be surprised at the growth it will make. I set the trees twenty-four feet each way, with the idea of thinning later when they begin to crowd. In this way I will get higher acre yields in the early years. When they reach maturity I will have them thinned down to forty-eight feet each way. As they reach heavy bearing the rate of growth will slow down and I will adjust the nitrogen to keep them from becoming too vegetative.

So far the only insects that have bothered me are caterpillars that ordinarily feed on wild maypops, or passion flowers. These caterpillars will defoliate a tree. The only tree that I have lost from winter-killing was one defoliated by the caterpillars early last fall. It may become necessary for me to spray for these worms if they become too plentiful.

I do not come before you as an authority on chestnut growing. I feel that to force myself to do my best I should plant enough trees to make me find out how to handle them. In the rush and bustle of peach and pecan growing if I had only a few chestnut trees I might decide that not much was involved, and neglect the chestnuts. I know that with two thousand trees already planted and some of them bearing I am going to make a great effort to make the project profitable. I have decided that chestnut growing has possibilities as a tree crop in my section, and is worth my time and effort. I know there are many problems ahead, but so did my father when he planted peaches and pecans many years ago. I am still meeting new problems with them each year. Problems go hand in hand with the fruit and nut business. It is the fellow who is willing to try to work them out who has a chance to profit. If I wait until all the problems are solved I will never grow chestnuts. The day that I decide that I know all the answers about growing peaches, pecans or chestnuts, is the day I start going broke. I have been badly bent several times while I was struggling to find an answer. Each year starts full of hope, with visions of a nice fat bank balance when the jobs are all done. Then the problems start and if I can lick enough of them, I come through with the right to see if I can't do a still better job next year, despite the risks of too much rain, not enough rain, hail, insects and diseases.

I have found that each year from 15 to 50 million pounds of chestnuts are imported from Europe. The same blight that destroyed our native chestnuts, is going full tilt in Italy and other European countries. If the blight runs its course as it did in this country, it will not be many years until we will not have chestnuts from Europe. I am going to grow some to fill this gap. In 1950 Dr. McKay sent me eight trees, four Meiling, two Nanking, two Kuling. Two Meiling and two Nanking to be planted together, two Meiling and two Kuling together. Each combination to be isolated so that the nuts produced would be of known crosses. These trees bloomed this spring and two of them set a few burs. Next year I hope to turn over to Dr. McKay nuts from these trees to be planted, and grown to fruiting age. I now have about one hundred and sixty grafted trees. I intend to fruit my seedlings with the hope that among them I will find trees superior enough to be given variety status. I will then top-work the rest to varieties. At present I intend to plant more trees each winter until I have at least one hundred acres of orchards. If and when the weevil moves in I will have the equipment on hand to spray, using the same equipment on peaches or pecans.

I would like to see this Association ask that more research on chestnut production be done by the U. S. D. A. It will not be done until we ask for it. The men in the department are not in position to do much asking for additional funds. It is the responsibility of groups like the N. N. G. A. and the Southeastern Chestnut Grower's Association. We are in need of more breeding and selection of new, and better adapted varieties. We need processing research, marketing research, and research in the field of production. We are not going to get it done until we insist on it good and strong.

This spring, at Fort Valley, Georgia, the Southeastern Chestnut Grower's Association was formed. We hold our convention in March and will be glad to have everyone interested in chestnut growing, marketing, processing or research, attend our convention. I think in time this organization will want to become affiliated with the N. N. G. A., to the mutual benefit of both. I will be glad to have any of you visit my orchards and show me how to grow chestnuts, I am constantly searching for information.

PRESIDENT MACDANIELS: We thank Mr. Wilson very much for his talk, and we think it does take a lot of courage to embark on an experiment of that kind.

In view of the lateness of the hour, unless somebody objects, we will adjourn until tomorrow morning at 8:30.

At 9:40 o'clock, p.m., the meeting adjourned.

TUESDAY MORNING SESSION

(Called to order at 8:30 o'clock, a.m., President L. H. MacDaniels presiding.)

Persian Walnuts in the Upper South

H. F. STOKE, *Roanoke, Va.*

My experience with the Persian walnut has been acquired in the Roanoke district of south-west Virginia. It is located 300 miles from the Atlantic seaboard and my trees are at an approximate elevation of eleven hundred feet. Roanoke is on the same parallel as Springfield, Missouri, and about thirty miles south of Rockport, Indiana.

This experience covers a period of more than twenty years with named varieties and seedlings of the species. I shall here attempt to present some findings that may be of some value to others similarly located.

For the sake of brevity I shall put the cart before the horse, the findings before the facts from which they are derived.

For the upper south and, in my opinion, for the middle west, late vegetating and blossoming is of prime importance for success with the

Persian walnut. No matter how vigorous, prolific and precocious the tree may be, nor how fine the nuts, the variety is worthless for anything except shade if the crop is destroyed by normal spring frosts.

In the second place is winter hardiness. This is of two kinds; resistance to extreme cold, and resistance to the wooing of warm winter days that starts premature activity, followed by a destructive freeze.

My experience with the Payne variety is a case in point. Having read some place of the vigor, precocity and heavy bearing of the new variety, then called the Payne Seedling, I secured some scions of it from its originator and worked it on a young black walnut. The variety was already making a name for itself in Northern California and Oregon, not only because of its bearing habits but for the superb quality of its nuts.

During the first few years it did well despite its early starting in the spring, and bore heavy crops; then disaster fell. One spring the tree failed to leaf out at the usual time. On examination I found that it had winter-killed back to five-year wood. The winter had been unusually cold, and the tree could not take it. Pruned back, the belated new growth did not fully mature before winter so in turn was damaged, a phenomenon that recurred from year to year. Exit Payne as a Virginia prospect.

An example of the other type of winter injury was that of my first Crath Carpathian. I secured scions of this variety from Rev. P. C. Crath in 1929. The parent tree had been growing and bearing in the vicinity of Toronto and was apparently fully hardy. The scions grew vigorously on the young black walnut stock on which it was worked, and completed their longitudinal growth early in July, giving ample time for the ripening of the wood before winter.

After several years I noticed the bark on the south side of the trunks dead from so-called sun-scald. Activity had been induced by the warmth of the winter sun, followed by freezing. After some years the wood was killed back to limbs the thickness of one's wrist, and this has been again repeated. The tree was hardy in Ontario, but not in Virginia.

The nut of this variety, which to me is the Crath, is much superior to the average Carpathian, and I think might be well worth while in the north-east and along the Great Lakes, but not in the upper South nor the Mid-West.

Besides their winter weaknesses, both the Payne and Crath start too early in the spring for my conditions.

Broadview and Lancaster both blossom here in mid-season and, since both have a rather long period of producing pistillate blossoms, they seldom fail to produce a crop when properly pollenized.

Franquette and Mayette, both highly recommended as being late vegetating and producing excellent nuts, have offered me some difficulties of another order. With Franquette the chief trouble has been to get a suitable pollenizer. Like the Mayette, its pistillate blossoms appear ten days or more after the staminate blossoms and self-pollination is not effected. I tried King, recommended as a pollenizer, but it was too early to be reliably effective. When Franquette is properly pollenized it, with Payne, is one of the heaviest bearers.

Mayette in Virginia produces a fine, healthy, vigorous tree, but it refuses to produce pistillate blossoms. A dozen nuts is an average crop for a tree that should produce a bushel. It, like Franquette, demands a late pollenizer, but the pistillate blossoms are simply not there. Neither of these two late varieties have ever suffered winter injury with me, nor have been damaged by spring frosts.

I will not attempt to go into detail regarding all the varieties and seedlings that I have tried through the years; Eureka, that ranks with Mayette and Franquette for lateness, but refuses to bear, apparently for want of pollination; Chambers that was recommended along with King for pollenizing the late bloomers but not fully successful; Breslau, with its huge nuts but slow growth, in addition to an assortment of Carpathian seedlings. Of the latter my Caesar is one of the more promising with its vigorous growth, large thin-shelled nuts and ability to pollenize itself in some seasons. Gilbert Becker has reported it passing through Michigan winters unhurt.

As matters now stand, I believe Bedford, Caesar and Lancaster have proven the most satisfactory varieties to date under my conditions, although some seedlings I have grown appear even more promising. Chief of these are several that I grew from open-pollenized nuts of the Lancaster, which I am here exhibiting.

You will note that the one I designate as L-2 is an extremely large nut, considerably larger than its seed parent which it somewhat resembles. L-8 is of somewhat similar type, but smaller. L-3 and L-6, on the other hand, are of entirely different type. Much smaller, they are smooth, thin-shelled and well filled, with kernels running 50% by weight and of high quality. They resemble their seed parent, Lancaster, not at all but in type are much nearer Bedford, their probable pollen parent.

Another one of these seedlings, L-7, resembles Caesar, its probable pollen parent, far more than it does its seed parent.

Some years ago I hand-pollenized several blossoms of Broadview, using pollen from my original Crath.

One of the seedlings from these hand-pollenized nuts resembles Crath much more than Broadview, the seed parent. I have it here as C x B 2.

Aside from the apparent profound influence of the pollen parent on the offspring, there is the unexplained fact at that with the exception of L-8, all these seedlings are later vegetating than the seed parents and any of the suspect pollen parents. Of the Lancaster seedlings L-2, L-3 and L-6 are fully as late as Franquette and Mayette, blooming well after the first of May. Inasmuch as there were no Persians producing pollen anywhere near that time I can only believe that these nuts were pollenized by the black walnut on which they were top-worked. I intend to plant some of these nuts, and expect to produce hybrids.

This brings up the enticing subject of breeding Persian walnuts adapted to one's own conditions. I have no suggestions to offer scientists, but offer the following for the benefit of amateurs like myself.

If your grounds are cluttered up with varieties, as are mine, ingratiate yourself to some friend who has an isolated young black walnut tree by volunteering to convert it to the production of Persian walnuts. Select two varieties whose characteristics you desire to blend and that will pollenize each other, and grow seedlings from the resulting nuts. You can check results in as little as four years by taking buds from the seedlings at two years and placing then on black walnut.

Creative work, this. You will get the thrill of your life—if you are that kind of a person—and may produce something well worth while.

Persian walnuts are self-pollenizing if pistillate and staminate blossoms occur at the same time, but such usually is not the case. Crath, Breslau, Caesar and King produce their pistillate blossoms some days before their staminate blossoms shed their pollen, while Payne, Lancaster, Broadview, Franquette and Mayette produce their blossoms in reverse order. Of all those I have tested only Bedford can be depended to produce both types of bloom simultaneously and certainly and fully pollenize itself.

It is enlightening to keep a record of the blossoming time of each variety relative to others, but dates should all be recorded for the same year. Warm, early spring induces early blooming; late, cool weather delays blossoming. By my records, Payne pistillates were receptive May 3 in 1935, April 28 in 1937 and March 31, in 1945, a variation of over a month. All varieties vary with the season, but the variation is greatest with the early varieties.

There has been little disease among my Persian walnuts except that in wet seasons leaves and nut shucks are sometimes attacked by a fungous blight. In the city there has been no insect injury worthy of note. In the country, adjacent to wooded areas, insect injury is sometimes serious. Pests include spittle bugs, stink bugs and other insects that attack young leaves and tender growth. These check the leaders and cause late multiple growths that may fail to mature and hence winterkill.

In such locations the butternut curculio also attacks and destroys the young nuts. Avoid wooded areas if choosing a site for a Persian walnut orchard.

The most destructive pest with which I have had to contend has been the large black-bird or purple grackle. Oddly enough they are much worse in the city than in the country. As soon as the young are grown, about the middle of June, they appear in flocks and attack the nuts of the Persian walnut. At first, before the shell has hardened, they penetrate the nut apparently for the nectar which is the substance of the immature kernel. When the shell can no longer be penetrated they continue to eat away the husk, which is equally fatal to the nut. This continues until late in July, when the squirrels take over. Fortunately squirrels are highly allergic to a bullet from a 22 rifle.

In pointing out some of the hazards encountered in growing Persian walnuts in the East the writer has not intended to be discouraging but helpful. Persian walnuts of good quality can be grown in this section; full understanding of the factors involved make it possible, I believe, to grow them successfully on a commercial scale.

Varieties of Persian Walnuts in Eastern Iowa

Ira B. Kyhl, *Sabula, Iowa*

There are a great many varieties of Persian walnuts, many of which originated in the region of the Carpathian mountains and other parts of Europe and a few varieties in the United States and Canada.

I believe that some varieties now grown in the United States and Canada which originated in Europe may have come from the same tree as they appear to have the same shape, thickness of shell and flavor. I have as many as four varieties that are identical.

The Persian walnut has always been my favorite nut. I started with 2 or 3 varieties and now have 35 or 40 varieties and 200 trees most of which are doing well. Some are superior in hardiness and vigor.

In eastern Iowa at 42 degrees N. latitude minimum winter temperatures vary from 25 to 32 degrees below zero. Usually the minimum is 12 to 15 degrees below zero, but last winter it was 25 degrees below zero for several days. Only the hardier varieties will endure -25 degrees without injury, but -12 to -15 does not injure any variety very much.

Schafer is my favorite variety and it was not injured at -25 degrees. I have several of these trees, some from seeds, some top-worked on black walnut and the others grafted trees from a nursery. It grafts easily, grows rapidly and bears a fine nut.

A top-worked tree of Colby withstood -25 degrees without injury and is one of the most vigorous trees I have.

Fifteen seedlings from Crath Mayette and Crath Franquette seeds from the late G. H. Corsan, of Toronto, Canada, are developing into very fine trees, but are not yet bearing.

One of the first varieties planted, Broadview, grew rapidly and produced nuts after two mild winters, but the several trees of this variety killed to the ground after the -25 degrees of last winter.

Crath No. 1, Crath No. 39, and Breslau grew well until last winter when they were killed. Three Breslau seedlings did not winterkill.

Rumanian Giant, the first tree I grafted, killed back somewhat, but is recovering. This variety produces the largest nut I have seen and it fills well.

Top-worked trees of other varieties that were not injured last winter are Crath No. 5, Crath No. 12, SG No. 5, Crath No. 29, Graham and Crath Special.

Seedlings in the nursery row that stood severe temperature are Carpathian D, NWF Nos. 1 and 3, FB O and FB OO, Fort Custer, Hansen, Jacobs and others.

MR. STOKE: Does the black walnut bloom at the same time that the Persian walnut blooms?

DR. MCKAY: It bloomed near the end of the receptive period.

MR. STOKE: That first experiment of yours was trying to pollinize the black walnut with the Persian, but the reciprocal cross may be quite different, as Jones proved with the filberts.

DR. McKAY: That could be. We have no large amount of data on the reciprocal cross. These cases where it is said that the black walnut pollinates the Persian regularly and is producing good crops of nuts, I would consider doubtful until I see the seedlings, their growth and characteristics. Yesterday Mr. Bolten asked the question whether or not some walnuts that have large nuts could possibly be tetraploid or polyploid. A number of years ago I examined the chromosomes of one of these large fruited varieties, and it had the same chromosome number as the others, namely sixteen pairs or thirty two.

The whole question of chromosome number in nut varieties and species is as follows. So far as we know, all of the species have a constant number within the genus except the hickories where we have tetraploid species and diploid species. All of the species of *Castanea*, as far as we know, have the same chromosome number, and all of the varieties within each species have the same number. In the Oaks, which are related to chestnuts, we have an extremely large genus in which there is a great constancy of number. The pines, and all other cone-bearing trees make up another very large group in which chromosome numbers are constant. Exactly the opposite situation is found in the related family of alders and willows where the chromosome number is very variable.

PRESIDENT MacDANIELS: Unless there is some special question or comment on this subject, we will go on to the next item.

MR. LEMKE: There was a panel discussion about four years ago, and they were talking about what nuts to grow, and one of the men said, "Before you offer a man a good nut, give him a good nut cracker." That's been on my mind for some time.

Commercial Production and Processing of Black and Persian Walnuts

EDWIN W. LEMKE, *Washington, Mich.*

Sometime ago a group of nut minded men associated with Spencer B. Chase announced their findings on the quality of the wild black walnut growing in the area of Norris, Tenn. Nuts were gathered from 151 wild walnut trees. After judging, the group came to the conclusion that only one tree had a flavor that was considered by their standards as good. It is these good nuts that caused the formation of the N.N.G.A. When we speak of the good nut it gives the word commercial an entirely different meaning. It by necessity excludes most of wild black walnut kernels processed by the large cracking plants of Kentucky and Tennessee. The large crackers are willing to pay better prices for the improved black walnut but were they to rely on this source of supply they could not stay in business very long.

To produce and process, I chose the Thomas and Ohio variety and I have met with some success. The black walnut can be made to bear in the first and second year after grafting but this is but a novelty feature. Jones from whom I purchased my trees, told me that the black walnut could be classed with the Northern Spy Apple for coming into bearing. This has proven true.

Commercial production of the improved black walnut is by its very nature small scale production. Because of this fact only small scale machines to process these nuts are feasible.

Since 1916 I have had time to reflect on the problem of the three basic machines needed. These are the huller, cracker and kernel picker. Fortunately for me I learned the machinist trade and had a machine shop at my disposal. I tried every way to hull the black walnut and finally accepted the commercial potato peeler as the best principle. I built several crackers and at last accepted the Wiley cracker as the best commercial cracker. The third machine is the picker which has yet to be assembled. This picker is copied after The Kenneth Dick machine with some variations in the separation process.

Let me briefly explain these three basic machines. As the nuts are gathered in the orchard they are brought to the huller in bushel crates. The huller is located in a separate room. This room has the floor depressed to catch the removed hulls that are flushed outdoors with the aid of running water. The cylinder of this huller is 30 inches in diameter and 14 inches high. It is made of 3/16ths boiler plate. Three inches from the bottom of the cylinder is a revolving disc smaller than the inside of the cylinder. The disc being small enough it allows a 5/8th opening around the inside of the cylinder. It is this opening that permits the hulls to drop to the floor. The nuts are held captive because there is no opening in the cylinder for them to leave until the discharge door is opened on the side of the cylinder. The cover of the cylinder has a 10 inch feed hole into which the nuts are fed. A 10 inch furnace pipe elbow runs from the hole to the serving trough into which the nuts are poured. A 10 inch pusher is used to shove the nuts into the huller and serves to keep the feed hole closed while the nuts tumble around. The disc runs at 250 RPM which is the proper speed to do a good job. While the nuts tumble around a stream of water is used to wash the hulls free from the nuts and force the removed hulls to the floor below. The disc is supported by a 1-3/8 inch diameter shaft that runs through the disc and is held central as it revolves in a flange containing a 3/4 ball bearing that fits into the end of the concave in the shaft. Up four feet from the disc is a link self aligning bearing that allows the shaft and disc to turn like a gyroscopic top. The shaft's pulley has 'V' belts connected to a 3/4 h.p. motor. I have hulled up to 40 bushels of clean nuts in 8 hours. The nuts after hulling are placed on drying trays indoors where temperatures are better controlled. The principal of this huller is that it separates the hull by centrifugal force. The hull drops down through the opening between cylinder and disc while the nuts riding on disc are discharged at right angles to the fall of hull. The machine is a separator.

The next basic machine is the cracker. This cracker is the Wylie cracker in principle and is made in Eugene, Oregon. Simply explained it could be likened to two pages in a book. One page is perpendicular while the other page is off the perpendicular about 7 degrees. The first page which is the anvil is fixed save for adjustments for nuts of varying size. The other page or hammer riding up and down through an inch and one quarter of travel is fixed to a crank below. Both of these pages or plates are heavy cast iron plates that are fluted and cause the nut to be cracked against these saw toothed flutes and while being cracked are revolved down through the plates. The plate moving at an angle forces the nut finally through a 3/8 inch opening where they fall into a rotary sieve. The sieve has three sizes of mesh. 5 mesh, 2 mesh and 3/4 mesh. The larger pieces go on through and are returned to the cracker. This cracker will crack up to 500 pounds per hour, and uses a 3/4 h.p. motor.

The last of the three basic machines is the picker. I have not yet built the picker but a number of the parts have already been machined and before long it will be a reality. The Kenneth Dick, picker, of Peebles, Ohio is the best for small orchards. It is essentially a separator using a conveyor belt which carries the cracked nuts to needles that pick up the kernels and deposit them on trays that at the timed moment accept the black walnut kernels. The discarded shells remain on conveyor and travel to the end and fall into a receptacle. After this process, further inspection becomes necessary but up to the present it is the best we have.

The black walnut is a messy nut to fool with but with the proper machines it soon becomes a pleasure to work with it. I can work all day hulling nuts and finish with clean unstained hands.

Processing the Persian walnut is a simple matter as compared with the black walnut. My Persian nuts are gathered and placed on drying trays. Most of the nuts fall free from hull and the stick tights are discarded as inferior. N.N.G.A. members need but write to the agricultural colleges in California, Oregon and Washington and a list of publications will be sent. One of the latest machines being offered is one that picks the nut from the orchard floor with a speed with which no human can compete. It has not only removed the back ache but the human back as well. The Persian walnut industry in the Pacific Coast states is big business.

There is only one organization that can and does disseminate the necessary knowledge and experience that will give the northern grown nut its proper place in the American diet. That is the Northern Nut Grower's Assn. You newer members have become heirs to knowledge based on the experiences of others which represents not only blood, sweat and tears but a lot of good hearty belly laughs. When one becomes nut conscious there is no turning back. It gives life a new approach and a finer meaning.

Black Walnut Processing at Henderson, Kentucky

R. C. MANGELSDORF, *St. Louis, Mo.*

MR. MANGELSDORF: Mr. Walker and Mr. McDonald are unable to be here today, and I don't know if I can fill their shoes or not, because I am not in the purchasing or processing end of the black walnut business.

We started this black walnut shelling operation a season ago at Henderson, Kentucky, with the idea of processing the nuts there and transporting the kernels to St. Louis for final processing and marketing. At Henderson, Kentucky we are located outside the city limit, and we have no fire protection, and as a result, the insurance rates on our building, storage sheds, and black walnuts in storage have been so high that we are looking around for possible plant location sites where we can reduce that expense of operation.

Another factor in our operation there is the transportation of raw material to our cracking site. If we have to transport black walnuts, which give an approximate 10 per cent yield, any distance, the freight adds materially to the cost per pound of the finished material. That is, if we have to pay 10 cents per hundred additional freight cost in transporting them from outlying districts to the cracking plant, that adds a cent a pound to the cost of the finished kernels. All such factors, have to be given weighty consideration, because our business is primarily concerned with making money for the stockholders. If we don't make money for the stockholders, they are not interested in seeing us continue the operation.

Mr. Walker and Mr. McDonald at the present time are out on a crop inspection trip and also making surveys of locations and availability of buildings or sites that might be more advantageous than the one at

Henderson, Kentucky. It may be that we will continue the operation there, making modifications in the building, which will result in lower insurance rates. At the present time, with the new crop coming on, we are in a chaotic state of affairs, because we just don't know exactly what's the best path to follow in our operation at Henderson, Kentucky.

Are there any questions?

DR. MCKAY: Will you tell us something about how you handle the nuts in your plant, how they are hulled and cracked, and so forth?

MR. MANGELSDORF: It's a similar operation to what Mr. Lemke described. The nuts are brought in in burlap bags by the farmers and growers and are put in storage in cribs. The plant at Henderson, Kentucky, was a popcorn processing plant, with a large crib under roof where the nuts are stored. After the moisture content is reduced somewhat, they pass through a tumbling drum to remove any of the extraneous hulls and other dirt that might be adhering.

After the nuts are completely freed of all this extraneous matter, they are passed through a series of cracking rollers with screens. The nuts are cracked, by passing between two rollers like a wringer then passed over a shaker screen, the free nut meats passing through the screen. The large material that comes off of the screen is then passed between more closely spaced cracking rollers and then further sifted and screened. Then the various materials that have passed through the screens are run through a Smalley picker. This is nothing more than metal pins on a series of fingers rotating on a roller that presses against a sponge rubber roller. The nut meats adhere to the prongs or points. The shells, not being penetrated by the points of the pins, are not picked up. Then there is a comb that picks off the adhering kernels from the picker prongs. That's the principle of most of the shelling operations of the black walnuts. I don't believe any major changes have been made in the processing of black walnuts in the last ten years.

DR. COLBY: How do you remove the hulls?

MR. MANGELSDORF: We try to buy only hulled walnuts, the farmer and the grower removing the hulls in a tumbler and selling to us only the dehulled walnuts.

The kernels are packed in cartons and shipped to St. Louis for final picking of remaining shells and off-colored nut meats and graded for color, size and quality. After this grading separation is made, they are either packed in our 4-ounce vacuum-packed tins or 30-pound bulk cartons which are then sold through the trade.

MR. WALLICK: What percentage of kernels do you get?

MR. MANGELSDORF: I think our operation at Henderson, Kentucky this past season for all of the nuts that were grown and gathered in this locality was about 9.48 per cent yield of black walnut kernels by weight.

MR. WHITFORD: Do you get any improved varieties, such as Thomas, Stabler or Ohio?

MR. MANGELSDORF: No. With most of the nuts that we gather in our marketing operation very little attention is paid to variety or source. We don't try to differentiate and store them separately, but everything is processed as it is brought together.

MR. MCDANIEL: Do you have any indication that you get a better quality nut from one county or one area than you do from another?

MR. MANGELSDORF: That is a question that I can't answer, because I am in the research and development end of the business, and have very little to do with the purchasing and marketing of the nuts themselves.

MR. LEMKE: What do you do when you strike a day that is very humid and the nuts start getting moldy?

MR. MANGELSDORF: That is a bugaboo. I always say you don't have to be nuts to be in the nut business, but it sure helps a little bit. All the nuts that I have ever had any dealing with seem to be very susceptible to mold growth. If the moisture content of the nuts is above a critical level, mold growth takes place in the shell at a very fast rate. The only thing we can do in a case like that is to get the kernels in to St. Louis and destroy the mold growth or spores on the surface before it can grow so that the fungous mycelium is visible to the eye. The black walnut and pecan, if you examine them under the microscope, all seem to have mold growth on the surface of the kernels. I am inclined to believe that the nut kernel is not completely sterile in the shell and that through some manner or means the mold spores have been introduced onto the kernel, because immediately after shelling examination of these nuts under a microscope, will show some fungous mycelium on the surface of the kernels.

DR. MCKAY: One comment is that the pellicle of a black walnut or a pecan, is very hygroscopic. It tends to absorb moisture readily, whereas the kernel itself, being high in oil, does not take up water readily. That, apparently, is why there may be evidences of mold growth on the kernel though it may not be actually penetrating. It is only superficial, growing on the pellicle of the kernel, not on the kernel itself.

MR. MANGELSDORF: Right.

DR. MCKAY: Black walnut kernels are outstanding in their resistance to heat and will get rancid very slowly under conditions of high heat—not humidity. For example, we had some nuts in our attic for two summers in a place where it gets very hot, yet dry. Those nuts are in very good eating condition today. I don't know about pecans.

MR. MANGELSDORF: That's very true of black walnuts. Pecans have to be carried throughout the season in our cracking operations under refrigeration, but the black walnuts we can store out in any shed with tin roof. The temperature gets very hot, and it seems to have no effect whatever on the edibility or rancidity of the nut kernel.

MR. STOKE: You spoke of storing the whole nuts in large bins. There you may have an extreme amount of mold, too, if the nuts are damp.

MR. MANGELSDORF: We try to have storage conditions such that air has free passage through the bulk of nuts. The mold and the yeast are there and when they start to grow, their metabolism throws off quite a large amount of heat. As a result the molding process is speeded up like a chain reaction, and before long the nuts will be worthless for shelling.

MR. MANGELSDORF: We had nuts until just a few weeks ago from our last season's gatherings. That's almost a whole year.

MR. SALZER: Can you tell me if the farmer is paid by the weight of the nuts, or does he receive his pay after the kernels are shelled out? Does he receive more money if it contains a higher percent of kernels?

MR. MANGELSDORF: He receives his pay on the basis of the whole nut that he delivers to the plant, and we try to exercise some control over the quality of the delivery. Samples are taken and cracked, and if most of the nuts are rotten or the quality is very low, we may reject buying that entire lot, or we may discount the lot of nuts a certain amount, depending upon the percentage of the nut meats that are salvaged.

MR. MURPHY: Do you pay a premium for cultivated nuts?

MR. MANGELSDORF: That I can't answer, but I don't believe that they have this past season. I wouldn't want to go on record as to that. There is a tremendous difference in the flavor of what we call the "eastern" black walnut in comparison with the California or western black walnut. We think that the flavor of the California walnut is not at all comparable to the eastern black walnut.

MR. MCDANIEL: You don't notice any difference, do you, between the Missouri and the Kentucky nuts?

MR. MANGELSDORF: No, not in my experience, but there is a tremendous difference in flavor between the eastern and western.

MR. ROHRBACHER: On what basis do you buy black walnuts?

MR. MANGELSDORF: I understand that each individual sale is an individual "horse-trading" deal, the price paid, depending upon the quality of nuts, moisture content, color and other factors. Of course, our aim is to buy the nuts as cheaply as possible and the object of the fellow selling the nuts is to get the greatest return that he can from what he has to offer. So we try to reach a happy medium in our dealings, and a lot of concessions might be made one way or the other with special lots that are offered for sale.

MR. WHITFORD: What sizes and grades of kernels do you have?

MR. MANGELSDORF: We have the large, medium, small and granules. Granules are very small pieces. Usually the prices paid for the nuts are not determined, actually, until the crop starts to move. Everybody has an idea what the market price will be for the nuts, but nothing is crystallized or brought to a focus until the first nuts are actually on the market. Then the nuts sold are examined as to quality, giving some idea of the future quality of deliveries that might be made in that section, and then prices can be established. As I say, it's a nutty business. I haven't grown very many gray hairs yet, but I expect to have many before I am through. And each new problem that arises in this nut business, when you reach a solution for it, invariably there are two other problems that are created, and if you are not wide awake, one of these problems can be much greater than the one that you just had a solution for.

MR. DAVIDSON: Do you know anything as to the bearing of black walnuts this year as compared to previous years?

MR. MANGELSDORF: Mr. Walker and Mr. McDonald are out at the present time making a crop inspection tour of the various localities, and I have had no report as to what the condition of the crop will be this year.

MR. WHITFORD: Which grades bring the highest prices?

MR. MANGELSDORF: The large particles of kernel demand a premium over the smaller sizes. That is one of the discrepancies in the shelling operation, that the material that costs us the least money to produce gives the largest returns. When you have small pieces, the operation of removing the last remaining shells and off-colored particles is much greater than with the large kernels. One large kernel amounts to considerable weight and you may have to pick up many small particles to represent the same weight.

PRESIDENT MacDANIELS: We appreciate very much your talk, Mr. Mangelsdorf.

One thing that interested me was your statement that having large pieces was an advantage. That question has been argued on the floor of these conventions a number of times and there have been those who claimed that the larger pieces were all ground up anyway and that the varieties from which you can recover large pieces were of no particular merit commercially.

The next paper is, "Nut Shells—Asset or Liability?", T. S. Clark of the United States Department of Agriculture, Regional Laboratory, Peoria, Illinois.

Nut Shells—Assets or Liabilities

T. S. CLARK, *Northern Regional Research Laboratory*,[1] *Peoria, Illinois*

ABSTRACT. The value of nut shells as materials for agricultural and industrial use is discussed. Problems of plant location, shell collection, processing, and hazards are considered. Applications and specifications are illustrated.

We are particularly pleased that the Northern Nut Growers Association is presenting this opportunity for a discussion of nut shell utilization. The Northern Regional Research Laboratory feels that it has played an important role in what is now becoming a new industry of increasing magnitude. For the benefit of those who are not already acquainted with the Laboratory, permit me to digress momentarily to explain briefly its organization and functions.

The Northern Regional Laboratory at Peoria, Illinois, is one of four large research laboratories established by an act of Congress in 1938 and placed under the administration of the Bureau of Agricultural and Industrial Chemistry. The function of these laboratories is to conduct research and to develop new chemical and technical uses as well as new and expanded markets for the farm commodities and byproducts of the regions in which the laboratories are located. The commodities studied at the Northern Regional Research Laboratory are the oilseeds, cereal grains and agricultural residues which include corncobs, stalks, straws, sugar cane bagasse, hulls and shells of nuts and fruit pits. Because of the great similarity in chemical and physical characteristics of the residues all research on these materials is conducted at the Northern Laboratory.

During the time that the Northern Laboratory has been actively investigating shell materials and other agricultural residues we have been in direct communication with operators of shell grinding plants; some of these have been visited. We have received numerous letters and calls for information and assistance in solving grinding problems, or in using the ground products. Through these contacts and our experiences we have learned much about the factors that lead to success or failure in this utilization. Ten plants are now producing a variety of ground shell products useful in both agriculture and industry.

When the Northern Laboratory was organized, only one plant, established originally by the California Walnut Growers Association, was grinding nut shells. This plant, following a number of operational difficulties and administrative changes, now processes 40 tons or more of shells per day and produces a wide variety of ground products including exceedingly fine flours for use in plastics and plywood adhesives. It has been said that this plant processes all of the English walnut and apricot pit shells and 80 percent of the peach pit shells available in California.

The Laboratory has attempted to determine the amount of shells and pits available commercially in different areas. Data of this nature has been obtained for the larger cracking plants but there are many small operations for which we lack this information. "Agricultural Statistics" compiled and published annually by the U. S. Department of Agriculture provide an excellent source of information regarding production and, in many cases, the disposition of farm commodities. For example, the production of pecans in 1951, presented by states, totaled more than 73,000 tons for the 10 states reported. However, no data were available regarding marketings in-shell, or the quantities remaining on the farms or in the orchards. Thus, the quantity of pecan shells actually available for processing can be determined only through surveys of cracking plants. Only limited information is available concerning black walnut shells and this has been obtained through the cooperation of shellers or crackers.

In some areas fruit pits, such as apricot and peach pits, accumulate at canneries or freezing plants. Similarity in character of the pit shells to those of the nuts permits their use in plants grinding nut shells. Thus, the supply of raw material in any area may be augmented by inclusion of fruit pit shells.

Collection of nut shells for grinding operations is a relatively simple procedure, particularly where grinding is done at a cracking plant. Where shells must be collected over large areas both rail and truck transportation are used. If fruit pits are considered, provisions should be made for removal of residual flesh or pulp before the pits leave the canneries. In the cases where the pits have been cut during processing of the fruits, the released kernels should be removed before shipping the shells. Pit kernels are valuable for their oil content.

Shell Use During World War II

The production and maintenance schedules set up during World War II resulted in the development and expansion of uses for ground shell materials. Fine flours from walnut shells were needed as extenders in plywood adhesives. Soft grits from various shells were used by the Army Air Forces in the air-blast method for cleaning airplane engines and parts. Grits were required for deburring metal stampings and flash-removal from molded plastics. These uses have expanded considerably to meet civilian needs since the war.

Grinding Nut Shells and Fruit Pits

As uses for ground shell products were developed the Laboratory sought advice of grinding equipment manufacturers for information on the design and construction of suitable grinding plants. Only limited tests had been made and data were not readily available in any published form. Consequently the Laboratory undertook an extensive study on grinding nut shells and fruit pits as part of its research on agricultural residues.

These studies were not limited to grinding only, but included methods of separation and classification based on physical characteristics of the raw materials; the relation of associated mechanical operations; a consideration of the hazards; the problems of labor, management, and merchandising.

A number of fires have occurred in plants grinding nut shells, corncobs, stock feeds, and similar materials. In most cases the causes of fire have been other than the grinding operation. From a consideration of the causes of fires a number of safety precautions have been developed. Good plant housekeeping is paramount. This is essential, not only because of influence of dust and dirt on the maintenance of motors and equipment, but because of the highly explosive nature of shell dusts. The U. S. Bureau of Mines has cooperated closely with the Northern Laboratory in evaluating the explosive hazards of the shell dusts.

Many of the present operators of shelling-grinding plants have benefited from the information and assistance available from this Laboratory. The cooperation of equipment manufacturers has aided considerably in extending the scope of the Laboratory's studies.

The Northern Laboratory has published bulletin AIC-336, "Dry Grinding Agricultural Residues, A New Industrial Enterprise" that summarizes the research conducted to date. This is the first time that such data on engineering and design has been assembled and published to cover this field. Copies of the bulletin may be obtained by addressing requests to the Northern Regional Research Laboratory, Peoria, Illinois.

Plants designed to produce at least 1-1/2 tons per hour of ground shell products will cost upwards of $60,000. A well-engineered plant of such size will require three to five men per shift. Among other factors, the working capacity of a grinding plant depends upon the quantity of shells available and the ability of the organization to merchandize its products. The plant should be located in an area in which at least 5,000 tons of nut shells or fruit pits are annually available at low transportation costs.

Uses of Shell Products

The more important uses for nut shell products, together with their specifications for particle size, are shown in Table 1.

Table 1.—Uses for ground nut shells and fruit pits

Applications	Size
Deburring, cleaning,	

burnishing and polishing | | | in metal stamping, electroplating and | No. 10 to No. 50 | | plastics industries | | | Soft-grit blasting | No. 10 to No. 30 | | Fillers for plastics and plywood adhesives | Finer than No. 100 | | Insecticide diluents and carriers | Finer than No. 140 | | Explosives | No. 10 to No. 100 | | Fur cleaning | No. 10 to No. 100 | | Poultry litter and mulch (almond and peanut) | 1/4 to 3/4 inch | | Fillers for fertilizers (almond and peanut) | Finer than No. 20 | + —————————————————————————————————————+ ——————————————+

Experience shows that no matter how nut shells or fruit pit shells are ground both under- and oversize particles will be produced. The hard, friable character of most of the nut shells makes their reduction to fine size particles less difficult than for tough materials, such as corncobs, or fibrous materials such as woods. Shells from almonds because of their bulk and very fibrous nature are somewhat less convenient to handle than other shells. Good business practice shows that sales outlets should be found for each fraction so that grinding expenses can be kept at a minimum.

Because there are some differences in physical characteristics of nut shells and fruit pits all shell products do not necessarily meet the same specifications, nor have the same uses.

Industrial Cleaning and Finishing

Oil, dirt, corrosion products, stain, paint, grease and the like can be removed from metal surfaces by air-blasting with soft grits prepared from shells of walnuts, pecans, peach pits, and similar residues. This method was developed originally for the Navy to use grits from corn-cobs for cleaning aircraft engines and parts. The method is inexpensive and foolproof because surfaces are cleaned without change of dimensions. No pitting or abrasion, such as produced by sand blasting, occurs. The method is particularly useful with mild steel, nonferrous metals, alloys, and parts that must be maintained at close tolerances. Modifications of the blast method are used in finishing molded plastics, metal die-castings, and machined parts. One manufacturer of precision instruments states that his company saves $100,000 a year in finishing parts with shell grits.

Many stamped metal articles and molded plastics are deburred, cleaned, burnished, and polished by tumbling in drums containing shell grits. Various grades of grits are required depending upon the nature of the pieces being finished.

Fillers for Plastics and Plywood Glues

The Laboratory has studied the use of shell flours for use in plastics and plywood glues. Many of these flours are now in regular commercial use. Flours for these applications are prepared in various grades, all finer than 100-mesh. Use of these flours not only improves the properties of the final products but also reduces the cost of the products. Molded plastics prepared with fine flour from English walnut shells have exceptionally fine surface finish.

Insecticide Carriers

The insecticide field provides a good outlet for shell flours. Flour from walnut shells was the first of this type of material to be used for this purpose. Often the active ingredient in a finished insecticide is present in quantities of less than 1 percent. Custom grinders should plan to recover the flour as a co-product of their operations rather than attempting to grind to flour alone.

Explosives

Large amounts of shell grits and meal are used as diluents in the manufacture of dynamite. Material for this use ranges in size from No. 10 to No. 100, the requirements of the individual manufacturers falling within much narrower limits as to size.

Fur Cleaning

Furriers have found that various ground shell products are very effective agents for cleaning furs. Size requirements for this purpose are broad, the limits being dependent upon the cleaning equipment maintained by the furrier. The natural oils present in some shell products are considered advantageous for this application.

Sundry Applications

Stock bedding, poultry litters, fillers in feeds and fertilizers, mulches, charcoal, tannin and abrasives in hand soaps are some of the other products that are prepared from nut shells. The shell products cannot be used interchangeably but must be selected in accordance with their chemical and physical properties.

I hope that the foregoing brief discussion has conveyed to you the potential value that lies in the piles of shells accumulating at the cracking plants, and that these accumulations can be converted from expensive wastes to profitable products.

FOOTNOTES:

[Footnote 1: One of the laboratories of the Bureau of Agricultural and Industrial Chemistry, Agricultural Research Administration, United States Department of Agriculture.]

The Propagation of the Hickories

(Panel Discussion led by F. L. O'Rourke, East Lansing, Mich.)

MR. O'ROURKE: I hope that we can have a rather stimulating session on hickory propagation this morning. Last year we had a session which was supposed to take in propagation of all nut tree species. However, we never got away from Chinese chestnuts. It was Chinese chestnuts from the start to the finish. The Program Committee this time thought that we should limit it to one group, and they chose the hickories.

I have compiled a review of all the literature pertaining to the hickories and passed it out yesterday afternoon. I hope that some of you have had a chance to read it and will have some questions to ask us this morning.

In order to really have some help, I am going to call upon Mr. Louis Gerardi of Illinois, Mr. Ferguson of Iowa, Mr. Max Hardy of Georgia, Mr. Ward of Indiana, and Mr. Wilkinson also of Indiana and Mr. Bernath of Poughkeepsie, New York.

The subject matter of the panel will be limited to the propagation of hickories, which includes the pecan.

Who has some questions that they'd like to bring up?

MR. SALZER: Which varieties will grow on fairly wet soil?

MR. O'ROURKE: That is a question pertaining to culture, rather than propagation, but we can still allow it. Which varieties—I presume you mean species, is that correct?—will grow on fairly wet soil? I think Mr. Ward has a little bit of black soil in that good, old state of Indiana.

MR. SALZER: I mean soil that doesn't dry well in the spring. I have one spot that's too wet for chestnuts.

MR. WARD: I wouldn't put any hickory nuts on it. You are going to find it is going to be very difficult for if the soil is the least bit heavy or wet, the hickory nut does not do well at all. In the Wabash bottoms there is a lot of this black soil that is overflowed every year, and some of the finest hickory nuts and some of the finest pecans that you can find in the country are there. Sometimes I have seen water marks on those hickory trees several feet from the ground in the spring of the year and sometimes in the summer, yet they come through with a good crop of nuts. Underneath it is a strata of gravel so that the soil drains out in a hurry.

MR. SALZER: This has subsoil drainage.

MR. WARD: The soil around Rochester is very heavy like what we call slashland type of soil here in Indiana, and where this occurs we find that the hickory nut does very, very poorly. I wouldn't advise putting them on such soils. The black walnut will grow a lot better in places like that.

MR. GERARDI: In Illinois we have that deep, black soil and we just call it plain gumbo. It's all filled-in soil, and I never have reached the bottom. It's at least 20 feet thick. And these swamp hickories—I think Reed was the one that called them swamp hickories—thrive there. They can be two months under water six foot deep, and still bear wonderful crops. You can get a wagon load of them in that mucky soil.

MR. CALDWELL: The hickory in New York State which will stand the most moist conditions is the bitternut hickory, and with that root stock you may be able to get some of the others through. The shagbark will withstand considerable moisture if it has deep soil. The bitternut does well on shallow soil or the soil that is made shallow by high water.

MR. O'ROURKE: The bitternut, then, will survive wet conditions. This is of interest as far as root stocks are concerned. I am wondering if anyone would like to report on the ability of the pecan to take wet soil conditions.

MR. WILKINSON: They will turn out all right if they have dry feet during the summer months, but they will not stand wet feet all summer.

MR. O'ROURKE: Will the bitternut do better, or would the mockernut?

MR. WILKINSON: I am not well enough versed on that to say. But the pecan, I have seen them stand under water for weeks at a time two or three times during the winter, water 20 feet deep and not affect them at all. But if they are around in a place where the water stands in July and August, they won't take it.

MR. O'ROURKE: Any other discussion on stocks that will take wet soil conditions? If not, let us take up Mr. Beckert's question: When do you take scion wood of the shagbark hickory? Who would like to answer that? Mr. Gerardi?

MR. GERARDI: The time I like best, the time it can be done in our particular area is the latter part of February. Leave it on the tree as long as you can before any sap rises.

MR. O'ROURKE: You would say probably 10 days to 2 weeks before the bud scales would break?

MR. GERARDI: That's right, before any growth begins.

MR. O'ROURKE: Any other comment on that? Dr. McKay?

DR. MCKAY: I want to ask the question about which there is difference of opinion. Do pecan seed have a rest period, and is there any difference between pecans and hickory in that respect?

MR. HARDY: I am not sure that I can answer the question exactly. Most pecans planted for seed have been allowed to dry before they are harvested, and it is general practice to stratify them either in sand for planting in the spring or planting them immediately in the fall. I am inclined to think that there is very little rest period in pecans and that if they were planted immediately from the tree that perhaps they would begin to grow almost immediately.

DR. MCKAY: I think that's true. The seed will germinate quickly. But can you plant dry seed any time during the winter?

MR. HARDY: Once they are dried I think they must go through after-ripening conditions.

MR. O'ROURKE: Do I understand you correctly that you do feel that the pecan must be after-ripened?

MR. HARDY: Yes, if permitted to dry.

MR. O'ROURKE: The work of Burdette in Texas a great many years ago has indicated that the pecan seed does not have a rest period. Mr. Wilkinson, what has been your experience in germinating pecan seeds?

MR. WILKINSON: I usually like to either plant or stratify soon after gathering, although one time I had some off the shelf of a grocery store in

March and got excellent results. One thing more about time of cutting graft wood. I never like to cut it for at least 48 hours after a freezing temperature, regardless of time. I would rather cut it in April with the buds green than to cut it in the first of March right after a freeze. I have had excellent results just this spring cutting extra graft wood with green buds on. But if you cut it within 48 hours after a freezing temperature, you might just as well throw it away.

MR. O'ROURKE: I am very glad you brought that out. Irrespective of whether it be pecan or hickory, I believe it would work the same, that the scion wood should be cut when it is moist, and that is not the condition after a freeze, when it is in very dry condition.

Let's get back to this seed propagation now. I am asking anyone here, can you throw any light at all on the need for stratification of pecan or hickory seed of any species.

MR. CALDWELL: I have read in several publications that hickories should be stratified over the winter period before planting for spring germination. I always find things a little bit different, so a year ago at the greenhouse I took seven different sources of seed of shagbark hickory, *Carya ovata* and one source of *Carya ovalis*. Some of those seeds germinated within three weeks from the time I put them in, and after a month and a half I had a full stand in all cases. I don't think that more than 2 per cent of the seeds failed to germinate. They were planted in warm greenhouse, with a minimum of, about 68 degrees at night and about 90 during the day. They were planted in a combination of peat and garden soil; no special care other than water. I have had no trouble since the seedlings have continued to grow, even though the seeds were planted only two and a half inches deep. So it may be that there is no need for stratifying hickories.

MR. O'ROURKE: Your experience is the exact duplication of Dr. Lelia Barton's of the Boyce-Thompson Institute. She found that hickory seeds germinated from three weeks, as you did, to a number of months, when put in a warm greenhouse. Apparently the difference in time is related to the thickness of the seed coat or possibly to an inhibitor in the pellicle rather than to any need for after-ripening. I think that Burdette in Texas also pointed out that thick-shelled pecans took longer to germinate than thin-shelled pecans.

MR. PATAKY: If you take a nut of any kind and let it dry and plant it, you will get quicker germination than if you plant it soon after harvest. I don't see any difference in taking a nut and planting it and stratifying it. If planted the rodents will get it, but if you put it in something all winter, it will be there in the spring. I don't see any reason for planting a nut in the fall, taking a chance of rodents getting at them. If you plant them in the spring, they come up so much quicker that the rodents don't have a chance to get at them. They got nearly all of mine that I planted in the fall.

MR. HARDY: A good many nuts don't have any rest period requirements. I think it probably is a matter of convenience as to the manner in which they are handled. I have talked with nurserymen in the South. If they get the nuts in the fall they may either plant them in the fall or stratify them over winter and then plant them in the rows in the spring. If they get them in the spring, they soak them for a day or two days in water before planting. Perhaps the dry nut is slow in taking up moisture direct from the soil, and they are primarily interested in getting a uniform stand of trees so that they handle it in such a manner that all the nuts will grow at the same time. And I believe many will agree that a dry nut planted in the spring will show considerable variation as to the time in which they appear above ground.

MR. O'ROURKE: The suggestion of soaking them in water a few days is well taken, because a great many have recommended it. Most folks recommend changing the water daily. By changing the water you replace the oxygen which would be in the water, and you also eliminate any toxic substances which may have leached out of the shells during the preceding 24 hours.

DR. MCKAY: I'd like to mention the reason for raising this question. Dr. Crane has the idea that there is no definite rest period in the pecan nut; if they are soaked in water they will sprout at any time.

I decided I would test that hypothesis, so I stratified one group of nuts of about four pounds. Another lot of four pounds I kept in the laboratory dry all winter long. Then I planted the two lots of nuts this spring together, side by side, in the cold frame. Today there is not a single seedling growing out of the dry lot, and there is a perfect stand in the group that was stratified.

To me that means that there is a definite rest period in the pecan seed. I don't see how you can get away from it.

MR. O'ROURKE: I am going to stick my neck out a little bit. I have absolutely no basis to make this statement, but it does give us something to think about. That is the greater the distance towards the north that certain species of plants may have migrated or disseminated, the greater the rest period requirement. That is a protective device for a species to persist in northern climates, because if it were not for this rest period, those seed would germinate in the fall of the year, and the young seedlings would be frozen out immediately. But by having the rest period requirement over winter, the seedlings do not germinate until the following spring, and the plant can persist. I am speaking now in general of northern plants. I am wondering if the pecan species in itself may not be variable in that the southern pecan does not need a rest period, and the northern pecan is beginning to develop the rest period requirement.

MR. HARDY: Mr. Chairman, I am inclined to think there may be some other factor entering into the picture there. A pecan carried through winter in a dry condition at normal room temperatures would be liable to develop quite a bit of rancidity by spring. Furthermore, nuts that have been held over so long in a dry condition may still be good and may germinate the second year. I'd hesitate to destroy that planting until next spring, and to my notion that does not indicate dormancy so much as it would possibly indicate the inhibition of growth by some other products developed during that storage period.

MR. O'ROURKE: You have brought up a very important point and something we should not neglect. It may be that drying to a certain degree will induce dormancy, a grievously overworked word, but you know what I mean. It may take two years for the seed to germinate, as Mr. Hardy has suggested. If you can leave them in that cold frame over this winter, maybe you can tell us next year just what happened.

MR. PATAKY: If we take nature's way, watch a squirrel plant a hickory or black walnut. He will bury it about an inch deep, and it will stay moist all winter long, the same as if it were stratified. But if you take a nut and store in a hot place you are going to slow up or kill that germ.

You can do that very easily in a chestnut. Take a little advice from nature itself in the locality where you are. If you are in the South, that nut can start growing in the fall, and it probably won't hurt it, but if you are in the North, you don't want to start a nut growing in the winter, because it's going to get winter killed.

MR. O'ROURKE: In all probability the amount of oxygen about the germinating seedling might be quite a factor. The shallow planted seed will have more oxygen available than deep planted seed, everything else being equal.

If we are finished with the discussion or germination of seeds, we can go on to the next question, that of a suitable root stock for hickory—and that could keep us here for two or three days. Have you had some experience, Mr. Ferguson?

MR. FERGUSON: We use the pecan and the shagbark as root stock for the hickory group. Formerly we have used some of the bitternut, but we do not use it any more. Some of the hickories will grow well on pecan, and some are not satisfactory at all. What they will do in old age is hard to tell. We have a few in the orchard down in Mr. Snyder's farm. I think we have Stratford on pecan, which is not satisfactory. Pecan grows too fast for the Stratford, and some way or other it just doesn't work.

MR. O'ROURKE: Are you familiar with Mr. Lassiter's stock work?

MR. FERGUSON: He has used the Rockville as an intermediate stock on pecan. The Rockville is a hybrid of the pecan and the shellbark.

MR. O'ROURKE: Mr. Lassiter sent us a letter in which he stated that he had a good variety of shagbark that when grafted on the Rockville intermediate stock produced much better nuts than on pecans alone. Is that due to the exceptional vigor of Rockville which apparently is a hybrid and may have hybrid vigor? Again, we can only guess. This interstock problem is a big problem. We now have some evidence that pecan is not always satisfactory for all varieties of hickory, although Mr. Dunstan at Greensboro, North Carolina, states it's been satisfactory for every variety he has worked upon it.

MR. HARDY: I am inclined to believe that root stocks and scion varieties worked in the north and grown in the north or worked in the south and grown in the south may not react the same.

MR. WILSON: I think you are right on that.

MR. O'ROURKE: Mr. Gilbert Smith's report of yesterday indicated a pecan was not satisfactory with him in New York State, and that may bear out the comment that Mr. Hardy has made.

MR. GERARDI: Well, I think that is true enough, myself. In southern Illinois I find that the bitternut hickory root for shellbark or shagbark don't seem to be satisfactory at all. With the shagbark on pecan, the variety of shagbark makes a difference. Some varieties of shagbark, and shellbark hickories seem to do all right, and then again others don't. It's going to need further study to determine what varieties will stand on pecans, what will stand on bitter hickories, or what will stand on regular ovata stock. I think that the nurseryman's wisest way is to use stocks of the same species as the scion and then he is on the safe side. Because the bitter hickory grows faster, the nurseryman may find it advantageous to grow the bitter hickory stock in preference to the other two.

MR. O'ROURKE: The bitter stock makes a hickory big enough to graft in two or three years.

MR. GERARDI: In two or three, and four or five for the shagbark. Shagbark or shellbark varieties on bitternut may grow for three or four years and then die.

The pecan does well on the bitter hickory and the bitter hickory on the pecan, but I have no reason to grow any bitter hickory because I don't like the nut. I think it's a waste of time to fool with it that way.

As far as the hybrid pecans are concerned, the pecan root is certainly the right stock to use on all hybrids. They grow very satisfactorily and bear well.

MR. WHITFORD: I have Gerardi and McAllister hybrids growing on pecan, and the Downing overgrows the pecan.

MR. O'ROURKE: To summarize some of this information that we have gathered this morning on root stocks, it seems that different clones behave differently on the same stock. That is true, we know, with other plants, such as apple. Instead of saying that shagbark is not compatible with pecan, perhaps we should say that the Davis or the Wilcox variety of shagbark is not compatible with a certain type of pecan. It's going to take years of effort to find out the truth of the matter.

MR. WARD: Sometimes you will find that a two-year-old scion, if you can get a dormant bud coming, is better than the matured wood from last year. I'd just like to get an opinion from some of the growers what they use for topworking stocks for grafting.

MR. FERGUSON: I think one thing quite important is to get scion wood that has a good layer of wood around the pith, whether one-year wood or two-year wood. At the base of the year's growth it will have a lot more wood in it. At the tip the wood around the pith is thin.

MR. O'ROURKE: Some years ago Dr. MacDaniels stated that a good scion may be made with the tip of the scion in the one-year wood and the base of the scion in the two-year wood.

Mr. Bernath at Poughkeepsie, New York, has done some bench grafting of hickory. Why other people have not done so, I do not know, and I'd like Mr. Bernath to tell us briefly just why he likes to bench graft hickory.

MR. BERNATH: I like it because I do my work in the wintertime under glass. I have no time in the spring to fuss with outside grafting. So if you gentlemen would like to hear it, I will tell you all about it.

Many years ago when I learned my profession, we had difficulty in finding a method to graft oaks. We finally did find a method that would take and which I have found successful with hickories.

The stocks are dug in the fall and stored heeled in earth. When I am ready to graft I put them on a table, along with the scion wood and start grafting. I use the side graft at the crown leaving a short spur above the graft. Leave them unwaxed and layer them in moss peat in a glass covered frame in the greenhouse with some ventilation. In three or four weeks' time, when the union has formed and just before the leaves come out, take them out and plant them in a cold frame outside. Of course you have to put glass on it to protect them from frost, as well as intense sun. Here you can use part peat and part soil. Leave them there for one year in those frames, with partial shade, until they get fairly high so they shade each other. They can then be set in the nursery row.

MR. O'ROURKE: Mr. Bernath, I know there are some folks here who are nurserymen and who are interested in the cost of production of a finished tree. Do you feel that you can produce a tree to transplant any height you want to select, five, six feet, so on, as cheaply according to this method of bench grafting in the greenhouse as if you bud it or graft it in a nursery row?

MR. BERNATH: That's a question. I have never kept a record of that. It is all right for a young man who is able to get down on his hands and knees and graft, but for me that wouldn't do.

MR. FERGUSON: What temperature do you use in the frames?

MR. BERNATH: About 65. Sun heat naturally will raise it. Care must be used to ventilate the frames in the greenhouse to prevent condensation soaking the grafts.

MR. FERGUSON: Do you carry higher temperatures for walnuts?

MR. BERNATH: All of them about the same. You follow the method just the same as nature. If you follow nature, you will never go wrong. But you have to watch out for fungus in the case, because if you have excessive temperature, the fungus disease will get in your case and ruin the whole thing.

MR. WARD: I presume, Mr. Bernath, when you set out a tree and get a hundred per cent stand it's going to reduce your cost.

MR. BERNATH: Yes, because you have a better take, because you have everything under control, moisture, heat, ventilation, and so on.

MR. BECKERT: Are the hickory stocks potted before you graft, or are you grafting bare roots?

MR. BERNATH: Hickory and oaks are bare rooted. They are too long to pot.

MR. SHESSLER: How many years are lost in this method of bench grafting compared with field grafting trees in the nursery row?

MR. BERNATH: Quite a few. The gentleman is right, if you graft outside where the tree remains, you get a big growth on it.

MR. SHESSLER: In other words, a tree grafted out in the field will have nuts on it three years sooner?

MR. BERNATH: Yes if you leave it where it is. But if you transplant it, look out for a large tree. It is likely to fail.

Bench grafted trees transplant easily. The roots are limited and little of the root system is destroyed.

MR. WILKINSON: I have been propagating for about 39 years, and I have grafted thousands of pecan trees in my nursery, and I have only a few trees growing from grafts. Budding is much more successful with me. Several times I have had up to a 90 per cent stand by budding.

MR. GERARDI: I have tried bench grafting but it sets you back three years in the nursery to get a tree of equal size compared to grafting in the nursery row. If you want a small tree, it's all right. And then again, it's your help situation. If you have got to set them out, they handle the grafts like brush, and I don't like that. Hickory is not hard to graft in the field. I think if you set 10 you get 9 to grow. For scions I go back on two-year wood and oftentimes on three-year wood where there are buds. I don't have trouble at all. With pecans, you have a little more difficulty, because the wood is more pithy inside and doesn't grow so well.

MR. BERNATH: With any tree, I don't care what it is, give me one-year growth, this year's growth, and I am going to have wonderful success. When you take the old wood you have to be sure that you have buds.

PRESIDENT MacDANIELS: This last discussion certainly shows that, there is more than one way to get results. The fact remains that all these different men are producing hickory and other trees by various different means of grafting and budding. They have their own techniques which worked. What there is behind it from a scientific basis we probably don't understand too well at the present time.

I now call on Dr. McKay to present his paper. Dr. McKay.

A Promising New Pecan for the Northern Zone

J. W. MCKAY and H. L. CRANE[2]

In late 1949 Professor A. F. Vierheller, Extension Horticulturist at the University of Maryland, College Park, obtained two small pecans from an exhibit at the Prince Georges County Fair, Upper Marlboro, Maryland, which he sent to the Office of Nut Investigations at Beltsville, Maryland. These nuts were very thin shelled and contained solid, well developed kernels very light in color and attractive. We gave them no particular heed until the fall of 1951, when the authors together with Professor Vierheller, P. E. Clark, County Agent of Prince Georges County, visited the tree on which they had been produced. We found also a number of other pecan trees nearby. All of them were on an old southern Maryland estate known as Brookfield. The present owner is John C. Duvall, whose address is Naylor, a small southern Maryland community located about 25 miles southeast of Washington, D. C. in the heart of the tobacco growing area.

Origin of the Duvall trees: The present trees probably grew from nuts sent to Maryland from the vicinity of Iron Mountain, Missouri, by a friend of the Duvall family named Mrs. Mary Medora Johnson. Mrs. Johnson had lived in Maryland as a neighbor of the Duvall family and when she moved to Missouri she apparently was so impressed with the native pecan that she sent nuts to her friends in Maryland for planting. This must have happened about 1850 since the oldest trees at Brookfield are estimated to be about 100 years old and Mrs. Johnson was a friend of John C. Duvall's grandmother. In terms of the human life span the trees are thus three generations removed from the time of planting, a time period which fits fairly well the estimated age of 100 years based upon size of the trees.

Description: The three largest trees are approximately equal in size and undoubtedly represent the original planting. The eight other trees are all smaller and could well have originated as seedlings of the original three. Five of the largest trees have been given numbers 1 to 5 and will be referred to by number. Duvall No. 1, 2 and 5 are the three large trees situated more or less in a circle surrounding the old mansion, each about 100 yards from the others. The smaller trees are located more or less between and around the larger ones, the old mansion being on a slight knoll in the center of the planting. The original dwelling of Brookfield is now crumbling ruins, part of the building being more than 200 years old, according to Mr. Duvall, who lives in a modern new country home across the road from the original mansion. The three large trees have a diameter at breast height of approximately 4 feet and all of them have a branch spread of more than 150 feet. They are 75 to 100 feet tall. All of the trees have very narrow and pointed leaflets characteristic of Texas and southwestern varieties, and they are remarkably free of insect pests and diseases.

The nuts from this group of seedlings are variable in size and appearance as might be expected of those from any group of pecan seedlings. However, one of the most striking characteristics of all the nuts is that the kernels are solid and well developed. This is an unusual characteristic for pecans grown in the latitude of Washington, D. C. In all of the varieties that are usually grown in this area none which regularly fill their nuts well are known. Another outstanding characteristic of all of the nuts produced by these seedlings is the bright, attractive color of the kernel. In fact, when the nuts of Duvall No. 1 are promptly harvested and dried in the fall, the kernels are almost white. Nuts that stayed on the ground 6 months during the winter of 1951-52 were harvested in late March 1952 and the kernels were still in good condition. Some of the nuts were on display at the Rockport meetings. Small size of nut is without question the chief undesirable characteristic of these trees. Duvall No. 5 produces the largest nuts of all the seedlings but they are so small that more than 100 are required to weigh a pound. Duvall No. 1 produces the smallest nuts and almost 200 are required to weigh a pound.

Past Yields: The one characteristic that sets these trees apart from all other pecan trees that we have observed in the Maryland area is that they yield heavy crops of nuts every year. We have known the trees only since the fall of 1951 but have observed two crops and Mr. Duvall has observed their performance for many years. In the fall of 1951 Duvall No. 2 yielded an estimated 8 to 10 bushels of nuts. Mr. Duvall harvested 3 bushels and he knew that 3 bushels were harvested by friends of the family. An unknown quantity estimated at several bushels was plowed under when wheat was sown shortly before we visited the tree in the fall of 1951. The tree had a heavy set of nuts in August 1952 and Mr. Duvall predicted that it would probably yield as much this year as last. He told us that the three oldest trees always have had annual crops of nuts except for 1 or 2 years when one of the trees failed to produce as much as usual. He could not remember which of the trees produced the light crops but he was certain that light crops were borne at only very infrequent intervals.

Sweeney Tree: The two nuts originally sent us by Professor Vierheller were produced by a tree growing approximately 200 yards from the nearest Duvall tree on a part of the farm recently subdivided and now occupied by a tenant named Sweeney. Mrs. Sweeney placed the plate of nuts on exhibit at the Prince Georges County Fair and from this plate Professor Vierheller procured the sample which he sent. Hence this tree has become known informally as the Sweeney tree. Its nuts are very long and pointed but in other respects resemble very closely those produced by the other trees. The

Sweeney tree is undoubtedly a seedling of one of the three large Duvall trees. This tree also has an impressive yield record, as Mrs. Sweeney said that she has harvested a bushel or more of nuts from the tree every year during the ten or more years that she has lived on the place. In 1952 the Sweeney tree was bearing a heavy crop of nuts.

Soil: The trees growing on soil that is classified as Sassafras fine sandy loam in the heart of the southern Maryland tobacco growing district. This soil type, one of the best agricultural soils of the area, is not generally regarded as one of high fertility. This soil is well drained and aerated and friable to a considerable depth, thus permitting the trees to root deeply. None of the trees are growing under crowded conditions since they are located around the margins of the building sites of the old homestead. The question now is whether grafted trees propagated from the best of the Duvall seedlings will yield heavy crops of well filled nuts that will mature early under other conditions of soil and climate in other localities. We are inclined to believe that some or all of these trees may represent a line of pecan genetically constituted to bear heavy crops of nuts every year under conditions in Maryland. If trees propagated from the Duvall trees will perform elsewhere in the northern zone there will be available for this area a new type of pecan that we feel will be distinctly worthwhile notwithstanding the small size of the nuts. Present varieties of the so-called northern pecan grown in the northern zone perform erratically at best and when many of the varieties produce crops the nuts fail to mature and fill properly.

FOOTNOTES:

[Footnote 2: Horticulturist and Principal Horticulturist, Bureau of Plant Industry, Soils, and Agricultural Engineering, United States Department of Agriculture, Beltsville, Maryland.]

The Hickory in Indiana

W. B. WARD, *Department of Horticulture, Purdue University, Lafayette, Ind.*

Mr. Charles C. Deam, forester, naturalist and botanist, in his book "Trees of Indiana," revised 1952, lists seven distinct types of hickory in the state and nine sub species. As Deam is approaching his 87th year (August 30), he makes this statement: "I thought I knew trees, and hickories especially, but at this time when I can hardly see and write I find there is a great need for reclassification." What is true in Indiana is no doubt true in other areas where *Hicoria* grows—each year new seedlings and hybrids are found that just step out of any previous description and a new tree may result or change the published data.

Some trees develop five leaflets, while others have seven and nine leaflets. The bark may be smooth, rough, scaly, or shag. The nuts will vary in size and form with a thin to quite thick shell. This, of course, applies to the seedlings as the grafted or budded varieties vary only with the location, season, and growing conditions.

The present classification, according to Deam, is as follows:

1. *Carya pecan*—Pecan.
2. *C. cordiformis*—Bitternut.
3. *C. ovata*—Shagbark and 2 sub species—*fraxinifolia* and *nuttali*.
4. *C. laciniosa*—Bigleaf Shagbark (Shellbark).
5. *C. tomentosa (alba)*—Mockernut—one sub species.
6. *C. glabra*—Pignut and sub species—Black Hickory.
7. *C. ovalis*—Small-Fruited Hickory and 5 sub species.

8. *C. pallida* }
9. *C. buckleyi* } —Minor species of lesser importance.

The hickory species thrive in Indiana, doing very well in all sections except in certain portions of the northwestern part of the state and on muck or sandy soils. The tree loves company or does well alone. When the hickory stands alone, the trees are well formed and make a good specimen tree. Many hickory trees are found growing in the river bottom land from Central to Southern Indiana with fewer trees found north of a line extending from Terre Haute through Indianapolis to Richmond. This southern area also contains the largest population of pecans. There are some woods that contain only pecan trees while a mile or so away no pecans are found but all are hickories and occasionally some woods contain both pecan and hickory. The trees in the woods areas, many of which seem to be the same species, produce a wide variety of fruits. When the trees are more closely examined there is a difference in the bark, the branch, the leaf, pubescence, shape of nut and shell structure. As there are all seedling trees in this particular woods, several outstanding trees have been checked and especially as to cracking qualities of the nuts. At harvest time a hammer is part of the equipment and the nuts are cracked at the tree and the tree marked for discard or further consideration.

Future Possibilities of the Hickory

The hickory nut has not reached the popularity of the pecan, although the hickory contains more protein and slightly less fat, carbohydrates, and calories per pound than the pecan. Where the pecan does not fruit, the better hickories, which are hardy, fill the need. The named varieties are good and trees are available from some nurserymen. The propagators have developed a few new crosses but man is far behind nature in this work. The many new seedling trees scattered all over the regions where the hickory grows require only propagation and distribution for wider acclaim.

The development of a new hickory is a long-time process, yet may be hastened by first planting the nuts for new seedlings and when the growth is mature to bud or graft the seedling on large rootstocks. When old trees have been top-worked it is only two or three years' time until the fruit develops and, if worthy of propagation, much time may be saved by this method.

Most of the hickories have either 32 or 64 chromosomes, except pecan which varies from 20 to 24 to possibly 32. The chances of making suitable crosses between the pecan and hickory are most difficult yet it appears that these chance crosses result from time to time as in the hican through natural cross pollination.

How extensive will be the plantings of the hickories is yet to be determined but it is a known fact that many people, especially north of the route of Federal Highway 40, prefer the hickory to the pecan. This may be due to the fact that from childhood the hickory was the local fruit. The fruit and tree hold great promise for the future. If the hickories are to be of commercial importance, the work must be done by all concerned and not left to a few eager individuals to carry on the work alone.

MR. MACHOVINA: Mr. Chairman, members of the Association, I hope you will bear with me if I run 30 seconds over. Perhaps I had better point out that my training is that of an engineer and not a botanist, hence this report on the Merrick tree is that of a layman. I have not bothered to go into detail on the various features of the tree, such as leaves, buds, and so forth, because I have slides which you will see afterwards.

The Merrick Hybrid Walnut

P. E. MACHOVINA, *Columbus, Ohio*

The Merrick hybrid walnut is a natural cross between Persian and black walnut and is distinguished from most other such hybrids by the good crops it usually bears. The tree is located in Rome Township, Athens County, Ohio, on property owned by Mr. M. M. Merrick a farmer and fruit grower.

In August, 1950, Mr. Merrick first described his "English" walnut to the writer and arrangements were made to view the tree. Most striking at first sight was the large crop of nuts. The general outward appearance of the tree suggested it to be pure Persian; however, upon closer examination, mixed parentage became evident. As a hybrid, the tree's history was a matter of interest and the owner was happy to supply what information he could.

Mr. Merrick purchased the property on which the hybrid is located, in 1921. A few years prior to this, the previous owner had planted six Persian walnut trees obtained from a nursery in northern Ohio. These young trees bore their first crop of nuts during Mr. Merrick's first year of ownership. It is known that the nursery owners were also proprietors of a commercial Persian walnut orchard located in the vicinity of Niagara Falls. With this combination of date and orchard location, it seems not illogical to presume that the six nursery trees were of the Pomeroy strain. From Mr. Merrick's description of the nuts produced by these trees, they appear to have been two each of three different grafted varieties. In the early nineteen-thirties, Mr. Merrick planted several nuts from the Persian trees and raised a number of seedlings. One of these seedlings, transplanted to its present location, is the subject of this discussion and is presumed to be a cross between one of the six Persians and a native black walnut. During the late nineteen-thirties, all of the trees, Persians and seedlings, with the single exception of the existing hybrid, were killed by an unusually hard winter.

The Merrick hybrid walnut, now about 20 years of age, is an extremely vigorous and healthy tree. Its height is between 55 and 60 feet and its spread nearly as great. Trunk diameter is at present about 12 inches at breast height. The location of the tree is very favorable, being near the crest of a high ridge and with protection from the northwest by the house. A chicken yard is near and the kitchen drain empties close by to supply moisture.

In nearly all aspects excepting the nut itself, the tree favors its pistillate parent. This is evidenced by the general shape of the tree, by the texture and color of the bark of limbs and twigs, and by the shape and color of the leaves, the buds, the flowers, and the nut hull. Hybridity is indicated by the (usually) eleven leaflets to the leaf stem, by the nut, and in the disintegration of the hull which, after falling, quickly changes into a most disagreeable, dark-brownish, semi-liquidlike mess. The nut itself is much more like a Persian walnut in appearance than a black walnut. The shell surface is slightly rougher and somewhat darker than most Persian nuts. The suture of the Persian parent is prominent. Black walnut parentage is exhibited by the thick shell, the interior configuration and in the flavor of the small kernel. Nut size varies somewhat with diameters ranging from 1 to 1-1/4 inches and lengths ranging from 1-1/4 to 1-1/2 inches.

The bloom, which is strikingly like that of pure Persian trees, is always profuse and precedes that of the surrounding native black walnuts by a week or two. In the two years during which the writer has observed the tree, the greater part of the staminate bloom has preceded the pistillate by several days. This was noticeably the case during the current year, and either this, or the rainy weather, has resulted in a small set of nuts which the owner states to be unusual. During the years observed, the tree appeared to be self-pollinating.

It is recognized, of course, that the Merrick hybrid is worthless as a producer of edible nuts. The possible value of the tree lies in opportunities it offers in being the forbearer of more worthwhile progeny. We know of the vast possibilities in hybridization. We know of the difficulties involved in obtaining nuts from controlled crosses between Persian and black walnut trees; and we know that seedling trees raised from the nuts of such crosses are almost always sterile. The Merrick hybrid, yielding good crops, offers possibilities both in crossbreeding and in the raising of seedling trees from the nuts of the tree itself. In the latter connection, Drs. Crane and McKay, of the U.S.D.A., requested several pounds of Merrick nuts for planting purposes this spring. The writer himself planted five such nuts, of which four germinated. Of the four trees, one died early in the season, while the remaining three have thrived. The heights attained by the three remaining trees thus far this season are 1, 2, and 3 feet, respectively. These trees have the general appearance of young Persian seedlings.

The only crossbreeding attempted thus far ended in failure when a storm destroyed most of the bags prior to application of pollen. Persian pollen was used on the few bloom remaining covered but, unfortunately, no nuts were set. The experiment will be continued. Also, the Merrick will be topworked onto producing walnuts, both Persian and black, in the hope of obtaining nuts from which interesting and perhaps better second generation hybrids can be raised.

An interesting point of conjecture on which to terminate this report, and one to which nut experts will likely give little credence, may be found in a statement made by Mr. Merrick and vouched to by Mrs. Merrick. The statement is to the effect that the nuts borne by the Merrick during its early years, that is, prior to the time the adjacent Persians were killed, were of much better quality, being more like Persian walnuts both in appearance and in flavor. We've heard of "pollen influence" with chestnuts. Did it occur here?

TUESDAY AFTERNOON SESSION

Producing Quality Nuts and Quality Logs

L. E. SAWYER, *Director, Division of Forestry and Reclamation, Indiana Coal Producers Association*

I was trained as a forester and having worked at the profession for nearly thirty years, my first thought of trees is for their utility in building or in cabinet work. In school we were taught that the fruit of forest trees was a by-product. Its economic importance was not emphasized nor was the possibility of establishing stands of some species specifically for the production of their fruit.

Through the years the value of the nut crop from some species has increased so that the fruit is now the primary crop and any wood materials that may be derived are the by-product. This production of valuable food and necessary materials of high quality for the building of quality furniture and interior finish is a combination that will work well together.

Black walnut, the most highly utilized of any of our native timber for furniture, veneer, and cabinet work is becoming increasingly more difficult for the mills to obtain in larger sized logs. Native chestnut, almost completely destroyed in our timbered areas by the chestnut blight, is in demand for interior finish. Pecan, which has had only a limited use in the past, is now enjoying a market for the manufacture of flooring.

The production of nuts from plantations or orchards of these three species will no doubt produce greater economic returns for many years after the initial planting than could be derived from the sale of the trees for the wood they contain. There will come a time in the life of any tree when it is no longer a profitable producer and should be replaced by a younger, more thrifty tree. When that time comes, the tree to be removed will have no economic value unless it contains products that industry can use. With the thought in mind that the wood from the tree is to have some future economic value the trunk of the tree should be kept free of all limbs to a height of about nine feet above the ground. The development of a large spreading top above that point will be desirable for nut production. The space below that top will give ample head room for maintenance work in the orchard and that clear length of trunk will produce a high quality log eight feet long. That is the minimum standard length normally used by the lumber industry. Some shorter lengths are utilized by the veneer industry but those lengths usually command a lower unit price.

The production of figured walnut could be combined with the production of one log per tree but it would take several more years to bring the trees to nut producing age. Mr. Wilkinson has successfully demonstrated that the figure of the Lamb Walnut does carry over through a graft or bud.

A double budding operation should not be difficult to perform. It would simply consist of budding the figured stock on the root at as low a point as possible, then when the figured growth has reached sufficient height, of budding again to the desired variety for nut production. This procedure would no doubt require a few additional years before the first crop of fruit would be harvested but it would produce an extremely valuable log when the tree is finally cut.

I would be remiss in my present job if I did not bring the revegetation program of the Indiana coal stripping industry into the discussion. That industry produces over fifty percent of the coal mined in Indiana today and is recovering coal that could not be mined by any other means.

In driving to Rockport many of you no doubt passed by areas of newly mined land, rough, barren desolate looking areas with no vegetation. They have the appearance of complete desolation and give the impression that those lands are forever lost. In that same vicinity you no doubt passed plantations of pine, or mixture of pine or Locust with our native deciduous species. Those too were mined areas that a few short years ago were just as desolate in appearance as the bare areas you saw. These plantations are the direct result of a reclamation program started by the members of the Indiana Coal Producers Association, a program that has attracted national attention.

The first record of an attempt at the reclamation of coal mine spoil is here in Indiana. In 1918, the Rowland Power Company, now owned by the Maumee Collieries Company, planted peach, apple and pear trees on mined land in Owen county. The records show that for a period of years the trees thrived and were good producers. Then, because the topography was rough and no spraying was done, disease and insects took their toll of the peaches and apples. Seedlings of the original apple and peach tree still grow on the area. The original Kieffer pear trees still stand and produce large crops of fruit.

In 1926, the larger, more far sighted companies began a definite program of reforestation of their mined lands under the direction of Ralph Wilcox, at that time assistant State Forester and fortunately our State Forester today. That voluntary program was carried on until 1941 when the Indiana Coal

Producers Association, the Association of the mining companies, sat down with representatives of the Indiana Department of Conservation, representing the state, and the Indiana Farm Bureau, representing the people, and drafted a bill which was enacted into law. This law required each company to obtain a permit from the state to operate and required that each company revegetate an area each year equal to 101% of the area they had mined. To insure compliance, a bond was required. This law remained in effect for ten years. In 1951, representatives of those same groups again sat down together and drafted several amendments to the original act. Some grading is now required where areas lie adjacent to public roads. Access roads must be provided and areas to be devoted to pasture must be graded so that they can be traversed with agricultural machinery.

Under this program, sponsored by Industry, the Farm Bureau, and the Department of Conservation, 79% of the area that has been mined to date has been successfully revegetated. The remaining 21% is a natural lag and represents lands newly mined or areas that have not weathered to the point where they will support revegetation. The demand for recreation lands and home sites where water is available is constantly increasing. At least 13% of the revegetated area is now being used for public recreation or for home sites. Near the more heavily populated sections the price commanded by mined territory containing good lakes often exceeds the value of the land before it was mined.

These lakes, formed in the final cuts and in low lying areas of the strip mines, furnish the only clean, clear water available for public recreation and fishing in the south western part of the state.

The reforestation being carried on under the reclamation program consists of planting several species of pines, as well as a large variety of our native deciduous trees. The older plantations are being used as a guide as the research started in the last eight years has not progressed far enough to give conclusive results on many points. Until the last few years the Agricultural Experiment Station has devoted little or no time to the problem of reclaiming strip mine spoil. The area of the state that is involved, less than 1/4 of 1%, has been too small to justify the use of their limited funds. However, since funds have been made available to that Station, through the Industry, to establish research fellowships, the Station has given whole hearted cooperation. The information being obtained through these fellowships and through work being carried on cooperatively with the Central States Forest Experiment Station is going to answer many of the questions on reclamation we have been confronted with.

Included in our reforestation has been a liberal scattering of black walnut. A breakdown of species is not available on much of the earlier work but since 1940, when accurate records have been maintained, we have planted 239,000 black walnut seedlings or seed. Initial survival is not high, averaging only about 50 percent but we still have a general distribution of seed trees that are providing a source of seed for natural reproduction. Trees from plantings made in 1927 to 1934 have grown well and we now have walnut trees over 10 inches in diameter and 60 feet in height. The average for all areas would probably not exceed 5 inches but individual trees have made remarkable growth. These trees are only seedlings, but they are bearing heavily and their fruit is sought by the local people.

In 1946 and 1947, budded stock of walnuts and pecans and seedlings of Chinese chestnut were obtained from Mr. Wilkinson and were set out on six selected areas. A wide variety of sites were picked and a wide variation in both survival and growth has been obtained. No special treatment was given the areas where the trees were to be planted nor were the trees mulched or watered after planting. Even under these rugged conditions we have a survival of over 60 percent of all trees. The walnut trees now range from 5 to 12 feet in height and the pecans up to 6 feet. The chestnuts vary in form from low spreading plants 4-1/2 to 5 feet in height and as much as 8 feet across to well formed trees 8 to 10 feet tall. Pruning on all three species to produce a clear butt log has been started.

Pasture seeding on areas high enough in available lime to support legumes is following a pattern laid down by three years of graduate study, financed by the Indiana Coal Producers Association, at Purdue and by work done by the Illinois Agricultural Experiment Station under a similar arrangement with the Illinois Coal Strippers Association.

Unfortunately, we have only a small portion of the spoil area in Indiana that is suitable for the development of improved pasture. Not over 10 percent of the area mined to date is good enough and that percentage will decrease. Modern operations are deeper than the early ones and are exposing more hard rock and shale. Fortunately, most of these areas can be reforested after three or four years. In exceptional cases less than 5 percent of the area mined the exposed materials contain large amounts of sulfides. These break down into acid that in some cases require ten to twelve years to leach out before revegetation can be undertaken.

The fact that these stands of trees established on raw spoil will produce merchantable timber has been proven. In 1951, an area was clear cut at the Enos mine in Pike county. The pines on this tract were planted in 1933-34. The products from that cutting, peeled posts and poles, were sold to the Indiana Wood Preserving Company at the rate of $335.59 per acre. An increase in value of $16.48 per acre per year.

Pasture, forests and fishing are not the only products. Game of all varieties is abundant in the worked out areas. One of the largest herds of white tailed deer in the state, now referred to as the strip mine herd, is located in northern Warrick and southern Pike counties. In the Indiana deer season of 1951, the first open season since 1893, the second largest recorded kill came from the strip mine herd. The Pitman-Robertson report of the Division of Fish and Game carries the following comment on deer from that area. "The superiority of the diversified range of the strip mine herd was reflected in above average weights and measurements in most age classes."

From the evidence at hand, there is every reason to believe that most of the mined area will again be highly productive forest land. It has completed the entire cycle of land use. Originally it supported magnificent stands of hardwood timber. This timber was cut and the lands devoted to farming. Poor management and erosion soon depleted the supply of top soil and many areas were abandoned to broom sedge, blackberries and gullies. Because it was close enough to the surface the coal has been removed and the areas replanted to many of the same species of trees.

With this reestablishment of the forest cover and the creation of the lakes in the final cuts, we can again have our forest resource combined with fishing, hunting and other forms of outdoor recreation, some areas of pasture and, I believe, others that can be profitably devoted to the production of nut crops and the by-product of quality logs for the veneer and lumber industry.

PRESIDENT MacDANIELS: If you ever think you are going to sell your logs for veneer or lumber, don't nail hammocks or other things on the trees. The metal is very soon buried and causes no end of difficulty. We will go to the next paper, which is, "Colchicine as a Tool in Nut Breeding," Mr. O. J. Eigsti, Funk Brothers Seed Co., Bloomington, Illinois.

MR. EIGSTI: Three years ago this project was conceived in a discussion between Mr. Best and myself. Then during the two-year period, all I did was turn over some Colchicine to Mr. Best. Mr. Best took the material, treated the trees and performed as well as any graduate student I had ever graduated in the 13 years that I was in university work. It is through his fine cooperation that we are able to start this project, and I look forward to this developing into a rather important nut breeding venture. But as you all know, it will take a long time. I have this paper written. It's only four pages double-spaced.

Colchicine for Nut Improvement Programs

O. J. EIGSTI and R. B. BEST, *Normal, Illinois, and Eldred, Illinois*

Colchicine (1, 2) as a plant breeders' tool is universally well known. Only limited use has been made of this technique for nut improvement. Early work was started by Dr. J. W. McKay, a member of the N.N.G.A., but numerous other problems demanded his attention and the Colchicine project was not carried to final completion. Other reports are at hand from Sweden and Japan but these results do not shed direct light on the problems under discussion today at Rockport, Indiana.

Colchicine, acting on cell-division, ultimately causes a doubling of the number of chromosomes within those cells in contact with the substance at the time of division. Such changes are transferred to succeeding generations by the hereditary chain familiar to plant breeders. Several species of nuts are among this class of plants with doubled chromosomal numbers, however, such duplications occurred in nature. A report on this phase was given at a recent meeting of the N.N.G.A. Therefore such excellent nut producing species as the pecan are naturally doubled types, called polyploids. We find numbers such as 32 representative of a polyploid situation.

Since colchicine is effective in doubling the chromosome number and that variations in chromosome number exist among species, the authors planned a series of experiments to determine the best methods of applying colchicine toward a nut improvement program. Seedlings of pecan were available and out of this experience a schedule is submitted that may be of use for other members of this association confronted with particular problems applicable to colchicine techniques.

The most satisfactory schedule for doubling the number of chromosomes is given in a number of steps as listed below.

1) Select expanding vegetative buds in the earliest stages of development.

2) Use seedlings or branches from mature trees.

3) Prune leaves and probe to the growing cone without damage to tissue.

4) Pack a small wad of cotton into the terminal point.

5) Soak this cotton by dropping .2% aqueous solution of colchicine on same.

6) Add glycerine to cotton to improve penetration of colchicine.

7) Place drop of colchicine on cotton morning and evening for four days.

8) Remove cotton wading from bud on 5th day.

9) If sufficient tests at hand, allow cotton to remain on some buds.

10) Try for at least one hundred buds treated.

11) Observe growth during first season and also next season.

12) If treated bud dies, watch for growth among lower laterals.

13) Evidence of changes appears in the new leaves, darker, thicker, greener.

14) Conclusive evidence of doubling rests with microscopic and anatomical analysis which is a task for trained technicians only.

The above procedures are suggestions for a start and everyone will wish to make changes suited to his particular needs. The concentration of colchicine need not be exact as in an analytical experiment in chemistry. One gram dissolved in 500 ml. water is an adequate and a sufficiently careful measurement. The local pharmacist or physician is well acquainted with colchicine in the practise of medicine since this drug is a standard for gout.

Effective use may be made from two specific areas of plant breeding. First, doubling of chromosomes changes sterile hybrids into fertile individuals. This is a promising field and whenever such hybrids are discovered, efforts should be made to apply the colchicine technique. Second, doubling of the chromosome number makes possible hybridization of individuals heretofore unsuccessful in such effort. In both instances germ plasm of wide genetic difference is incorporated into a new propagating breeding stock. In the case of the sterile hybrid transformed into fertile individuals, no counting of chromosomes is necessary because restoration of fertility is evidence of changes in the chromosomal makeup. However, the second type of experiment requires microscopic analysis.

There are a number of fundamental research problems in the plant sciences associated with the treatment of plants with colchicine. From horticultural subjects such as the apple,(3) pear, cranberries,(4) and grapes, it is obvious that periclinal chimeras will be of prime importance in analysis of results in treatment of nut trees. Following the treatment of a growing point with colchicine the outer layer of cells may be doubled by colchicine but the lower layers may remain unchanged. Or a reverse of this situation may obtain, and even other types. Since the formation of pollen takes place from a certain layer it is very important that such specific layers are changed. The course of plant breeding can be altered by these kinds of changes. To our knowledge, no investigations of periclinal chimeras have been made with nuts, following treatment with colchicine.

Specific experiments were conducted at Eldred, Illinois in the spring of 1951 with seedlings of pecan. The cooperation of the R.B. Best Farms and Nut Plantation made this project possible. Several types of treatment were tried. Out of this experience the above schedule listed in 14 steps was developed. Other details may be obtained by contacting the authors direct. Observations of the new growth in 1951 and 1952 were made and the shape of leaves, color, texture and general appearance suggest that doubling of chromosomes has been induced. Up until the present time, no microscopic analysis has been made but this is a contemplated step and facilities are at hand to complete this work.

While this paper is not a completed research, the authors hope that the presentation of technique will aid and stimulate interest in this new approach to nut improvement. In such instances where certain members may have a particular problem such as a true hybrid-sterile as a result of hybridity, it is hoped that the suggestions given in the above pages may lead into a new field of improvement. There are rewards in store for the plant breeder willing to master this new technique, but the mastery requires careful study and diligent work.

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PRESIDENT MacDANIELS: The Resolutions Committee for this meeting is: John Davidson, chairman, and Dr. Rohrbacher working with him. If you have anything in mind that should be brought up in the resolutions, see one of these two men.

The next paper is:

An Early Pecan and Some Other West Tennessee Nuts

AUBREY RICHARDS, M.D., *Whiteville, Tenn.*

MR. RICHARDS: There came under my observation in the latter part of last summer a seedling pecan tree growing in the city limits of my home town. It seemed that this tree had been growing unnoticed for possibly 50 years, judging by the size of the tree. The outstanding thing about this tree and what called it to my attention was a patient who came into my office complaining with a backache from picking up pecans on the 20th day of August.

I wrote my friend, Mr. J. C. McDaniel, about this pecan, and when he visited me during the Christmas holidays I gave him a sample. The only thing that he could say bad about the pecan was that it was slightly on the small side. I know personally that at least three or possibly four bushels of good quality nuts were harvested from that tree, most of them on the ground by the 20th of August.

In my section the Stuart pecan, which we use more or less as a yard-stick, was ripe the latter part of October, and we thought that possibly this tree, since it had undergone an unusually low temperature the winter before of 20 below zero, might have possibilities.

But let's dispense with this pecan and say that we believe in the old adage that one raindrop doesn't make a shower. It has a fair crop this year, and they are just as green as my Stuarts now.

There is another tree that originated in West Tennessee which Mr. McDaniel chose to call this nut "Rhodes heartnut." This tree is 7 years old from a dormant bud on a 2-year-old black walnut seedling growing on my back yard. It bore two clusters its second growing season, and since that time it has borne annually, the crops increasing in proportion to the size of the tree. This year's crop consisted of 88 clusters of nuts, with an average nut count of 10.2 nuts per cluster, giving a total of almost 900 nuts on this 7-year-old tree.

There is one more figure I'd like to give you. The count of clusters compared to the number of terminals we had this spring is better than 90 per cent clusters. I have a few bud sticks here cut from green water sprouts. That's the only kind I can find a sprout on. I brought them up to Mr. McDaniel. If anybody can talk Mr. McDaniel out of a bud he wanted to try, but I don't really know what plans he had for these bud sticks. The 7 or 8 other varieties of heartnuts I have growing don't have any that have clusters like the Rhodes.

Scab Disease in Eastern Kentucky on the Busseron Pecan

W. D. ARMSTRONG, *University of Kentucky, Princeton, Kentucky*

MR. ARMSTRONG: Mr. Chairman, ladies and gentlemen: It is nice to be here at the Northern Nut Growers meeting. This is my second session. I attend all the pecan and nut sessions in the country. I have attended Georgia-Florida Pecan Growers Association and Oklahoma and Texas Pecan Growers Association.

These plates that I have contain some of the Busseron pecans affected with pecan scab. The disease has shown up in Southeastern Kentucky, about a hundred miles southeast of Lexington, a hundred miles west of the Virginia line, and about a hundred miles north of the Tennessee line, on a straight line west of Roanoke, Virginia.

These trees were planted in bottom soil, rather well drained, and they made a rapid growth. In the original planting there were two Green River pecans, one Major, one Busseron and two walnuts, a Stabler and a Thomas.

About 1946 we noticed that all of the pecans on the Busseron were like these that we have here—did not mature, completely covered with scab fungus and dropped off the tree. The shells were so thin that you could just crush the whole pecan, hull, shell and all with no meats in them. The Major tree right beside it and the two Green River trees had none of this trouble, and they have none of it as yet. And each year now that this Busseron tree has borne there, practically all of the nuts have been like this.

At the time we located this disease first in 1946, I sent samples to the U.S.D.A. at Washington and also to the Southeastern Pecan Laboratory at Albany, Georgia, and Dr. Cole, there identified it as pecan scab.

I reported the presence of the disease to Mr. Wilkinson and to Dr. Colby and they were surprised to see the disease on Busseron in any location, and particularly that far north.

In the south this disease frequently affects Schley, Delmas, Alley and Van Deman and some others. Formerly the trees were sprayed with Bordeaux Mixture. I think they are using Zerlate now. It's a problem to be reckoned with. It occurs on the nuts and on the leaves, and it is carried over winter on the stems and the one-year shoots.

Further News About Oak Wilt

E. A. CURL, *Illinois Natural History Survey, Urbana, Ill.*

In 1951 a review of the oak wilt situation was given in a paper, "Present Status of the Oak Wilt Disease", at the Forty-Second Annual Meeting of the N.N.G.A. at the University of Illinois. The following report is aimed at bringing up to date the present known distribution of the oak wilt disease, recent developments in scientific research on the disease, and possible control measures.

The oak wilt disease is caused by the fungus *Chalara quercina* Henry and is characterized by a very noticeable bronzing and wilting of leaves that drop prematurely. Brown streaks are usually present in the outer sapwood. These symptoms may be seen from June to September or until normal autumn colors of the foliage develop.

More than 30 species of oak are known to be susceptible to the disease. Other susceptible genera of the family Fagaceae are Chinese chestnut, *Castanea mollissima*, golden chinquapin, *Castanopsis chrysophylla*, tanbark oak, *Lithocarpus densifiora*, and *Nothofagus* from South America. The red and black oaks seem to be most susceptible and are often killed within 6 weeks after infection.

Distribution

During the past few years the oak wilt disease has spread with such rapidity and destructiveness among valuable forest and shade oaks in parts of the eastern half of the United States that its seriousness is now well recognized. At present oak wilt is known to be in the following states: Wisconsin, Iowa, Minnesota, Illinois, Missouri, Indiana, northern Arkansas, eastern Kansas, southeastern Nebraska, Ohio, Pennsylvania, West Virginia, northwestern Virginia, western part of North Carolina, eastern Tennessee, northeastern Kentucky, western Maryland and southern Michigan. Aerial surveys for 1952 are not yet complete, but there are indications of extensive new infections in Pennsylvania, Ohio, and West Virginia while the other states show a moderate increase in the number of infections.

The first case of oak wilt in Illinois was seen in Rockford in 1942. Today 54 of the 102 counties in the state have oak wilt areas. The disease is present in both the extreme northern part and the southern-most tip of the state. Practically all wilt areas in the southern half of Illinois consist of 5 trees or less that appear to have died within the last 4 years, indicating a recent spread of the disease southward. A similar condition exists in southern Missouri and northern Arkansas.

Developments in Research

In 1942 a report from the Wisconsin Agricultural Experiment Station revealed that the oak wilt disease was caused by a fungus, and research programs were started early in Wisconsin and Iowa. Neighboring states were quick to follow as surveys showed a wider distribution of the disease. Now almost every state in which oak wilt occurs is taking part in efforts to learn more about the disease and its causal agent so that practical control measures may be applied before the spread of the disease gets out of hand. The National Oak Wilt Research Committee at Memphis, Tennessee, supports in part an intensive oak wilt research program in coordination with several midwestern universities and with the U.S.D.A., Bureau of Forest Pathology.

Until recently the causal fungus of oak wilt was known only in its asexual or imperfect form living in the sap stream of infected trees. The most important question to be answered now is how the fungus spreads over long distances from diseased to healthy trees. Before this could be accomplished, however, we had to know how the fungus escapes from the inside to the outside of diseased trees where it can be exposed to agents of dissemination.

In the late summer of 1951 clearly visible mycelial mats of the oak wilt fungus were found in Illinois under the loose bark of wilt-killed trees. These mats were usually located beneath cracks in the bark; thus, they were exposed to the outside air and to visiting insects. Most wilt-killed trees contain beneath the bark numerous insect larvae of wood and bark boring beetles. Larvae were frequently found in direct contact with mycelial mats of the fungus. Larvae of the two-lined chestnut borer, *Agrilus bilineatus*, were most abundant, but larvae of species of the families Scolytidae and Cerambycidae were also present in large numbers.

In addition to the mycelial mat under the bark there was often present a thick dark pad usually in the center of the mat. It is not known yet what part this pad plays in the life history of the fungus but we do know that it is produced by the same fungus which causes oak wilt.

We also found in Illinois that the oak wilt fungus often develops into visible mats from chips of bark and wood that have been chopped from wilt-killed trees and allowed to lie on the moist forest floor. This should be remembered when considering sanitation as a partial means of controlling the disease.

In 1951 the sexual or perfect form of the oak wilt fungus was produced on laboratory media in Missouri by crossing different strains of the fungus. The sexual form is recognized by the appearance of microscopic, black, short-beaked fruiting structures or perithecia that are filled with sticky ascospores. This sexual form is a species of *Endoconidiophora*.

The sexual form of the fungus was first found in nature in Illinois in the autumn of 1951. The perithecia are produced on the mycelial mats beneath the loose and sometimes cracked bark of diseased oaks. Both the ascospores of the sexual form and the endospores or conidia of the asexual form will cause wilt if the spores are injected into oak trees.

From the foregoing information it is apparent that several methods by which the disease might be spread over long distances are possible. First, and what seems to be most probable, is transmission by insects. Adult beetles, such as the two-lined chestnut borer, which emerge from dead trees in the spring and feed on the leaves of healthy trees might transmit the spores of the fungus. Other insects might feed on the fungus mats that are exposed through cracks in the bark and carry both the sticky ascospores and conidia to other trees. Additional agents that must be considered are woodpeckers, squirrels and air currents.

Besides searching for the vector or vectors that spread the disease other important studies are in progress. Among these is the consideration of chemotherapy as a possible means of controlling oak wilt. For our purpose, plant chemotherapy may be defined as the control of disease by chemicals which are introduced into the plant. According to Dr. Paul Hoffman of the Illinois Natural History Survey, a number of chemicals have shown promise in curing small diseased oak trees when treated in a very early stage of the disease. In one instance, trees that were inoculated with the oak wilt fungus then treated with chemicals 2 years ago are still alive. The most promising results were obtained by injecting the chemicals into the soil where they are taken up by the roots and by applying chemicals directly to the foliage in a spray. Trunk injection showed least promise because of the limited distribution of the chemicals through the tree.

The use of chemicals for curing wilt-infected trees is still in the early experimental stage and is not yet recommended as a practical control measure.

In 1949 Wisconsin workers demonstrated the local spread of oak wilt through natural root grafts. They found that the poisoning of a single healthy tree with sodium arsenite often killed as many as 15 other trees nearby, indicating that their roots were connected.

Recently the results of experiments in Wisconsin explained in part what causes the leaves of diseased trees to wilt. When a tree becomes infected it is stimulated to produce tyloses or swellings in the vessels of the wood. Therefore, the flow of water from the roots to the tree top is restricted and the leaves wilt and die. It is also known that the fungus itself produces a toxin which might be responsible for the actual killing effect on the tree.

In Illinois experiments are being conducted with insects in relation to the spread of oak wilt. Insects of various species are collected from wilt-killed trees and allowed to run over or feed on laboratory cultures of the oak wilt fungus. The insects are then caged on parts of healthy trees to feed on the leaves. A single red oak treated in this way contracted the disease and died. This shows that the disease can be transmitted by an insect.

Controlling the Disease

The spread of oak wilt in local areas may be stopped by preventing the underground movement of the disease from tree to tree through natural root grafts. This can be done by (1) poisoning all healthy trees within 50 feet of diseased trees, (2) cutting a ditch 30 inches deep with a small trenching machine between diseased and healthy trees to sever root connections or (3) severing root connections with a tractor drawn plow on which a knife blade is attached. Unfortunately the use of such heavy equipment is not practical in rocky and hilly areas. Chemicals used for killing trees are sodium arsenite and ammate. Ammate is safe to use but does not kill trees as rapidly as the other poison. In some localities 2,4,5-T used as a trunk spray has given satisfactory results in killing small trees.

If infected trees are left standing mycelial mats with their numerous spores develop under the loosening bark. It is therefore advisable to cut and burn all parts of diseased trees as soon as possible after symptoms appear.

A combination trenching and eradication program was started in the summer of 1950 in the Forest Preserve District of Cook County in Illinois. According to Mr. Noel B. Wysong, Chief Forester, 2 newly wilted trees were found in the Forest Preserve in 1948, 72 trees in 1949, 141 trees in 1950, and 96 trees in 1951. The count for 1952 is not complete but a continued decrease in the number of new infections would indicate good control.

There is no information on resistant species of oak. In very rare cases, however, trees have been observed to recover after showing symptoms in the early spring.

Future Outlook

Among the many things that we need to know yet about the oak wilt disease and its causal fungus one is outstanding. How does the disease jump from one infection center to healthy trees 200 yards, 2 miles or even 100 miles away? Although spread through root grafts may be controlled by severing root connections, the value of such a control measure is limited as long as the agent or agents responsible for long distance spread remain unknown. The discovery of other methods of spread might result in the development of control measures that are cheaper and less drastic than those known at present.

A great deal remains to be done and research is increasing in the various states concerned. There is reason to believe that oak wilt can be checked before it reaches devastating proportions comparable to chestnut blight which wiped out our American chestnuts.

MR. SLATE: What is the origin of the fungus? Is it a native fungus, or imported?

MR. CURL: Yes, it is a native fungus, as far as we know.

MR. SLATE: Any evidence that the fungus is mutating to make more virulent strains?

MR. CURL: That's something that hasn't been found yet. There are several strains of the fungus, what we call strains, because they will form the sexual stage, and a strain alone will not. There is not too much known about that yet, the strain business.

MR. GRAVATT: Just a word. We had a conference in Beltsville all day Sunday about the recent developments on the oak wilt. There has been very extensive spread in Pennsylvania, Ohio, West Virginia and Maryland this year. We are very much alarmed about the situation. The Chinese chestnut is very severely affected. We have learned that in Missouri. One year there were three Chinese chestnuts killed by the fungus, the next year 60. The oak wilt is a serious threat to the chestnut orchards.

Life History and Control of the Pecan Spittle Bug

STEWART CHANDLER, *Associate Entomologist, Ill. Nat. History Survey,*
Urbana, Ill., Consulting Entomologist, Southern Illinois University

Since it was a year ago that this subject of spittle bug was first brought to the attention of the Northern Nut Growers Association, it might be well to review briefly the high lights of that report. I told you at the annual meeting at Urbana, something of the life history. There are two broods, one appearing in June and one in July. The adult is a small sucking bug about an eighth to a quarter inch long. The species at that time was uncertain but now has been determined by specialists in that group as *Cercoptera achatina*

Germ. This insect, I reported, is not the same as the one occurring on meadow and other field crops, not only the species but the genus being different. The distribution was found to be in every area where pecans are grown. As to its importance I pointed out that in Illinois it had become very serious in the past three or four years, apparently causing a marked reduction in crop. Control measures were directed against the nymphal stage, which is protected by the spittle which the insect emits continuously while feeding. Three insecticides were tested at Anna, Illinois, Lindane, parathion, and tetra ethyl pyro phosphate, known as TEPP. Lindane proved to be approximately 95% efficient, parathion roughly 60% and TEPP about 10%.

In 1952 the work was resumed in the orchard of Conrad Casper near Anna, Illinois and was begun at the Richard Best place at Eldred, 175 miles northwest.

In 1952 five phases of the work with pecan spittle bug were undertaken as follows:

1. A study of the importance of the pecan spittle bug.
2. The hibernation of the insect.
3. Life history and occurrence of the various stages and broods of the insect in relation to nut development of the pecan.
4. Control measures.
5. Varietal susceptibility to the insect.

1. Importance of the insect

Hibernation Studies

To learn to what extent if any the insect reduces the crop of pecans, terminal shoots from trees sprayed the previous season with three different materials were compared with the unsprayed check. These are shown in Table 1.

Table 1. Pecan spittle bug effect of 1951 sprays on terminal shoots in spring of 1952

Treatment	Dead shoots per hundred
Check	87
TEPP	62
Parathion	17
Lindane	4

Since these terminals shoots later develop most of the nuts it would appear that the pecan spittle bug is responsible for much of the loss of crop under these heavy infestations.

It was planned to follow this up with later examination of nuts, and this was done with the assistance of Mr. J. C. McDaniel, but unfortunately it was found that this was the off year and the crop was very small, so we could not definitely settle that point. This will be a job for the future.

2. Hibernation studies.

In August of 1951, I introduced adult bugs into a cage placed over a branch of an unsprayed pecan tree for the purpose of determining whether there was possibly a third brood. Finding none the branch was removed and examined to study the hibernating eggs and the egg slits in which they were layed. The slits were not over a quarter inch long and frequently in pairs.

Eggs were deep enough that they were rarely seen without opening the slits. Many slits were found containing egg shells, presumably from the previous brood, but possibly from a season earlier as the slits are corked over.

Following this study branches were cut from the sprayed and unsprayed blocks and gone over very carefully to find the numbers and location of the egg splits and the numbers containing live eggs and egg shells. Each split would contain as many as 5 or 6 eggs. Table 2 show their numbers and locations, and Table 3 the effect of sprays on numbers of live eggs.

Table 2. Pecan Spittle Bug Location of egg slits in branches

	Diameter of branches, inches				
	1/8 to 1/4	3/4	3/8	1/2	1/2 to 1 inch
Live eggs	2	9	3	1	0
Egg shells	5	42	94	23	0

Table 3. Pecan Spittle Bug Effect of 1951 sprays on number of eggs Examinations made March 4, 1952

Treatment	Inches wood examined	Number of Slits with live eggs	egg shells
Check	508	10	63

TEPP 795 5 25

Lindane 478 0 13

3. Life history and correlation of stages of insect and nut development.

It was soon found that the pecan spittle bug was putting in its appearance earlier according to the calendar than in 1951 so an effort was made during the season to correlate insect life history and nut development during the season. Table 4 give some of the principal points in both.

Table 4. Pecan Spittle Bug and Nut Development Anna, Illinois, 1952

	Insect	Date	Tree

Egg stage Apr. 24 Catkins 1/2 to 3/4 inch First nymphs May 5 Catkins 1 to 1-1/2 inch Many nymphs and spittle May 12 Catkins 2 to 3 inches Fruit buds Peak hatch May 20 Female flowers Spittle drying June 2 Nuts developing 1st. 2nd brood June 27 Hatch mostly over July 7 Spittle drying July 26

Another phase of life history which is of practical importance is the increase of second brood over first. Records were made both at Anna and at Eldred in unsprayed blocks at approximately the peaks of occurrence of nymphs and spittle, and are tabulated in Table 5.

Table 5. Pecan Spittle Bug Infestation, first and second broods, 1952 Number of spittle masses per 100 terminals

	First brood, June	Second brood, July
Anna	41	62
Eldred	23	50

This table shows an increase of approximately 50% at Anna and 100% at Eldred. It is thought that a 3 inch flash flood which occurred at Anna might have reduced the first brood infestation somewhat after the counts were made and been responsible for no greater increase and possibly that the heat and drought in both places might have resulted in a reduction. Be that as it may the total infestation was not as severe in 1952 as in 1951.

4. Control.

First Brood Sprays

It was originally planned to spray in both places but at Anna the owner sprayed all but the 1951 check block with parathion early and the infestation was reduced to the point where later hatch did not build up to a sufficient point that good results could be observed so no spraying was done at Anna till the second brood. At Eldred two materials only were available, Lindane and Dieldrin.

At Eldred we had two difficulties in spraying. One was the type of machine with which I was not familiar and the other the inaccessibility of some of the trees. The machine is probably more fitted for field crop work than for large trees. It is called a Mechanical Aresol Generator, manufactured by the Hessian Microsol Corporation of Darien, Conn. The engine is a Wisconsin Air cooled motor made in Milwaukee, Wisconsin. The machine was mounted on a platform and transported in the orchard on a truck. Two fifty gallon barrels constitute the tank. Due to the nature of the machine and to lack of agitation only liquid materials can be used in it. It uses a much smaller amount of material than I had been accustomed to, and my first job was to learn to what extent the materials must be concentrated to compensate for the small output and how to get a comparison with the amounts used in regular orchard sprayer. In concentrate tests on fruit trees we arrive at this by judging the number of gallons which a tree would normally receive with a standard sprayer. There was little background to go on with nut trees and the problem was further complicated by the arrangement of trees which were not planted but grafted in their original positions in the woods. A clump of trees which could not be approached individually might have to receive not much more material than one tree which could be hit from both sides. Sizes of trees also varied. It was decided to use only 25 gallon lots of material and even this small amount sprayed from 55 to 65 trees of varying sizes. It was soon seen that the tops of the moderate and large sized trees were not covered very well. For the first brood sprays at Eldred about six times as much material per 100 gallons was used as had been successful at Anna the previous season. The results are shown in Table 6.

Table 6. Spittle Bug Control, Eldred, 1952 First brood, sprayed May 23, examined June 9

Treatment	Amount in 100 gallons	Spittle masses 800 terminals
Dieldrin	1 gal. of 18-1/2%	18
Lindane	1 gal. of 20%	27

Check ——— 189

It will be seen that the reduction over the unsprayed blocks was about 90% with Dieldrin and 85% with Lindane.

For second brood sprays at Eldred materials were increased to about 8 times normal in hopes of getting better results. In this test 10 trees were selected in each block that could be reached moderately well and sprayed separately before the entire block was sprayed. Records were made the day before spraying, 3 days after spraying, and 10 days after spraying. Four materials were available, making five blocks with an unsprayed check. The results of these sprayings are given in Table 7.

Table 7. Spittle Bug Control, Eldred, 1952 Second brood, sprayed July 18

Treatment	Amounts in 100 gallons	In 200 terminals July 17	July 21	July 28
Lindane	6 qts. of 20%	123	24	2
BHC	10 qts. of 11.7%	98	11	0
Dieldrin	6 qts. of 18-1/2%	130	19	9
Toxaphene	8 qts. of 58%	107	16	3
Check	———	99	98	47

Due to the natural reduction in the check by July 28 most attention probably should be given to the July 21 examination. This table shows approximately 92% reduction from Lindane, 87% with BHC, 85% from Dieldrin, and 85% from Toxaphene on July 21.

At Anna trees are all very big, from 50 to 75 feet high. They are planted in rows. A regular orchard sprayer was used with 600 pounds pressure using one gun and sprayed from the top of the rig. Approximately 25 gallons was used per tree. As will be noted the dosage was much smaller than at Eldred, and for ordinary use these are probably the proper dosages. Table 8 gives the results of these tests.

Table 8. Pecan Spittle Bug Control, Anna, 1952

Treatment	Amounts in 100 gallons	In 200 terminals July 10	July 14	July 22
Lindane	1 lb. of 25%	214	1	1
BHC	2-1/2 lbs. of 10%	244	5	9
Dieldrin	1 and 1/3 pints of 18-1/2%	148	3	5
Toxaphene	1 qt. of 31%	146	22	21
Check		61	47	20

The reduction in the check block July 14 may be due to proximity to the sprayed block which was not true in Eldred. This check was small. Table 8 shows on July 14 an approximate reduction of Lindane 99%, BHC 98%, Dieldrin 98%, and Toxaphene 85%.

From these tests in both places it appears that we have a choice of three very good materials, Lindane, Benzene hexachloride called BHC and Dieldrin, and for that reason we can ignore the less efficient material, toxaphene.

At Eldred, since first brood sprays were applied in a sizeable area records of infestation were made shortly before time to spray for the second brood to determine whether the first brood spraying would eliminate the need for second brood spraying. However, the infestation was found to be practically as great in this area as the unsprayed part of the woods. It appears that the control was not good enough to allow this. In part this was due to failure to reach the tops of the trees. Records were made in the lower parts.

5. Varietal susceptibility.

At Anna where there was a limited number of trees, the orchards were plotted on paper and location of each tree with variety indicated records were made of each tree separately, in hopes that some varietal susceptibility would be shown. There is nothing very clear in this respect except that of the varieties in the Casper orchard, Butterick, Busseron, Indiana, Posey, Stewart, Osburn, Major, Green River, the Indiana and Posey may be a little more heavily infested than the others. At Eldred for the second brood infestation, the variety of each of the 10 record trees was reported, but there were so many varieties and they did not occur often enough in the five plots to make variety infestation data reliable. However, the rather high average on the Indiana variety did seem to corroborate the findings at Anna.

There was some foliage burn in two of the record trees in the Dieldrin plot at Eldred, both being the variety Rockville. Another tree in another part of the plot was also found to be burned and also found to be the same variety, so it appears that this may be particularly susceptible to spraying especially in this concentrated form such as we used. There were no Rockville trees in any of the other plots, so we have no way of knowing whether the Lindane, BHC or Toxaphene would have done the same or not.

PRESIDENT MacDANIELS: The next paper, the last paper of the afternoon, is Control of Insects Injuring Nut Trees, by Howard Baker, U.S.D.A. Bureau of Entomology and Plant Quarantine, Beltsville, Md.

MR. BAKER: Mr. Chairman, members of the Northern Nut Growers Association: It is a great deal of pleasure to be back here speaking before a group of nut growers. Back some years ago my first assignment to a station of which I had charge was an investigation to count insects in Louisiana and Eastern Texas, so it is a pleasure to be back before a group of nut growers.

Insect Enemies of Northern Tree Nuts

HOWARD BAKER, *U.S.D.A., Agr. Res. Admin., Bureau of Entomology and*
Plant Quarantine

The small number of requests for information on insect pests of northern tree nuts received in the Bureau of Entomology and Plant Quarantine is a strong indication that such pests are of little concern to northern nut growers. This is fortunate, because intensive, all-season spray programs, such as are necessary to produce most other crops without serious losses due to insect injury, are laborious and expensive and not always as effective as desired. However, as your acreage is increased and as your trees become older and larger, insect problems are likely to increase in number and intensity and require more of your thought and attention.

A somewhat similar situation prevailed in the pecan industry at one time in the South. I well remember the statement of one of the larger pecan growers in Louisiana to the effect that all the pleasure of growing pecans would be gone the day he had to start spraying to control insects and diseases. Only a short time later it became necessary for him to initiate a regular spray program. He still took great pride in growing pecans, however. It is well, therefore, for you to watch your trees closely for insect damage and keep informed concerning the habits and control of the species that show up in your plantings or in those of your neighbors.

Because of the scattered nature of the northern nut industry, the small size of most plantings, and the more pressing demands for information on the control of pests of more intensively planted crops, it has not been possible for the Bureau of Entomology and Plant Quarantine to give attention to many of the pests of northern nuts. A great deal of work has been done on the pests of pecans in the South, and some work on those that attack filberts and chestnuts. In addition, some of the pests with which you are concerned, or others similar to them, are receiving attention in connection with studies of pests of tree fruits. The results of these studies will give you up-to-date information applicable to your particular problems.

The timely use of insecticides is the most effective means of combating most injurious insects, but if spraying is not possible, other methods can often be used to prevent or reduce damage. A great many new insecticides have become available during the last six or seven years. Work with them has resulted in the development of treatments effective against a number of pests for which there was formerly no known means of control and markedly more effective treatments for the control of others. It is my purpose to bring to you as much of this new information as is applicable to your problems.

Leaf-feeding Caterpillars

The fall webworm[3] and the walnut caterpillar[4] are the leaf-feeding caterpillars most commonly reported as attacking northern tree nuts.

Fall webworms[5] are the insects usually responsible for unsightly webs on or near the end of the branches of the trees during the summer and fall.

They enlarge the webs as they need more leaves. When nearly full grown they scatter to complete their feeding. The full-grown caterpillars are a little more than an inch in length and are covered with long black and white hairs. They spend the winter in cocoons in trash on the ground or just below the surface of the soil. There are two broods a year in many areas, the second usually being the more numerous.

Control can be obtained by applying a spray containing 3 pounds of lead arsenate with an equal quantity of hydrated lime (to prevent possible injury to the foliage), 2 pounds of 50-percent DDT wettable powder, or 2 pounds of 15-percent parathion wettable powder per 100 gallons of water. Apply the spray when the caterpillars are still small. Follow the precautions furnished with each package. Parathion is a particularly dangerous material to use. If you are not equipped to spray or have only a few trees, you can control this insect by removing the webs from the trees with a long-handled pruner or a long bamboo pole with a hook at the end.

The walnut caterpillar feeds in groups, or colonies, and commonly eats all the leaves on small trees or on certain limbs on large trees. The winter is spent in cocoons in the ground. The moths appear late in the spring or early in the summer and lay masses of eggs on the underside of the leaves. From time to time as they grow, the stout, black caterpillars go down to a large limb or to the trunk of the tree to molt, or shed their skins. After molting they return toward the ends of the branches and resume their feeding.

This insect can be controlled with the same spray treatments that are recommended for the fall webworm, and also by crushing or burning the caterpillars when they are clustered on the lower limbs or tree trunks.

Pecan Phylloxera[6]

Swellings called galls sometimes appear on leaves, leafstalks, succulent shoots, or nuts of the current season's growth of hickory and pecan. These galls are caused by small insects known as phylloxera, which are closely related to aphids, or plant lice. Several species are involved, but only one, known as the pecan phylloxera, causes serious damage. It causes twigs to become malformed, weakened and finally to die, and destroys the crop on the infested terminals. The insect passes the winter in the egg stage in protected places on the trees. The young appear in the spring about the time the buds begin to unfold.

The phylloxera can be controlled by spraying the trees thoroughly with a mixture containing 3/4 pint of nicotine sulfate plus 2-1/2 gallons of lime-sulfur or 2 quarts of lubricating-oil emulsion to 100 gallons of water during the delayed dormant period or by the time buds show about an inch of green. Sprays containing 3 pounds of BHC (10-percent gamma) or 1-1/4 pounds of 25-percent lindane wettable powder per 100 gallons are also effective, and their use is increasing. Other materials have given good control when applied about the time the buds begin to swell. They are 36-percent dinitro-o-sec-butylphenol liquid, 3 quarts per 100 gallons, and a mixture of 40-percent dinitro-o-cyclohexylphenol powder, 2 pounds, and lubricating-oil emulsion, 5 quarts, per 100 gallons of spray. Do not use the dinitro materials after the buds begin to open.

Twig Girdler

A stout, brown beetle about 1/2 inch in length, known as the twig girdler, [7] often cuts off the twigs of hickory, pecan, and many other trees in the late summer and early fall. The larvae spend the winter in the cut twigs, which are gradually broken off and fall to the ground. Injury can be reduced by collecting and destroying the fallen twigs before the larvae complete development the following spring. Recent work on pecans in Florida indicates that most injury can be prevented by applying a spray containing 4 pounds of 50-percent DDT or 3 pounds of 15-percent parathion wettable powder per 100 gallons of water. Three applications appear to be necessary, the first when the injured branches are first noticed, usually sometime in August, and the second and third two and four weeks later. When handling parathion be sure to follow the precautions on the package.

Weevils and Curculios

Weevils and curculios are small, hard-shelled, grayish to brown beetles about 1/4 to 1/2 inch long, with stiff, slender snouts or beaks. They feed and lay eggs in the nuts and/or shoots of many kinds of nuts, including hickory, walnut, pecan, chestnut, hazelnut or filbert, and butternut. There are a number of species, but most of them attack only one kind of nut. The species usually called weevils most often lay eggs and injure the nuts from the time the meat begins to form until it is mature, whereas the group known as curculios generally emerge and cause most serious damage during the early part of the growing season, when the new shoots are developing and the crop starts to set and grow.

The chestnut weevils are probably the weevils best known to most of you. E. R. VanLeeuwen, of the Bureau of Entomology and Plant Quarantine, has added much to our knowledge of these weevils in recent years. Two species, the small chestnut weevil[8] and the large chestnut weevil,[9] are commonly present together and cause similar injury. The small chestnut weevil appears as an adult over a period of about 6 weeks beginning near the first of May in the vicinity of Beltsville, Md., but it does not lay eggs until about the middle of August. The larger species does not emerge until about the middle of August and begins to lay eggs soon thereafter. Eggs are laid in the developing nuts, and injury is caused by the feeding of the larvae therein. Most of the small weevils require two years to complete development, and most of the larger weevils but one year.

Some control of these weevils can be obtained by collecting and destroying the infested nuts before the larvae leave them to enter the soil. Better control can be obtained by spraying the trees with DDT. Apply a spray containing 4 pounds of 50-percent DDT wettable powder per 100 gallons of water (3 level tablespoonfuls per gallon) 30 days before the first mature nuts are expected to drop, and make two additional applications at intervals of 7 days. If you are not equipped to spray, you may obtain some control by treating the soil under the trees with ethylene dibromide at a depth of 5 inches. Make injections at intervals of 1 foot in each direction and also in the center of each square formed by these injection holes. Place 1 milliliter of 40-percent ethylene dibromide or an equivalent quantity of another dilution in each hole. Make the application in the fall immediately after the nuts are harvested and close the injection holes by pressing with the foot. The soil should preferably be loose to a depth of 5 inches.

The pecan weevil,[10] also known as the hickory nut weevil, often causes heavy losses of pecans and most species of hickory. Two or three years are required for the insect to complete its life cycle, but some specimens reach maturity every year. Adults emerge from the ground from the middle of July until early in September, according to locality and seasonal conditions. Injury is of two types—(1) that resulting from attack before the shell-hardening period in July and August, causing the young nuts to drop, and (2) that resulting from attack after kernel formation, the kernel being destroyed by the developing larvae, or grubs. Egg deposition in the nuts usually begins late in August.

To control this weevil spray the trees twice with 6 pounds of 50-percent DDT or 40-percent toxaphene wettable powder per 100 gallons of water.

Make the first application when at least six weevils can be jarred onto a sheet on the ground beneath any tree known to have been infested in previous seasons, and make the second 10 to 14 days later. The first application will be needed sometime between the last week in July and the first week in September. If the soil is hard and dry, it will delay emergence of the weevils. If you are not equipped to spray, you can reduce weevil injury about 50 percent by jarring the limbs of the trees lightly and gathering the weevils on a sheet during the period of emergence. The dislodged weevils will remain quiet on the sheet long enough to be picked up and destroyed. Begin jarring about the last week in July and confine it to two or three trees until the first weevils appear. Then jar all trees at weekly intervals until about the middle of September, when egg laying will have been largely completed.

The butternut curculio[11] attacks native butternuts and introduced nuts of a similar type. It passes the winter as an adult in trash or other shelter it can find in the vicinity of nut trees. It is a small, hard-shelled, rough-backed snout beetle. Late in the spring it makes its way to the trees, and lays eggs in the young shoots. On hatching, the young larva penetrates into the young shoot or leaf stem or nut and feeds there, causing the leaf or nut to dry up and fall off. Upon completing development in the fallen leaf or nut, the mature larva enters the soil. After a month or so in the ground the adult emerges, feeds on the foliage for a while, and then enters hibernation. There is but one generation a year.

The black walnut curculio[12] is similar to the butternut curculio in seasonal history, but it attacks principally the fruit of the black walnut and butternut, apparently preferring the former.

The hickory nut curculio[13] is much like the preceding two species, but it attacks chiefly partly grown hickory nuts, causing a heavy dropping in midsummer.

The hickory shoot curculio[14] attacks chiefly the shoots of various kinds of hickory. The damage is seldom of much importance except to newly transplanted trees. On pecan it attacks the unfolding buds and shoots. Pecans most commonly attacked are those that are uncultivated or are adjacent to woodlands containing native pecan and hickory trees.

For many years these curculios have been controlled by spraying the trees soon after growth starts with lead arsenate, 2 pounds per 100 gallons, plus an equal amount of hydrated lime. One or two additional applications may be needed as new growth appears or as the nuts increase in size. Recent experimental work indicates that BHC or lindane may be more effective for controlling these insects. A spray containing 3 or 4 pounds of technical BHC (10-percent gamma) or 1-1/2 to 2 pounds of 25-percent lindane wettable powder per 100 gallons, applied when the buds show from 1/4 to 1 inch of green growth or when jarrings show adults are present, has given fairly good control.

Walnut Husk Maggot

The walnut husk maggot[15] attacks black and English walnuts, butternuts, and a few other nuts. The feeding of the larva, or maggot, in the husks impairs the quality of the kernels, discolors the shell, and often causes the shells to adhere to the nuts. It causes the most damage to English walnuts. This insect hibernates in the pupal stage in the ground. In midsummer it transforms to the adult fly stage, leaves the soil, and flies to the nut trees. After 1 to 3 weeks the flies lay eggs in the husks of the developing nuts. The eggs hatch in a week or 10 days, and the young maggots burrow within and throughout the husks of the nuts; they mature in the fall.

The walnut husk maggot can be controlled by spraying the trees with lead arsenate or cryolite the latter part of July and again 3 to 4 weeks later. Use 2 or 3 pounds of lead arsenate plus an equal quantity of hydrated lime or 3 pounds of cryolite per 100 gallons of water.

Filbert Moth

The filbert moth,[16] a serious pest in some filbert orchards in Oregon, also causes some injury to chestnuts. Adult moths begin emerging toward the end of June and lay their eggs singly on the leaves beginning early in July. The newly hatched larvae tunnel through the husk and feed between the husk and the chestnut shell before entering the nut. This feeding produces a gummy substance, which causes the husk to adhere to the nut. The larvae may tunnel into the center of the kernel or excavate an irregular cavity in the side. They reach maturity about the time nuts are ripe, and then leave the nuts and construct cocoons in the soil in which to pass the winter.

Control can be obtained by spraying the tree with lead arsenate or DDT early in July. Use 3 pounds of lead arsenate or 2 pounds of 50-percent DDT wettable powder in 100 gallons of water.

Mites

Two general types of mites sometimes damage nut trees, eriophyid mites and spider mites. The most important eriophyid mites are the wormlike gall mites and bud mites, most of which overwinter in the buds and cause deformities of the buds and leaves and otherwise limit their development. The spider mites may overwinter in the egg stage on the twigs or as adults in protected places on or beneath the trees. These mites feed primarily on the foliage.

The filbert bud mite[17] is occasionally of economic importance as a pest of filberts in Oregon and has been of some concern recently in New York. It attacks the leaf and flower buds and catkins. Infested catkins become distorted, rigid, and brittle, and yield no pollen. In Oregon this pest has been controlled with 3 gallons of a dormant oil emulsion or 6-1/2 to 8 gallons of liquid lime-sulfur in water to make 100 gallons of spray just as the buds are opening. Related species of similar habits that attack walnuts have been controlled with 9 or 10 gallons of liquid lime-sulfur in water to make 100 gallons of spray applied at the time the buds break or soon thereafter.

The feeding of the spider mites on the foliage of infested trees causes it first to have a bronzed or scorched appearance, and later to dry up and fall. These mites frequently become abundant following the use of some of the new organic insecticides, such as DDT and BHC, which destroy their natural enemies and perhaps have other effects on the trees favorable to mite activity. The European red mite, which overwinters on the trees in the egg stage, can be controlled by application of 3-percent oil-emulsion spray in the late-dormant period. The two-spotted spider mite and related species, as well as the European red mite if it is not controlled with the dormant spray, can be controlled with a spray containing 1 pound of a 15-percent parathion or 1-1/2 pounds of a 15-percent Aramite wettable powder per 100 gallons. Apply the spray before many leaves show the typical bronzing or leaf scorching. If the infestation is heavy, a second application may be necessary in about 8 or 10 days. Be sure to follow the precautions on the container, especially if you use parathion.

PRESIDENT MacDANIELS: We greatly appreciate your care in getting this thing together, and we know it is going to be a great help to us when we get it printed as a matter of reference.

MR. O'ROURKE: I'd like to ask Dr. Baker if insects are getting stronger or if the chemicals are getting weaker. I refer to the rates of application. Formerly we were told that one-half pound of parathion for one hundred gallons and one pound of DDT would control almost all insects. I note the rates are going up.

MR. BAKER: That's true, particularly with parathion. The first year that we tested parathion on any scale we thought a quarter to a half a pound would control mites for 30 days or more and would control curculio for 20 or 30 days, but the next year we used it we found that was a little optimistic. It seems that each year since we have had to use more of it or use it more often, or with mites, particularly, there are a number of instances where it just doesn't control them at all.

Two years ago that came to notice in the Wenatchee area of Washington on apples. Mites in a certain orchard just couldn't be controlled with parathion. A year ago the area in the Pacific Northwest where that was true was extended and included several orchards of the Yakima Valley. This year it also includes orchards in the East, in New York. We have seen an orchard where two pounds of parathion and a hundred gallons of water just didn't have much effect on the mites, and we have had to use other materials. We hear of instances of codling moth on apples where DDT doesn't seem to be as good as it was in the beginning. I have talked with some of the people working on the problem, and they find that there is quite a difference between different brands of some of these insecticides. Possibly that is the answer.

MR. MACHOVINA: After spraying for shuck maggot with DDT do you encourage the presence of mites?

MR. BAKER: It's very possible that you might. That has happened where DDT has been used. With some of our work with chestnut weevils, mites seem to be a little more abundant where we used DDT. We have had reports of this happening in California where they used DDT on walnuts. So it is a possibility, and that's why I brought into the paper a little information on the control of mites.

Session closed at 4:15 o'clock, p.m.

FOOTNOTES:

[Footnote 3: *Hyphantria cunea* (Drury).]

[Footnote 4: *Datana integerrima* G. & R.]

[Footnote 5: *Clastoptera achatina* Germ.]

[Footnote 6: *Phylloxera devastatrix* Perg.]

[Footnote 7: *Oncideres cingulata* (Say).]

[Footnote 8: *Curculio auriger* Casey.]

[Footnote 9: *C. proboscideus* F.]

[Footnote 10: *Curculio caryae* (Horn).]

[Footnote 11: *Conotrachelus juglandis* Lee.]

[Footnote 12: *Conotrachelus retentus* Say.]

[Footnote 13: *Conotrachelus affinis* Boh.]

[Footnote 14: *Conotrachelus aratus* Germ.]

[Footnote 15: *Rhagoletis suavis* Loew.]

[Footnote 16: *Melissopus latiferreanus* (Wlsm.)]

[Footnote 17: *Phytoptus avellanae* Nal.]

TUESDAY EVENING BANQUET SESSION

We will now have the report of the Resolutions Committee.

MR. DAVIDSON: "To Royal Oakes, Chairman of the Program Committee, and to J. Ford Wilkinson, the City of Rockport and its hospitable people, the Northern Nut Growers Association extends its grateful greetings to you and to your loyal helpers, mentioning only a few; that is, Mrs. Negus, Mr. and Mrs. Sly, Mr. Richard Best, a group of people who say little and who do much, our very hearty thanks to you and to your helpers. We have had a splendid meeting, good attendance, good fellowship and tomorrow a good field trip.

"RESOLUTION: The sincere and grateful appreciation of this Association is hereby tendered to J. C. McDaniel, who has so faithfully and fruitfully served it as Secretary for five years. Your creation of new avenues of service, such as *The Nutshell* is sufficient evidence of your resourcefulness in a difficult and most important office.

"RESOLUTION: Be it resolved, that this Association instruct its Secretary to communicate the following action to the responsible agencies of Federal and State authorities in all areas where the oak wilt disease is present or threatens:

"'The oak wilt disease threatens severe damage to our eastern and southern oaks and Chinese chestnut trees. Recently reported spread of the disease in Ohio, West Virginia, Maryland and Pennsylvania indicates a very serious and critical situation. All state and federal authorities are urged to take prompt and appropriate action before it is too late.'"

All NNGA members are asked to write to their state and federal senators and representatives urging immediate preventive measures against the spread and for the eradication of the oak wilt disease. Please write those letters. They are important.

"To Dr. Deming, greetings and congratulations from your Association on the occasion of your 90th birthday, September 1, 1952. May your years continue to be golden and happy. May our organization deserve in the future the gifts of inspiration and accomplishment that you have had so large a part in giving it in the past."

"To Dr. J. Russell Smith: The Northern Nut Growers assembled at Rockport send greetings and best wishes to you. We miss you this year and hope to see you at Rochester, New York, next year."

"To Mildred Jones Langdoc. Mildred: We have missed you at our meeting. Your absence is noted by all who know you. May the illness in your home be short. May we see you and your family in Rochester in 1953."

"RESOLUTION: On behalf of the members of the Northern Nut Growers Association the Secretary is asked to send our affectionate greetings to two well-loved, absent members, Mrs. C. A. Reed and Mrs. G. A. Zimmerman: 'Best wishes to you both for speedy recovery of good health and with our hope to see you next year.'"

PRESIDENT MacDANIELS: Is it your pleasure to adopt these resolutions all at once, or do you wish to separate them? I take it that you wish to adopt them, all at the same time, and to that end a motion to accept the report of the Resolutions Committee and to adopt the resolutions and to send the greetings would be appropriate.

The report of the resolutions committee was accepted unanimously.

MR. MCDANIEL: Before this meeting convened we planned a bud wood exchange at the convention. Mr. Gerardi and I brought some buds, and Mr. Richard brought a few of the Rhodes heartnut. We have persimmons, some buds of the new Crandall apple, and a few sticks of Chinese and hybrid chestnuts. They are for anyone who would like to experiment with them.

PRESIDENT MacDANIELS: Next year at Rochester we are going to have opportunity for putting on a considerable exhibit of nuts, and I think that it would be much to the advantage of the Association, if we could have an outstanding exhibit there where there is a good chance to have a large number of people see the exhibits and become interested. To that end I think that all of us who have nut trees bearing this fall, should save some samples with extra care; that is, clean them up, make them look attractive and have them on hand ready for the exhibit next fall.

A good sample for exhibit should be about 10 or a dozen for black walnuts and the Persian walnuts and perhaps 20 to 25 for the hickories and the smaller nuts, the hazel, particularly. I think that we have a good chance next year to forward the cause of the Association, and certainly having these exhibits will be much to our advantage.

At this time, towards the end of our session, it is our usual custom to elect our next year's officers. Before going on with that election, I would just like to say that I personally, as president of the Association during this year, wish to thank all of the other officers who have worked with me. It has been a pleasure to work with them and with the committee chairmen, and I think the meeting here at Rockport and the work during the year attest to their effective service.

The Nominations Committee report. For president next year, Mr. R. B. Best; for vice-president, George Salzer of Rochester, New York; for Treasurer, Carl Prell of South Bend, Indiana, who continues in the office; and for Secretary Mr. Spencer Chase of Norris, Tennessee.

The slate presented was elected unanimously.

A nominating committee consisting of Max Hardy, Gilbert Becker, George Slate, Dr. William Rohrbacker, and Ford Wilkinson was unanimously elected for 1953.

PRESIDENT MacDANIELS: I will now call upon our newly elected president to come forward. It is usual at these meetings for the retiring president to present the gavel to the incoming president, and here it is. This gavel is made of pecan wood presented to the Association by Mr. T. P. Littlepage, who was born in this locality. I hope you will have as much fun and pleasure as president of the Association as I have had. It's all yours.

MR. WILKINSON: That gavel was made from the wood of a pecan tree. Mr. T. P. Littlepage planted the nut when he was 14 years old on a piece of land that he inherited as a boy. I cut the wood and sent it to him in Washington to have the gavel made of it.

Chestnut Breeding

Report for 1951-1952

ARTHUR H. GRAVES[18] and HANS NIENSTAEDT, *Connecticut Agricultural Experiment Station, New Haven, Conn.*

Weather Conditions

Two serious enemies of the chestnut, if we disregard parasitic organisms, are drought and extreme cold. The winter of 1950-51 was unusually mild—scarcely cold enough to freeze the ground. The precipitation was plentiful during the winter months so that the water table was sufficient to tide over a slightly dry June and a much more serious drought in September and early October. But the latter dry period came when the nuts were matured, or nearly so.

The winter of 1951-52 was again mild except for a short cold spell at the end of January, with plentiful precipitation up to the first week of June, and then a long drought with the driest July since 1944. However, the heavy rainfall of August, 8.69 inches,[19] made amends for this, and with the normal rainfall of 3.48 inches of September, prepared the trees to endure the long drought of October and early November. This serious drought,[20] which resulted in disastrous forest fires filling the air with smoke over much of the New England States, came late, however, after the nuts were nearly matured, some of the early kinds being ripe as early as the first week in September.

The excessive heat of July, in which month occurred the greatest number of days on record with a maximum temperature of 90 degrees or above, was probably the chief cause of somewhat smaller results from our cross pollination work. There is evidence, indeed, that for effective fertilization, considerable heat is needed, but not the extreme temperatures that occurred during this period.

In spite of the mild winter of 1951-52, the attacks of *Cryptodiaporthe castanea* (Tul.) Wehmeyer caused considerable twig blight, especially on our crosses of *Castanea mollissimax seguini*. This is not surprising since *C. seguini* comes from a warmer region in China, but why these attacks should occur during a mild winter is a puzzle. Evidently other factors, such as the drought of the preceding fall, entered in.

Hybridization in 1951 and 1952

A total of 2400 hybrid nuts was harvested in the 1951 season and 1690 in 1952. This compares with the 1259 nuts reported for 1950. The increased production over past years can in part be ascribed to a concentration of the efforts on a fewer number of different crosses; while 103 were made in 1950, the total was 77 in 1951 and 80 in 1952. The pollinations followed the same general program in the two seasons, the emphasis being on the Chinese x (Japanese x American) hybrids. This is our most promising timber tree hybrid, and it seems worthwhile to test it on a somewhat larger scale under forest conditions. Therefore, some of the best early crosses have been repeated, new parent trees are being tried and selected hybrids intercrossed. Back-crosses to the native chestnut with the CxJA hybrids were made in an attempt to improve the form of the hybrid.

Another cross which has attained some importance in the last years is the hybrid between Japanese chestnut (forest type, from U.S.D.A.) and S-8, the latter being a hybrid between Japanese chestnut and *C. pumila*, the common chinquapin. This cross has a high degree of resistance and a sufficiently good form to make it a possible timber tree (Fig. 1). It is also a fairly good nut bearer with nuts which ripen early, perhaps due to the influence of the chinquapin parent (Fig. 2). Selected individuals of this hybrid were intercrossed, and some crossing with the native chestnut was done.

In the last two seasons the total harvest from some older Chinese trees (26 yrs.) was recorded. The best tree yielded 25.0 lbs. in 1951 and 28.2 lbs. in 1952; on other trees the yield varied between 15 to 22 lbs. The average size of the nuts varies considerably from year to year on the same tree. On one Japanese tree the average weight per nut was 5.6 g. in 1951 and 14.5 g. in 1952; on a Chinese tree the same values were 7.7 g. and 15.1 g. Other trees showed a 20-40 per cent increase in the average weight per nut in 1952 over 1951. This seems to indicate a marked influence of the climatic conditions during the latter part of the growing season on the weight of the nuts. A long-term study of this relationship might yield some interesting results.

[Illustration: Fig. 1. Hybrid of S-8 and *Castanea crenata*, U.S.D.A, forest type, 18 years old. About 35 ft. high. Good forest type and also good nut bearer. Blight resistant. Sleeping Giant Chestnut Plantation, Hamden, Conn. Photo by Louis Buhle, Brooklyn Botanic Garden. Sept. 26, 1952.]

[Illustration: Fig. 2. Fruiting branches and nuts of S-8 x *crenata*, Sleeping Giant Chestnut Plantation. About 1/2 natural size. Photo by B. W. McFarland, Conn. Agric. Expt. Sta. Sept. 8, 1952.]

Grafting

A considerable amount of grafting has been done since 1949 and the results have been good. Two year old Chinese transplants are usually used as rootstocks and all grafting is done in the field. The best results have been obtained where the rootstock plant was transplanted one year prior to the grafting. The simple splicegraft, or the bark or rind graft are used, depending on the size of the scion compared to that of the rootstock, the latter technique being used when the stock is considerably larger than the scion. There is some evidence of incompatibility; thus, scions from Chinese trees, or hybrids that show a dominance of Chinese characters, give a higher percentage of takes when grafted on Chinese rootstocks than scions from the native chestnut, or from hybrids between Japanese and native chestnut. Some indications of incompatibility between European and Chinese chestnut in grafts have also been encountered where scions received through the cooperation of Dr. C. Schad, Centre de Recherches agronomiques du Massif Central, France, and Count F. M. Knuth, Knuthenborg, Denmark, were used, but in some cases these grafts were successful. Topworking, using the veneer crown graft, has been quite successful as long as sufficient sap drawers are left on the stock (Fig. 3).

Inarching

The senior writer has already explained in detail (2) the simple method by which blighted chestnut trees can be restored to health and vigor by cutting out blighted areas in the bark, painting them over, and inarching or ingrafting one or more basal shoots into the healthy bark above the lesion. We do this work from mid-April to mid-May, and make a systematic canvas of all the trees in all our plantations, inarching all those where if is necessary or might be advantageous. Each operation requires only a few minutes. Last year we put in many hundreds of inarches, altogether, which later showed nearly 100% "take".

Owners of chestnut orchards should take advantage of this method of keeping valuable nut-bearing trees, although with cankered areas, in healthy, vigorous condition.

We believe that, in cutting out the diseased bark, it is advisable to cut out also a few of the outer annual rings of wood (of course tangentially), especially if the canker is one of long-standing, since we know that the fungus eventually penetrates the outer rings of wood. Since that is true, the canker might enlarge later on from this same source of infection. Further it may also be possible for spores or bits of mycelium to be transported upward in the sap stream and cause new infections higher up in the tree. A thorough painting of the cut surfaces should go far toward remedying this situation.

One can usually judge the extent of damage caused by the blight by the number and vitality of the basal shoots, a large number of basal shoots indicating a heavy attack. However, if the roots have been severely injured, perhaps by short-tailed mice, as sometimes happens, no basal shoots appear, in which case the tree is doomed.

If no blight is present, but one or more basal shoots appear (sometimes due to shrubby ancestors), it is advisable to inarch these as an insurance against possible trouble in the future.

This inarching process has not received the attention it deserves. There is absolutely no reason why, if this method is followed, there should be *any* death from blight in resistant hybrids or in Japanese or Chinese chestnuts, barring, of course, cases where roots are attacked by mice (or *Phytophthora* in warmer regions). Those of our trees in Connecticut which have been blighted have continued in health and nut-bearing ever since we began the inarching method in 1937 (Fig. 4). If the inarches become blighted, they can themselves be inarched, as shown.

[Illustration: Fig. 3. Veneer crown grafting on chestnut. Photo by B. W. McFarland, Conn. Agric. Expt. Sta. May, 1952.]

Research on Blight Resistance

[Illustration: Fig. 4. Japanese-American Chestnut, 21 yrs. old, showing inarching begun 15 yrs. ago. Original trunk, long since dead and now rotting, shows in center. Kept alive and vigorous because valuable for hybrid vigor and future breeding. Sleeping Giant Chestnut Plantation, Hamden, Conn. Photo by Louis Buhle, Brooklyn Botanic Garden. Sept. 26, 1952.]

A study has been made of the factors that cause the Chinese and Japanese chestnut to be resistant to the Endothia canker, and a close correlation was found between the tannin content of the bark and the relative resistance of the three species, i.e., Chinese, Japanese and American chestnut. The total tannin concentration in the bark of the Asiatic species is only slightly higher than in the American, and native trees can be found with as high a concentration as is found in the Asiatic. A similar overlap in resistance does not occur and it is therefore clear that the total tannin concentration as such cannot account for resistance. There is, however, good evidence that the tannins in the Asiatic species, as a result of the way in which they are bound to other colloids in the cells, are more soluble than in the American species. This, of course, would have a marked bearing on the effectiveness with which the tannins could check the spread of the parasite. Furthermore, it has been found that the types of tannins in the three species differ. In the American and Japanese species they are a mixture of catechol and pyrogallol tannins, while they appear to be pure pyrogallol tannins in the Chinese species. Considering the specificity of the enzyme systems of fungi it is quite possible that different tannins show different degrees of toxicity to a certain fungus. The following hypothesis has been suggested to explain the relative resistance of the three species: In the American chestnut bark the concentration of the available toxic tannin never reaches a level where it can stop the advancing parasite. The tannins in the Japanese species, although of the same type as in the native tree, are more soluble and reach a level toxic to the fungus. In the Chinese trees all the tannins of the bark belong to the toxic pyrogallol groups, and this, combined with their high solubility, results in the high degree of resistance in this species (4).

The information available at present regarding the formation of tannins in plants is not conclusive. In some plants, apparently, they are formed in the leaves, and the presence of carbon dioxide and light is required; in other plants the tannin concentration can increase when the plants are grown in darkness (5). A more general formation of tannin in tissues with a high metabolic rate throughout the plant has also been suggested (3).

It would be important to know the centers of origin of the tannins in the chestnut, their translocation, and whether they are translocated through or over graft-unions. In other words, will a susceptible scion when grafted on a resistant rootstock become more resistant because antibiotic substances formed in the roots of the resistant rootstock are translocated into the scion?

From a number of older grafts of non-resistant Japanese-American hybrid scions on Japanese or Chinese rootstocks it appears that this indeed might be the case. These grafts, some of which are 16 years old, appear to be more resistant than the original hybrid tree, even if not as resistant as the rootstock.

This would indicate the possibility that the antibiotic substances are produced in the roots and translocated into the scion. However, the possibility still remains that the compounds are formed also in the leaves and translocated to the base of the tree. To clarify this whole problem an experiment with Chinese-American grafts in different combinations is under way. Preliminary results show that antibiotic substances are formed in upper parts of the plants, but that they are not translocated downward across the graft union. Thus it was found that Chinese branches grafted on two year old American seedlings remained resistant, without the American seedlings showing any increase in resistance. In future experiments the upward translocation will be studied in detail on grafts of American scions on Chinese seedlings.

Some Abnormal Conditions

1. *Sterility*

Sterility occurs quite commonly in interspecific hybrids either because the chromosomes fail to pair in meiosis or because the parent genes when brought together in the hybrid interact in some way deleterious to the formation of sex-cells. Furthermore, cytoplasmic sterility is likely to occur in a wide cross.

Sterility has been encountered in several instances in American x Chinese and Japanese x American hybrids. In most cases it is a case of pollen abortion only; either anthers fail to develop completely as shown in Fig. 5, B, or the anthers develop but are much reduced in size and contain no functioning germ cells.

Pollen sterility is not sporadic in a given individual: it is uniform throughout the flowering branches. The individual flowers are arranged on the catkin axis as in the normal flowers (Fig. 5). But when the flowers open, a hand lens reveals 3-5 tiny, membranous perianth-segments for each tiny flower, whitish in color, and more or less connected at their bases. A minute rounded mass appears in the center of the flower, perhaps primordia of abortive stamens, but this does not develop further. The catkin begins to take on a brownish color and at length the whole catkin, in case it is staminate, drops off. If it is androgynous, the staminate part drops off, or withers.

These male sterile trees appear to have a normal, sometimes excessive, development of the females, and are quite prolific nut producers. Information on the occurrence of female sterility in the hybrid trees is incomplete, but the indications are that at least partial sterility is frequent.

[Illustration: Fig. 5. A. Normal androgynous catkin (female flower at base); B. Androgenous catkin with sterile pollen. From Sleeping Giant Chestnut Plantation, Hamden, Conn. Photo by Mary Alice Clark, Conn. Agric. Expt. Sta. July, 1949.]

2. *Triploid Hybrid*

In 1934 we produced a cross of Chinese and American chestnut which proved to be unusual in several respects. The leaves are enormous—9 inches to 1 foot in length, and 4 or 5 inches in width. The hybrid is not particularly blight resistant but more so than its American parent. It died back from the blight about 1940 and the present tree has developed as a shoot from the old roots. The growth is rapid and vigorous. The flowers appear normal, but we have never been able to make a cross with its pollen, nor to effect fertilization of its pistillate flowers. It may be triploid, that is, with 3 sets of chromosomes instead of the normal double set, and this would account for its barrenness.

In the spring of 1952 some of the vigorous shoots of this tree were successfully grafted on shoots from an old stump of Chinese chestnut, using the veneer crown graft method. The scions had not been taken when dormant, but were transferred directly from the tree to the stock in late April. This grafting was done in order to impart greater resistance, if possible, to the CA hybrid by means of the roots of the Chinese stock.

3. *Systemic Defect*

Since the early 1930's we have seen occasional individuals with abnormal foliage—somewhat mottled, usually curled and often misshapen. Thinking that a virus might be the cause of this trouble the senior author tried grafting some of the shoots on to healthy stocks. The grafts were in no case successful because the scions were too weak. Finally he succeeded in grafting a branch from an affected tree on to a branch of a normal individual. The only result was an increased vigor of the healthy branch. This year he rubbed juices from leaves of such an abnormal individual on to wounded healthy leaves, without result. Moreover, such sick individuals, although growing for years close to healthy trees, have never communicated the malady to their neighbors. Growth is comparatively slow, and there is much dying back or dying out of the slender branchlets.

The evidence indicates that this is *not* a virus trouble, but a systemic defect, probably caused by chromosome aberration or gene abnormality. It is significant that this trouble occurs only in hybrids. Such trees never flower. We have known four such cases, two of which are now dead. Similar types appear in other species as inherited deviations from normal.

Insect Injuries

A heavy attack from the spring canker worms developed in 1951, but spraying with DDT on May 24th prevented serious damage. No outbreak of canker worms appeared in the spring of 1952. The Japanese beetle has been very little in evidence. The principal bad actors are the mites, *Paratetranychus bicolor.* Although barely visible to the naked eye, the effect they produce of whitening the leaves is conspicuous, especially on the Chinese chestnut and its hybrids. These insects overwinter in egg form on the surface of the bark. Last winter they were so numerous on some of the trees that the bark had taken on a red color—especially on smooth-barked trunks just below a branch. An application of "Scalecide" on April 21, while the trees were still dormant, followed by two heavy applications of "Aramite" (6-7 lbs. per acre) on June 13th and 27th, gave good control for the rest of the summer. Spraying with DDT for weevils was done on August 18th and September 3rd in 1952 with good results.

Cooperative Hybrid Chestnut Plantations

In 1947 the first hybrid chestnut plantation under forest conditions was made in cooperation with the U.S.D.A. Bureau of Plant Industry, Division of Forest Pathology. The plantations are made in order to test the hybrids under normal forest conditions and different climatic conditions. In general, each plantation consists of about 100 trees, 50 U.S.D.A. hybrids and 50 Connecticut hybrids. The trees are planted at a 10' by 10' spacing, and the overstory is girdled at the time of planting in order to give the plants better light conditions without causing an abrupt change in the microclimate of the forest floor—a method developed by Dr. J. D. Diller of the Division of Forest Pathology (1). Ten plantations at 9 locations have been established since 1947. These are listed below:

No. of Plots	Location	Year Established
1	Edward Childs Estate, Norfolk, Conn.	1947
1	Tennessee Valley Authority, Norris, Tenn.	1947
1	Table Rock State Park, Pickens, S.C.	1948
1	Antioch College, Yellow Springs, Ohio	1948
1	Upper Perkiomen Valley Park, Green Lane, Pa.	1949
1	So. Ill. Univ. Fish & Wildlife Service, Cartersville, Ill.	1949
1	Russ State Forest, Decatur, Mich.	1951
2	Nathan Hale State Forest, Coventry, Conn.	1951
1	Ouichata Nat'l. Forest, Hot Springs, Ark.	1952

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Connecticut State Ownership of Sleeping Giant Plantations

On April 11, 1951, at a meeting at the "Little Red House", Sleeping Giant Mountain, the lands on the Sleeping Giant Mountain, Hamden, Connecticut, about 10 acres, on which about 1500 chestnut trees are now growing, including nearly every chestnut species known to science, and many valuable, blight resistant hybrids, were formally deeded over to the State of Connecticut by their owner, the senior writer of this report. The meeting was attended by officials of the Sleeping Giant Park Association, the Connecticut State Park and Forest Commission, The Connecticut Agricultural Experiment Station, and the Yale School of Forestry. The transfer to the State was made with the understanding that The Connecticut Agricultural Experiment Station would continue the chestnut breeding work. The whole region is now undergoing a fairly rapid housing development, and in the ordinary course of mortal events this plantation would have been divided into building lots within the next few decades. The State ownership will obviate this, and The Connecticut Agricultural Experiment Station sponsorship will assure a continuation of the breeding work.

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FOOTNOTES:

[Footnote 18: Also of The Division of Forest Pathology, U.S.D.A., Plant Industry Station, Beltsville, Maryland.]

[Footnote 19: Records furnished by the U.S. Weather Bureau at New Haven,
Conn.]

[Footnote 20: October, 1952, was among the six driest Octobers on record. These were: 1879, 1892, 1897, 1916 and 1924. From U.S. Weather Report, New York City.]

Effect of Vermiculite in Inducing Fibrous Roots on Tap-Rooting Tree Seedlings

HERBERT C. BARRETT[21] and TORU ARISUMI[22]

When seedlings of nut trees and other tap-rooted species are transplanted from nursery to orchard, the percentage of survival in often quite low. Perhaps the chief reason for this failure is the marked and pronounced tendency of most tap-rooted plants to produce little or no fibrous, branched roots in lieu of the long, straight, and seldom branched tap roots.

The common practice of undercutting seedlings during the dormant season to induce a branched root system requires additional labor, and often results in reduced growth and vigor during the following season. The use of hardware cloth or other close-meshed wire is effective, but this method also has the disadvantage of being relatively expensive for the nurseryman.

Preliminary work carried on during the past two years has shown that with certain nut trees and other tap-rooted plants, it is possible to induce fibrous roots by growing such seedlings in vermiculite. The methods and results of this work are presented in this paper.

Material and Methods

Seeds of black walnut (*Juglans nigra*), Persian walnut (*Juglans regia*), Chinese chestnut (*Castanea mollissima*), pignut hickory (*Carya glabra*), shellbark hickory (*Carya laciniosa*), shagbark hickory (*Carya ovata*), pecan (*Carya illin*), pawpaw (*Asimina triloba*), and three persimmons (*Diospyros kaki*, *D. lotus*, and *D. virginiana*) were stratified in moist sawdust for three months at a temperature range of 35 to 40 degrees F. After this period of stratification the seeds of each species were divided into three lots and planted in flats 25 x 26 x 6 inches containing one of the following media: (1) sharp sand of the type used in potting soil, (2) potting soil, and (3) vermiculite. Seeds were kept moist with ordinary tap water and allowed to germinate and grow in the greenhouse. When the seedlings had grown two or three true leaves, they were carefully removed from the medium and examined for the type of root system developed.

Results

In the first eight species listed in Table 1, the differences between branched and tap-rooted seedlings were quite pronounced. The few tap-rooted seedlings growing in vermiculite medium showed some laterals and were less strongly tap-rooted than those in soil or sand. Pawpaws in soil and sand media were practically devoid of laterals, and their fibrous root system in vermiculite was not as pronounced as with the walnuts, hickories, and pecans. Of the species studied, the persimmons

Table 1.

Species	Sand		Soil		Vermiculite	
	Number of plants					
	Tap rooted	Fibrous	Tap	Fibrous	Tap	Fibrous
Black Walnut	20	3	24	2	0	39
Persian Walnut	15	2	13	1	0	15
Chinese Chestnut	35	6	32	7	3	37
Pignut Hickory	19	0	22	0	3	16
Shellbark Hickory	9	0	8	0	0	13
Shagbark Hickory	27	0	25	0	2	28
Pecan	21	0	23	0	0	15
Pawpaw	102	0	140	0	20	85
D. kaki	6	2	5	3	0	10
D. lotus	20	11	18	7	0	30
D. Virginia	16	0	20	0	0	14

showed the least tendency to produce tap-rooted seedlings. Typical branched or fibrous-rooted seedlings grown in vermiculite are illustrated in Figure 1.

[Illustration: Fig. 1. Seedlings grown in vermiculite medium. Left, *Juglans regia*; right, *Castanea mollissima*.]

Summary

The chief difficulty encountered in transplanting several nut tree and other commonly tap-rooted seedlings is thought to be due to the lack of a branched root system. The methods and results of a fairly simple technique of inducing fibrous roots, that of growing seedlings in vermiculite, have been presented.

FOOTNOTES:

[Footnote 21: First Assistant in Plant Breeding, University of Illinois, Department of Horticulture.]

[Footnote 22: Formerly Half-time Assistant in Plant Breeding, University of Illinois, Department of Horticulture.]

Eastern Black Walnut Survey, 1951

H. F. STOKE, *Roanoke, Va.*

The Northern Nut Growers Association, at its 1950 Annual Meeting, adopted a resolution directing that a survey covering the eastern American black walnut, *Juglans nigra* be conducted during the ensuing year, and that the services of the State and regional Vice-presidents be utilized in making the survey.

In carrying out this mandate fifty questionaires were sent out, and 37 replies were received. Of these, 33 were from the States, including the District of Columbia, three were from Canada, including British Columbia, Ontario and Prince Edward Island, respectively, and one was from Belgium.

From these replies, as compiled, it is apparent that the natural range of the American black walnut may be defined approximately as follows:

Beginning at the Atlantic seaboard at Massachusetts Bay curving slightly northward then westward across northeastern New York to Toronto and on westward across lower Ontario, Lake Huron, Michigan, Wisconsin and Minnesota, in which state the line curves south-westward, crossing about the northwest corner of Iowa. From this point the line runs approximately south across the eastern parts of Nebraska, Kansas, Oklahoma and Texas. As the line approaches the Gulf of Mexico it turns eastward, crossing the southern parts of Louisiana, Mississippi, Alabama and Georgia, back again to the Atlantic.

The natural range of the black walnut may be said to have been limited on the north by winter cold, on the west by lack of sufficient rainfall and on the south by a winter climate too mild for the required dormant rest period. Where these limitations are removed the American black walnut appears to do well far out of its natural range.

In its native state it seemed to thrive best along water-ways and in hollows among the hills and mountains, though it was also to be found on the uplands wherever the soil was fertile and other conditions favorable. The overflow of streams undoubtedly did much to distribute and plant the seed, aided always by the ubiquitous squirrel.

Twenty-nine of the States reported the trees as thrifty and bearing well-filled nuts. Eastern Maryland reported the trees as thrifty but the nut crop light. Michigan reports the nuts as having been well filled formerly, but poor in recent years. West Virginia makes a similar report, and attributes poor crops to the presence of anthracnose, a fungus disease of the leaves causing early defoliation.

The nut crop of the wild trees appears to be ungathered to a large extent, taking the country as a whole.

Eleven states report whole husked nuts being marketed in a limited way and six report the marketing of home-produced kernels. Prices for the whole nuts are quoted as low as $2.00 per bushel, with a top of $5.00 per bushel for Kansas-produced named varieties.

Accurate statistics as to whole nut and kernel production are not available.

Tennessee reports black walnut cracking plants, as follows: One each at Lebanon and Morristown, and three located at Nashville.

A West Virginia report estimates the State's kernel production at $200,000 per annum. A cracking plant in St. Louis is reported as processing 1-1/2 million pounds of whole nuts annually, for which it pays 5-1/2 cents per pound. Other cracking plants reported are one at Stanford, Kentucky, one at Broadway, Virginia and one or two in West Virginia, location unstated. No statement was received as to the amount of business done by these. A new one is starting operations at Henderson, Kentucky in 1951.

Production of black walnut kernels as a home industry has languished since the Federal ruling that the kernels must be pasteurized as soon as produced. Most of such kernels are now consumed locally, so as not to run afoul of inter-state regulations. No epidemic has, as yet, been traced to such local use.

A question designed to disclose what named varieties give the best results in the various localities was not very effective. Replies usually came in the form of lists of varieties being planted with little definite indication as to the ones that have proven superior.

As might be expected, Thomas led the list by being mentioned 15 times. Elmer Myers was listed 9 times, Stabler 6, Ohio 6, Mintle 3, Snyder 2, (New York and Tenn.), Sifford 2, (Kentucky and Kansas), and the following one each: Adams, Grundy, Korn (Michigan); Rohwer, Vandersloot (Kansas); Sparrow, Victoria, Homeland (North Carolina); Ten Eyck (New Jersey); Creitz (Virginia); and Impit (British Columbia).

A study of the geographical distribution of the preferred varieties fails to produce any significant conclusions as to the varieties best adapted to any specific state. Doubtless Thomas heads the list because it has had the longest and largest distribution. A New York state survey gave Thomas the preference 9 times, Snyder 7, Myers 4, Ohio 2, and one each to several other varieties. A similar survey in New Jersey gave Thomas preference 2, Stabler 2, Ten Eyck 1 and Ohio 1.

One New Jersey correspondent reported Ohio as "excellent", another listed Ten Eyck as "fair", and a third reported Thomas as "terrible".

One Kansas producer reports Thomas his best and Ohio his worst. Another Kansan reports the exact opposite.

Pennsylvania reports Ohio as best, Stabler as worst. Her neighbor to the east, New Jersey, rates Stabler highly, as does Ohio, immediately to the west.

The notable leaf-disease resistance of the Ohio variety is worthy of the consideration of planters in districts where early defoliation causes poor filling of the nuts.

For a late comer, the thin-shelled Myers makes a strong showing, which may be significant. It is worth watching.

Until there is wider planting and production of the named varieties, it will not be possible to name the varieties best adapted to any specific state or location, in the opinion of your reporter.

The possibilities of profit in planting black walnut orchards have not been determined.

From Pennsylvania comes the report that of the several black walnut orchards planted twenty-five years ago, only three are now being given care.

A ten-acre orchard at Wharton, Md. that, presumably, was being given special care, is reported as nearly all dead—"too much commercial fertilizer, or the wrong kind."

The report on several small West Virginia plantings is submitted as "inconclusive".

The main general interest at present appears to be the planting of the better walnuts on home grounds and on the farm. Twenty-four states reported such use, with varying degrees of interest.

Considering that the black walnut is our finest cabinet wood, and one of the best in the world, forestry planting may be truthfully said to be lagging deplorably.

The state of Pennsylvania has shown some interest and made some small plantings.

Ohio has done some planting. The Sunny Hill Coal Company of New Lexington, Ohio, is reported to have planted 5000 seedlings.

In Indiana Ford Wallick has reported the planting of 14 bu. of seed, the seedlings to be budded later to the Lamb curly walnut. Tennessee and West Virginia report small plantings.

Kansas reports some interest in planting walnuts on lands that have been destroyed for agricultural purposes by strip coal mining.

As a whole, the forestry plantings of the walnut of the future, as of the past, appear mainly dependent on the untiring squirrel.

There has never been an adequate supply of walnut timber since pioneer days when walnut logs were rolled together for burning in the clearing of land, or split for fence rails, nor is an adequate supply in sight for the future.

In producing districts buyers are always ready to pounce on the owner of any walnut tree of marketable size. Prices paid are usually much lower than the real value of the timber, partly because the stand is so scattering as to prevent the use of efficient means of logging and transportation.

Of all the agencies tending to destroy the black walnut, war is the most devastating. The superb qualities of the wood for the making of gun stocks causes the country to be combed more and more closely by buyers in each succeeding war.

However, from the standpoint of human interest, the picture is not wholly dark. It is perhaps too much to expect that private enterprise will enter into the long-time investment necessary for extensive forestry plantings, but the states can and should do so in connection with their park and forestry programs. As already indicated some few states are working in that direction.

Of perhaps more immediate concern and value are the possibilities of interesting the 4-H clubs and similar organizations of youth in making home and farm plantings. Refreshingly encouraging is the following excerpt from the report of the Arkansas state Vice-president, Mr. A. C. Hale, a vocational instructor of Camden, Arkansas.

"When a student comes into the class of vocational agriculture in the ninth grade I try to get him to plant some black walnuts so they will get big enough to graft while he is in high school. The use of this method is helpful in getting many trees started. By grafting one or more of the Persian walnuts, interest is also added."

"One way that has helped me get people started with a tree on the home grounds is to pot a few sprouted nuts and when a neighbor is sick take a seedling walnut instead of a flower. I usually go back to help with the transplanting of it."

Such practical methods, if widely used, would bring far more valuable results than any legislative program.

The Virginia Polytechnic Institute is showing some interest, and conducted a field clinic in top-working the walnut in the Shenandoah Valley area in the spring of 1951. County Agents have become interested, and a county-wide Black Walnut Contest will be held at Harrisonburg, Va., Nov. 9 and 10th of this year, in which VPI is collaborating. It is hoped this idea will spread.

On Prince Edward Island, just off the Canadian east coast, there does not appear to be enough summer heat to mature the nuts, though the tree is grown somewhat on home grounds.

In the fruit-growing sections of British Columbia the black walnut appears quite at home, trees of a diameter of from three to four feet being reported at Chilliwack, in the Fraser River valley. J. U. Gellattly also reports the walnut at Brooks and Medicine Hat, Alberta.

Confirmation of the ability of the black walnut to stand extremely low temperatures is to be found in a letter of Aug. 22, 1951 from W. R. Leslie, Superintendent, Dominion Experiment Station, Morden, Manitoba, as follows:

"Black walnut is doing fairly well in such places as the Provincial Horticultural Station, Brooks, Alberta, (P. D. Hargrave, Supt.), and at Portage la Prairie, Winnipeg and Morden, Manitoba. Apparently the black walnut enjoys a heavier soil than the butternut (or white walnut). The white has been more widely planted than the black. The Manchurian seems hardier than either and is the most rapid grower of the three *Juglans* on test here. However, the two natives usually give us a fairly abundant crop of nuts."

"Our source of black walnut was from around New Ulm, Minnesota; the butternut came from around Sault Ste. Marie, at the lower end of Lake Superior. I am not aware of either indigenous species being native closer than the points mentioned."

Belgium reports the black walnut as thriving in door-yards and along roadways, where the nuts are mentioned as a menace to traffic.

In conclusion it is urged that friends of conservation and a sound economy should lend their every effort to the extension of black walnut plantings. Some progress has been made since the days of pioneer plunder, but much remains to be done.

Thanks are extended to all those who have contributed to this survey.

Crath's Carpathian English Walnuts in Ontario

[23]P. C. CRATH, *129 Felbrigg Ave., Toronto 12, Ontario*

Introduction

The English Walnut (Juglans regia) in England is known as Persian walnut. Some think that the nuts originated in Persia. The primeval forests of English walnut trees, which in many places cover the southern as well as northern slopes of the Caucasian Mountains show that Caucasia is the country of the origin of those trees.

But in the Western Carpathian Mountains in Europe geologists had excavated ancient walnuts in the salt rocks of the pits of Weliczka. In some places of the Eastern Carpathians walnuts could be found in a wild stage; and of course domesticated walnuts flourish in every Ukrainian orchard from the northern slopes of the Carpathians up to the southern banks of the Pripet River, and all over Ukraine as far as the Don. But there they could not be found in a wild form.

Walnuts in such countries as Italy, Spain, France are probably of Persian origin.

Since Canada was discovered by Cartier European settlers have many times tried to introduce the southern European walnuts in to the New World, but without success. Only in California, along the Ocean's shore, Europeans succeeded in acclimatizing some, as they think, "English Walnuts"; though in reality the California Walnuts are halfbreeds.

In Old Ontario the people enjoyed the local wild black walnuts, butternuts and hickory. Up to the present English Walnuts are imported into this Province.

When in 1917 I settled in Toronto and found that even in the southern part of the Province, so rich in different fruits, no English Walnuts grew there, I was amazed.

In my old home in the Ukraine walnut trees were as common as elms in Ontario. And I have found that the Southern Ontario climate is warmer than the climate of Kiev or Poltava regions in Ukraine.

It has seemed to me that English walnuts from the Carpathian region should thrive well around Toronto.

My Experiments

In my old home I have heard gardeners say: "Where apples grow, walnuts will grow there also." And around Toronto there I have seen nice apple orchards producing splendid fruits. The Ontario apple trees withstood winter colds well, and that fact encouraged me to try to plant English walnuts from Ukraine in the neighborhood of Toronto. At the end of the First World War Ukraine revolted against the Russian Empire and at the same time she was fighting for her independence with Poland.

At that time my father's family lived in the city of Stanyslaviv at the northern foot of the Carpathians. I asked my sister to send me as many local English walnut seeds by mail as she could. Giving such an order to my sister I expected that the nuts would arrive not later than the end of October, just in time to be planted before the freeze up. This was in 1921.

I remembered from my boyhood that planting of English walnut seeds was surrounded by some mystery. It seemed to me that people in Ukraine regarded it as a very difficult matter to cultivate walnut trees.

Being under such a notion myself I asked a horticulturist how long the germination power of a walnut seed would last. He told me that it could prevail in a fresh walnut not longer than a week. He advised me in order to prevent walnuts from drying to dip them in melted parawax. Following that information I wrote my sister to parawax the walnut seeds before sending them to Canada.

Owing to the Polish-Ukrainian war at that time the shipment of the walnut seeds got to Toronto not late in the Fall, as had been expected, but in February when the farm land around Toronto was frozen. And the worst of it was my sister did not parawax the nuts!

Being sure the kernels were dead I allowed the children to do what they pleased with them. But before they cracked the last one my wife advised me to plant a dozen of the nuts in our flower pots, as she said, "for fun". I did it. Other nuts the children destroyed, and in spite of my sorrow and anguish in two weeks the walnut sprouts came up in the pots. Everyone of them came up, proving that you do not need to protect walnut germination by dipping the nuts into melted parawax.

From the flower pots the walnut seedlings were transplanted that spring of 1922 into our city garden at 48 Peterboro Ave., Toronto.

At least a thousand of the kernels of several varieties were thus destroyed and I was obliged to wait until another fall when the *Juglans regia* nuts were sent again by my sister. They came also late in the winter and were dry as pepper.

In the spring of 1923 I took the walnut seeds of the second shipment to the farm of my friend Mr. M. Kozak located a couple of miles north of the Scarboro Golf Club. There I soaked them in water in a tub for five days and then planted in rows 1-1/2 ft. apart, row from row, and the nuts 6 inches apart nut from nut and two inches deep. In a couple of weeks nearly every nut produced a sapling. I kept them well cultivated the whole summer, and in the Fall the seedlings were from six to eight inches tall. The nuts on the Kozak farm were of different varieties; some were small, some large, some were round, some oblong, some paper-thin-shelled, some hard shelled; some varieties had sweet kernels, some had a little slightly bitter taste, some were flat. According to their variety the bark of the seedlings, some of them at least, was shiny brown, while other varieties had their bark shiny dark green, light gray, light green.

Now I have known how to produce walnut seedlings. Then another worry came—could the seedlings stand the Ontario winter? They had stood the winter of 1925-28 very well. Only the tops of those were spoiled, which were injured by buffalo tree hoppers.

It seemed that the regular Ontario caterpillars did not like the sap of the English walnut foliage. But the worst enemies of the Carpathians was the bacterial disease. The leaves and young shoots curled, turned black, being infested by the disease. In such a case the spraying is needed.

Acquaintance with the Vineland Government Experimental Farm

Somehow, but very soon after I started my experiments with English Carpathian Walnuts in Ontario, Mr. James Neilson, the nut specialist in the Government Experimental Farm, Vineland, Ont. discovered me. By him I was introduced to the late Mr. G. H. Corsan of Islington, Ont. who was known as a prominent nut grower in Ontario. In the year 1924, when we met the first time, Mr. Corsan already was interested in the culture of black walnuts and butternuts, in hickories, pecans, hicans and filberts. Soon I transferred my English Carpathian walnut nursery to Corsan's place at Islington. Mr. Corsan, with a great deal of enthusiasm broadcasted my Carpathians all over the American continent, but under different names: English Walnuts, Persian, Russian, Carpathian, etc. Soon we were joined by a third walnut enthusiast Mr. L. K. Davitt, a teacher in a Toronto High School.

Prof. C. T. Currelly the Founder and at that time the Director of the Royal Ontario Museum of Archeology in Toronto, also became interested in my walnut experiments. Then later on some other prominent Torontonians followed us and the Nut Growers Society of Ontario was organized.

Americans also became interested in the Carpathian walnuts. First among them was a graduate from Cornell University, a farmer near Ithaca, N. Y., Mr. Samuel Graham. Mr. George Slate of the Geneva Experiment Station was one of the first Americans who early got interested in the Carpathians.

There in the States is the Northern Nut Growers Association. Following Mr. Corsan I also became a member of the Association.

My Research in English Walnuts in Ukraine

From the year 1924 until 1936 I spent most of my time as a Presbyterian missionary in Western Ukraine, which was then under Polish occupation. From time to time I used to come to Canada on furlough. Every time, coming from Ukraine, I brought also a box or more of Carpathian English Walnuts for planting.

Then I liked to tell Dr. Palmer, the Director of the Vineland Government Experimental Farm about my research in walnuts in Ukraine.

In Western Ukraine my headquarters were in the city of Kolomyja, Province of Galicia, at the foot of the Eastern Carpathians. Thus I was in the center of the culture of the Carpathian walnuts.

Though my circuit was very large (Provinces of Galician and Volynia) and there was a time when I served 30 congregations, nevertheless I had a little time also to study the English Walnuts in their native environments.

Before starting the research in that country I decided for myself what in my conception should be the ideal English walnut. I have come to the conclusion that the nut should be of large size, thin shelled, its kernel well filled up, being of a pleasant sweet taste; inside of the nut there should be no partitions, thus allowing the kernel to roll out unbroken.

Then I printed questionnaire blanks for each individual nut tree to be examined. Beside the above mentioned questions I added:

What is the name and address of the owner of the tree, and its location?

How old, tall and thick the trunk of tree is?

How many pounds of the nuts the tree yielded that year?

In what kind of soil does it thrive?

What enemies attack it?

What fertilizer, or manure, has been used in the particular case, or none?

Is there in the nuts, leaves and bark any sign of cross-pollination?

Regarding the grafting and budding I found that the local nut-growers had not the slightest idea how to go about it. They also did not care to prevent their walnut trees from cross-pollination.

Soon I found that there in Galicia alone could be found several hundreds of varieties of Carpathian English walnuts. Anyway till 1935, I sent to Toronto 200 varieties of the Carpathians.

Some of those English Carpathian walnuts were 2-1/2 inches long, or five nuts to a foot; others were only one third of an inch. Some very small Carpathians produced nuts in clusters, like grapes. In some Carpathians it was possible to detect cross-pollination with Asiatic walnuts by their harder shells, by partitions, by the shape of nuts, by the construction of the leaves and their odor, and in some cases by the color of bark.

By kernels all the Carpathian halfbreeds are English walnuts, differing group from group by the taste. I remember that only in 1898 in the bourg of Loubni, and in 1933 in the City of Kolomyja I came across two trees which resembled our black walnut. In both towns some people used to live in America, and coming home they could bring with them some American nuts.

In the region around Kossiv I came across groves of American black walnuts and butternuts. Those trees were planted there by the Austrian Government 75 or so years ago. Of course they did not cause all the hybridizing I mentioned above. Maybe the Asiatic nuts were brought in Eastern Carpathians when the Tartar hordes crossed the mountains in the region of Pokouttia (Kossiv) in the year 1242.

Not far from Kossiv, westward, in the village of Kosmuch in the Carpathians 2500 feet above sea level I found English walnut trees of small size (15 feet tall, 6 inches thick) with light gray bark, producing 2 inch long nuts of speary shape, like our Canadian butternuts but of English Walnut shells and kernels. The kernels were tasty. There was no question but that they were halfbreeds, English plus Mongolian nuts.

There in Kosmuch, not far from the historical Tartar Passage, through which in 13th century Ghengis Khan hordes invaded the Danube plains, in winter the temperature falls to 45 degrees below zero. Owing to the hardiness of the strain and pleasant taste of the nuts I picked up about 10 pounds of them to be tried in colder parts of Ontario, (and some of them already are bearing north of Toronto and true to the type.)

I called the nuts Hutzulian Pointies, as they grow in Hutzulia the country of the Ukrainian Mountaineers.

The year 1936. My last trip to Western Ukraine

In Ontario farmers were slow to grasp the idea of cultivating my Carpathian English walnuts. Either they did not believe the English walnuts could thrive in this Province, or waited till my trees would start to bear. Nevertheless some thousand of my seedlings were planted here and there all over Ontario and smaller quantities in the Maritime Provinces, Manitoba and Alberta. The late Sir Wm. Mulock hired Mr. Corsan to graft with the Carpathian scions tops of many of his black walnut trees in Orillia, Ont. Fred Gaby, the engineer who built the Ontario Hydro, ordered through me from Ukraine 50 to 12 feet tall Carpathians of bearing age and planted them on 10 acres near Cooksville. Ont. Prof. Currelly has bought 25 acres near his estate west of Pt. Hope, Ont. for my use in experimental work. The late Col. McAlpyne planted one thousand of my yearlings on his estate at Fenelon Falls, Ont. Two young farmers, Papple Bros., in the Georgian Bay region also started an English Carpathian walnut orchard. In 1935 I moved my Carpathian walnut nursery from Islington to Prof. Currelly's estate, and

Mr. L. K. Devitt sold his lot of the trees through the Dominion Seed Co., Georgetown, Ont.

In the States, Mr. Carl Weschoke, a manufacturer in St. Paul, Minn., who in the year 1935 was elected the President of the Northern Nut Growers Association, also got interested in Carpathians. His son-in-law about that time started a walnut nursery on their estate some 30 miles east of St. Paul. That 1936 year Mr. Weschoke sponsored my expedition to Northeastern Poland (Northwestern Ukraine) to find the geographical line north of which English walnuts do not thrive in Europe.

My expedition was successful. I discovered that northward from the Pripet River, which flows from west to east toward the Dneiper, English Walnuts could not be found. If I had come across there some English seedlings nearer to the Lithuanian boundary and the Baltic Sea shore, they would have been planted there recently and not before the year 1924.

Farther north, though there English walnuts do not thrive, around the Lake Peipus I came across filberts not as bushes but as large trees. Every fall peasants in that district go in the woods and bring bags of filberts for winter use.

Such filbert trees I found also in the Carpathian mountains near the Ukrainian settlement of Vizhnytza in the Province of Bukovina.

West of the town of Sarny and south of the Pripet I came across a grove of 18 ancient English walnut trees. In the year 1648 when Ukrainian Hetman Bohdan Khmelnytzky led a war against Poland those trees already were 70 years old, and they still were bearing in 1936 when I visited that region. Indeed their limbs were broken and they presented a sad sight, but they proved how long the Ukrainian English walnut could live. The seeds of those ancient trees I also shipped to Mr. Weschcke. Beside that I brought to my sponsors thousands of selected walnut seeds, seedlings and scions.

My English Carpathian walnut tree in the back yard of 48 Peterboro Ave. Toronto, Ont., being planted out there from the pot in the spring of 1922 started to produce nuts in 1929. The nuts were exactly to the type: oblong, pointy, inch and a half long, the shell semi-hard, partitions large, the kernel of pleasant taste. It started to produce female bloom when it was 4 years old, but till 1929 there were no catkins of male bloom.

The crop of the nuts, that year and following years was usually carried away by marauding black squirrels.

Other people who got from us the Carpathian English walnut seedlings reported that their plants also started to bear the seventh year or around that. But the Papple Bros. reported that they had a case when a seedling produced by them straight from the Carpathian walnut bore a nut in the second year of its life. On the other hand there were cases where some Carpathian English seedlings, as well as grafted ones, still produce no nuts though they are 15 years old and over.

I think the cause lies in the soil. On the gravelly hills over Ithaca, N. Y. Carpathian walnuts are slow to bear, even being grafted. The undersoil in the valleys 6 miles north of Pt. Hope, Ont. is not favorable, not only for English walnuts but even for native black walnuts, though very favorable to hickories.

On another hand, north-east of Toronto and near Unionville at the place called Hagerman Cornor on the farm of Mr. M. Artymko there is an orchard of 27 Crath's Carpathian English walnuts over 18 years old, each fruiting now every year. The trees are 25 feet tall, 5-6 inches thick, situated on a knoll of clay, well drained soil, lying open toward the northwest. When the trees were younger they were subject to attacks of the bacterial disease and their barks were cracked by frost. Now the trees are in nice shape, no trace of the bacterial disease injuries and the frost's scars disappearing. Some of those trees produced a bushel of the nuts each.

Among Artymko's trees there is a tree bearing the walnut of giant type, and the tree—Hutzulian Pointie. The success of the Artymko's farm lies probably in the soil and its high elevation.

There in Toronto Mr. T. H. Barrister, has in his backyard two Carpathian English Walnuts, producing nuts of the giant size—five nuts to a foot. The bacterial disease had touched them slightly, and the tree never has been sprayed.

We should expect that the Ontario Agricultural College at Guelph would find out what is the best soil for English walnuts and what fertilizer to be applied for them. Chicken wire fences should protect the walnut orchard from squirrels and the trees should be sprayed against bacterial disease.

About walnut trees bearing and fertilizer—let us return to their native abode in the Carpathians. There in the village of Peestynka I have come across a large English walnut tree 40 feet tall and about 36 years old which, as I was informed by the people there, never fruited till the First World War. During the war an Austrian horse squadron had put a stall around the tree. The horses well manured the soil around there and since that time the tree was bearing nuts regularly and abundantly when I saw it in 1936.

At Last Success!

The year 1951 should be regarded as the final establishing of the culture of the Carpathian English walnuts in Ontario. The three decades of experimentation have passed leaving a splendid result. The fact is established that the Carpathian English walnuts have become aclimatized in South Ontario. This fall I had an opportunity to examine my walnut trees at many points in the Province. Everywhere I have seen the tree bearing. In Toronto in many a backyard, in Thorold South, in Welland, in Port Colboren, in Islington, near Port Hope on Prof. Currelly's estate, around Scarboro, Ont. and so on, the Carpathians are in good shape and all are bearing.

The more the trees mature, the better they look. On the average they are 20 years old, 20 feet tall and 6 inches thick.

The summer of 1951 in Ontario was more cloudy than usual, and it caused the Carpathian walnuts in this Province to turn out smaller than their size, should be about one quarter smaller.

The people who knew Carpathian English walnut trees in Galicia agree that in Ontario the Carpathians grow more slowly than they do in their native land.

It is not in Ontario, but on the University Farm at Madison, Wisconsin, one of our Carpathian trees is nearly 40 feet tall and bearing. In Galicia I had seen many a Carpathian walnut tree as high as 60 feet.

Polish Government Interested in My Activity

During the time of my activities, in the town of Kessiv, there used to live a famous physician, Dr. Tarnawski. Outside of his clinics he was much interested in the welfare of the country. My activities could not be hidden from his sight. "What does that "American" see in our nuts? Are there in America no nuts?" he asked. Soon I was introduced to him. It was in the fall of 1934. He was not well and in bed at that time. He liked to talk with me about the walnut culture and wished to know why I was collecting the nuts, scions and seedlings for Canada. And then it seemed to him impossible that there in Ontario and the northeastern states English walnuts were not yet cultivated. Then I turned his attention to the fact that in Poland they know little about their own trees. My challenge awoke him to activity, and through his intervention Starosta, the county governor, planted the first twenty-five acres with walnut seedlings along the south side of the highway leading from Kessiv to the town of Kooty.

Dr. Tarnawski wrote also an article to a horticultural magazine on English walnuts on what he learned from me.

When in the fall of 1936 I was going back to my home in Toronto, Dr. Tarnawski wrote about me to the Department of Agriculture in Warsaw introducing me to the minister. I had an opportunity to give a talk on the Carpathian English walnuts in the presence of many horticulturists in the Government Experimental Farm at Skieerniewice near Warsaw.

Late in 1936 I came back to Canada and till the Second World War continued to cultivate the Carpathian walnuts and other horticultural material brought by me from Western Ukraine.

The Second War cut me off from my field in Europe.

A decade and a half has passed. The Carpathians have been acclimatized, have grown, and have been bearing nuts in Ontario. When such success has been achieved, it seems that there in Canada all the enterprise is forgotten. Of course, the Carpathian walnuts could not advertise themselves—they are "dumb critters."

In the States the situation with the Carpathians is entirely different. Interest in them is growing steadily, and as I said previously the American nurseries have already put the Carpathians on the broad market.

In 1950 at the annual meeting the Northern Nut Growers Association made me an Honorary Member of the Association.

In 1951 the Association held a contest and the "Crath" Carpathians won most of the prizes.

Culture of Crath's Carpathian English Walnut Trees

1. *Propagation by seeds*

Pick up the largest and heaviest nuts from a certain tree. Dry them in a windy place, but not in the sun. Gather the nuts into a jute bag and hang for the winter in a dry and cold place protected from squirrels.

Around May 14th put the nuts into a vessel with lukewarm water, soak about one week.

Prepare a bed of rich soil manured previously with horse manure. The land should not be of a wet kind. Plant the nuts in rows, 6 inches nut from nut, and two feet, row from row. Protect your nursery from squirrels.

In a week or two the nuts should come up.

Keep the nursery free from weeds. It will protect the seedling from the buffalo tree hoppers. If the signs of the bacterial disease are detected spray the seedlings at once.

For the first winter leave the seedlings as they are in the field. The next spring dig them up, every one. Cut off the leading root of each plant and transplant the seedlings again in rows a foot apart seedling from seedling and two feet row from row.

The amputation of the leading root causes the seedling to grow up instead of down and will make them start to bear nuts earlier.

In Europe instead of cutting off the walnut seedling's main roots they put under them a flat stone, or start in an earthen pot.

The next spring the walnut seedlings are ready for the permanent planting. Being permanently transplanted they should be cultivated at least two or three years.

Whitewash the walnut trunks in the late fall to protect bark from bursting by the winter sun. Put a screen around the trunks to protect them from mice and rabbits. Though, if a walnut is gnawed by rodents do nothing about it, the tree will produce a stalk—a new one—from the root.

2. *Propagation by Grafting*

Take Canadian black walnut seedling, one or two years old early in the spring, if you have a greenhouse and can graft them one inch above the root line, tie up with raffia, cover with melted parawax and put in boxes covering each row with light soil mixed with the moss. After 20th of May when the danger of frost is over transplant in your nursery.

The grafting of walnuts should be called a barking method. Cut off the upper part of the stock horizontally. Split the bark with your grafting knife as much as needed and lift up the bark as far as the wood and insert the scion. Tie up with raffia and do the rest as said previously.

The top grafting on the large Canadian black nuts gives good results also.

3. *Budding*

We bud the walnuts in the middle of August. Regular "T" cut has to be done, the bud put in and wrapped with raffia. Then it should be covered with parawax and left for a couple of weeks. After that time the budding should be examined and the raffia removed. If the leaf by the bud remains green it indicates that the grafting is successful.

The next spring, cut off the upper part of the stalk about two feet over the bud. You will tie up to it the budded shoot, which by the fall might be up to 6 feet high.

Spraying and cultivating is required as has been said above.

Owing to the fact that the budded plant in its first year continues to grow deep into fall and in many cases its upper part does not harden well, wrap the budding with straw for winter.

4. *Harvesting*

In the Carpathian Mountains when they gather the walnuts in the fall they mash them down with a very long and quite thin hazel sticks. Doing that they beat off the thin tops of the walnut branches. They say such an operation causes a better crop of the nuts next season.

5. *Giant Walnuts and their problems*

Some giant walnuts on the same tree have sometimes small kernels or withered ones. In the Carpathian Region they do not know what to do with such a problem.

It seems to me that we in Canada have to solve it. Maybe it is because of the bacterial disease, or it may be a lack of the proper fertilizer.

In Warsaw I have seen the giant walnuts sold not being dried.

6. *Reforestation with the Carpathian Walnuts*

Crath's Carpathian English walnuts could produce for Canada a very valuable forest and in shorter time than other trees do. We should always remember that in the Caucasian Mountains there are huge walnut forests. Some trees are of primeval age. Before the First World War English buyers often paid a Caucasian farmer from 5,000 to 10,000 rubles for a tree.

Walnut Wine

There in the Town of Kooty Mrs. Babiuk, a good wife of a local burgher told me about the walnut wine as follows:

"In my girlhood in this region there raged an awful epidemic of cholera. Many people died. But those who drank the wine made of green English Walnuts did not die."

The recipe that she gave me is as follows:

Take equal parts of walnuts in which the shells are not yet hardened, and the same quantity of sugar. Cut each green walnut in half a dozen parts, mix them with the sugar. In a couple of days the juice will be extracted by means of the sugar and ensuing fermentation which continues about one month. In two months it is ready to be consumed.

On my return to Canada I made wine from the Canadian black walnuts. The color of the wine was dark brown and quite pleasant. It stops stomach ache.

Also we should not forget the walnut oil and the use of walnuts in confectionary.

Walnut Candies

Take equal quantities of walnut kernels and honey. Mix. Boil, watching that the honey does not over-run. Mix with a wooden spoon. In half an hour cool to see if the honey has turned into taffy. If not, boil longer. When it is ready put upon a wooden board, with a spoon. When cooled the candy is ready.

FOOTNOTES:

[Footnote 23: Mr. Crath died late December 1952]

Nut Tree Plantings in Southeastern Iowa

ALBERT B. FERGUSON, *Center Point, Iowa*

Last year on our return from the Nut Growers Assn. tour, Mr. Snyder and I stopped to see the Schlagenbusch Brothers and their nut plantings. We thought at the time that it would be profitable to the Association to have a report on their work. Mr. Snyder and I went down a month ago to visit them again.

Sidney and Carl Schlagenbusch live in the southeastern part of Iowa. The walnut orchard is on high land overlooking the Mississippi River bottom. The ground was formerly oak and hickory timber. Most of their other plantings are near the farm buildings which are just below the higher ground.

The first planting of the walnut orchard was made in 1928 and was completed 8 or 10 years later. It consisted of 205 trees. Later additions have been made. There are about 325 grafted trees in the orchard at present, most of them of bearing age. The trees are spaced 50 feet by 50 feet in staggered rows. Some of the branches are beginning to touch. The diameter of the larger trees is 18 inches. The orchard is in grass which is not grazed close. The larger portion of the orchard is the Thomas variety. They have a selection of their own which was first in the Iowa contest a few years ago. I thought it outstanding, but they consider it a little small.

The nuts are gathered in a wagon and run through a corn sheller, then cleaned in a device they made themselves. The nuts are then floated and dried. Over half of the crop is cracked and sold as kernels. They have been getting around a $1.20 per pound in Fort Madison. No crop to date has exceeded a thousand dollars in value.

They also have several hickories and hybrids. The shellbark variety, Wagoner, is outstanding—the best I've seen. It is large, thin shelled, cracks easily, and is of good quality. A small tree grafted on shagbark is bearing well. They have the common varieties of pecans, a few chestnuts, a few English walnuts, Japanese walnuts and hybrids. The Winkler Hazel has not been very productive with them.

They had several trees of Stabler, which were not satisfactory so they cut the trees off close to the ground and put 6 or 8 bark grafts in the stump. They saved the largest one as the main trunk and taking a graft or a large sprout from the opposite side of the stump, inarching it into the main trunk two or three feet up. This prevents the wind from blowing the graft off of the stump. It also makes it possible to utilize the strength of the roots from the opposite side of the stump. They had several trees worked this way which are now of good size.

In addition to caring for their large farm, nut orchard and a choice herd of Hereford cattle, Carl has found time to do some breeding work with Oriental poppies from which he has made some very choice selections. They have also worked with several other perennials. Sidney and Carl Schlagenbusch are true horticulturists by nature and are fine folks.

On the way home from this recent trip, we stopped to see Corliss Williams near Danville. His brother Wendell Williams, located the Winkler Hazel, before the first world war in which he served and never returned. We saw a Persian walnut, 25 or 30 years old, in Mr. Williams front yard. It was a U.S.D.A. introduction from Russia. It seems to be perfectly hardy, bears well and is of excellent quality. The shagbark hickories are plentiful in his locality. He has top-worked 200 or more, many of them to Burlington, which is productive and fills well with him.

Rockville as a Hickory Interstock

HERMAN LAST, *Steamboat Rock, Iowa*

As a nut-grower I am afraid I have been over-rated; I make my living tilling the soil and dabble in my nut grove only when I can find a few moments to spare—in fact all I know about nuts and nut-grafting, I owe to my good friend, Edgar Huen. I shall always remember that balmy May morning 25 years ago when Mr. Huen came over with a kit full of hickory scions, and suggested we go out in my pasture and do some grafting. In that bag were Stratford, Rockville, Des Moines, Marquette, Hagen and Monahan.

We grafted all that day—that is Mr. Huen did the grafting and I watched him. Today these trees are living monuments of our work.

The only tree of these varieties that has ever borne enough nuts to feed a squirrel is the Stratford.

Meanwhile I have been doing a little grafting myself. I acquired a few pecans for understocks but the only variety that was congenial with pecan as far as I knew was Rockville, but it produced no nuts—it was just a nice tree to look at.

One spring my brother-in-law who lives just across the line in Missouri sent me some shellbark scions from a tree in his pasture. I grafted these scions on a pecan and they took off like a house on fire. This variety proved to be a rugged individual and bore every year but the nuts were no good—all cavities like a true shellbark.

Then one spring morning I grafted some of these shellbark scions on Rockville; the grafts took and I soon noticed a transformation. The grafts had blended with the understock and the offspring was different from either parent. The best part of the new hybrid was that it bore abundantly and the nuts are of fine quality.

To those who have some young Rockville trees for top-working, I can furnish a limited amount of scionwood of this shellbark which I have named my Super X, it being so rugged and hardy.

To me the grafting of trees is a noble work. Someone has said that he who plants a tree is a true lover of his race and I don't know of anything that will live longer in the memory of our children and those who follow in our footsteps than a row of hickories laden with nuts.

A Fruitful Pair of Carpathian Walnut Varieties in Michigan

GILBERT BECKER, *Climax, Mich.*

I would like to tell you briefly my experience with the difficulties of Persian walnut pollination. It took 8 years before I got any nuts, although they had nutlets time and again! It was after I had Crath #1 bearing, that all proceeded to fruit, and then heavier every year, until 1951 when the freeze of November 1950 eliminated the nuts.

Crath #1 has done so well that I feel it well worthy of being a commercial prospect for us. The size and shape are so attractive. (The accuracy of the numbering was once questioned by Mr. Stoke, so I do not know if it is the same No. 1 that others have had from Crath. This was named by Prof. Nielson. It definitely is not Broadview, as Stoke at first thought.)

My Crath #1 had over four bushels of hulled and unhulled nuts (as they are picked up, after shaking) this fall. It was grafted on black walnut in 1938.

At my folks' place I planted a grafted Crath #1, and a Carpathian "D", side by side. There are no other Persian walnuts near, and they have always had nuts, since they started to bear. I feel that this is a proper combination. I do not know whether the blooming periods overlap.

Suggested Blooming Data to be Recorded for Nut Tree Varieties

J. C. MCDANIEL, *Univ. of Illinois, Urbana, Ill.*

Such experiences as Mr. Becker's (extracted from a letter to me) are well worth knowing, and we need similar information for several years and at different locations, for all the promising Persian seedlings and new varieties. I would suggest that all of us who have them flowering in our plantings (even if only one tree) make an effort in 1953 to record as much as possible of the phenological data on them. A form such as the following might be used, for flowering, fruiting, and related data.

Year: 19_____ Location: ____________________

Data by: _______________ First freeze previous fall: (Date) _______________

Minimum temperature previous winter: _____ deg.F. on (Date) _______________

Last killing frost this spring (Date) ______________________________

Variety	Age	Date	First	End	Date	Nuts	Yield	Remarks
(or	of	from	catkins	of	pistils	harvested		
seedling	tree	new	shedding	shedding	appear	(date)		
No.)	or	growth	(date)		receptive			
	graft	scion						
1.								
2.								
3.								

Under "Remarks" could be recorded such information as the distance and direction to trees furnishing pollen in the period when a given variety has sticky appearing pistils, the abundance of pollen shed, apparent winter killing of catkins, etc. The list of items could be expanded, if desired, but it is thought that those included here are among the most important in determining the potential performances of varieties and variety combinations in specific climates. A compilation of such data for a period of about three years, supplemented with data on the nuts themselves, would be of very practical value as a basis for selecting varieties most promising to plant or propagate. The same data form would be applicable to other walnuts, hickories, pecans, and filberts, and perhaps to a lesser extent with chestnuts.

Note on Chinese Chestnuts

HARWOOD STEIGER, *Redhook, N. Y.*

My earliest Chinese chestnuts are ripening. Stoke Hybrid is earliest and the nuts are so attractive, too bad they are not better in quality. It is an exciting time here as there are always a few seedlings that are ripening for the first time. Honan, which ripens later, has been one of my best grafted trees. One of my seedlings has very large nuts, very early ripening, nuts are now falling, and it is prolific, nearly every burr has from two to three large to very large nuts. The quality seems good. We like the large nuts as they are easier to peel and we like them boiled and served as a vegetable. The boiled nuts keep well when frozen. I think this tree is superior to any of my grafted and named varieties.

Scott Healey—An Obituary

Scott Healey was born December 3, 1881, in Wheatley, Ontario, Canada, and came to Otsego, Michigan, in 1904. He married in 1908. Mr. Healey was a chiropractor for a number of years.

In 1921, Mr. Healey and his cousin, Lewis Healey, formed the Healey & Healey Lumber and Coal Company, in Otsego, which they operated together until a few years ago, when Mr. Healey retired due to ill health.

Mr. Healey was a director of the State Savings Bank in Otsego for many years. He was a member of the first Baptist Church in Otsego.

He became interested in nut culture while the late Professor James A. Neilson was nut specialist at the Michigan State College. Mr. Healey planted a nut orchard of about eighty grafted nut trees in 1933, which Professor Neilson helped him plan. Most of the trees were black walnut varieties, chiefly Thomas. However, there were some Ohio, Stabler, Allen, Crietz, Stambaugh, Ten Eyck, and Rohwer trees. There were also some filberts, several Chinese chestnuts, and some heartnuts he had raised from seed. One nice tree of the McCallister hican makes good shade, but has never borne any nuts. He did some topworking in a large black walnut tree in the backyard, where he got a Persian walnut to grow.

Mr. Healey was very much interested in nut culture, and had planned on having a nut grove for a hobby to keep him busy when he retired.

Mr. Healey joined the Northern Nut Growers Association in 1933. He and his wife attended the Battle Creek meeting one year later. They also attended the Rockport, Indiana meeting in 1935, and the one at Geneva, New York in 1936.—"The rest of the time he couldn't go or was in too poor health to go."

They sold their home, with the nut planting, to a young couple, Mr. and Mrs. Lewis Lovett, in 1948, moved into Otsego; and retired.

Mr. Healey died, January 18th, 1952 at their winter home in Port Richey, Florida. Surviving are his wife, Mabel, and one son, Virgil.

GILBERT BECKER

A Letter from Dr. W. C. Deming, the Only Living Charter Member of the Association

Northern Nut Growers Association,

Dear Old Friends:

The 42nd Annual Report has recently come to me. Think of it, the 42nd Annual Report! How familiar to me are a great many of the names of the officers and members! I can even recall the very features of many of them. I am myself now ninety years old and practically house-bound. Though yesterday, a day almost like summer, I did take a taxi and a drive through the park amid the brilliant foliage, with Miss Dorothy Hapgood, who by the way is a member of our association a thing with which I may have had something to do. Recently I was in the Veterans Hospital at Newington for a couple of weeks. The doctors called it "*polycythemia*", the direct opposite of "*anaemia*", did 10 phlebotomies taking 5 pints of blood which they said they used for transfusions on ward patients, much to my gratification. I now have in, or had put in me, a dose, of radio-active phosphorus P32 which, they assure me will be getting in its good work for the next three months. Nothing like being up to date, even if valetudinarian.

You have made me Dean of the association. In the beginning Clarence Reed was always back of me with his abilities and vast fund of information. Although I believe I am, by virtue of my office, exempt from dues and entitled to the annual reports, I wish my five children to be at least once represented in the membership. I append their names and addresses:

Hawthorne, the eldest, is with the Gen. Electric Co. in New York. I don't know what he does but presume that with the other New York millionaires he is busy accumulating wealth. This hint may guide you in soliciting alms for the association some day. His home is in Hamilton Lane, Larien, Conn. But I don't know if he knows a nut from a lunatic. He has two kids, one now preparing for Korea. God preserve him.

Benton is already a member. He has a few acres in the town of Avon, Conn. where, among the rocks and the native rattlesnakes and copperheads he tells me he has Chinese chestnuts growing. Recently he got two of the copperheads. He is an energetic chap. He rises at 4 a.m. and drives the several miles into Hartford where he broadcasts from 7 to 8, for people's breakfasts, I suppose, and is released at 10 a.m. He has just contracted for a television program once a week in New Haven.

Olcott is a consul in the U.S. Embassy in Tokio, transferred from a similar position in Siam. If there is something you want from Japan I guess he is your boy. Mention my name! He has a lovely wife and three children.

Una King, my elder daughter, whose husband was killed in an accident, interviews VIP's on the same radio station as brother Ben.

Joan Howe (Mrs. Paul) and her husband, who is in a bank in New York, live in my old home on Umpawaug Hill, Redding, Conn. She writes of having had a crop of black walnuts from one of the trees I planted. I've forgotten all the others there may be there. Nothing of value I guess. Joan has two daughters. Ben has a son and daughter.

That makes five children I'm responsible for and they have acknowledged the eleven grandchildren for me. I want you to make four of my children (Ben is already ensnare) members of the association, for which I will enclose a check for $12.00 (if I don't forget.) (The many typing mistakes of this letter are due mostly to the age of the machine, not mine.)

My two sisters who live in our old home in Litchfield and who are close behind me in years, recently sent me a handful of nice chestnuts, Chinese, from a tree 40 feet or more high in our backyard. They have to divide them, very unequally, with the squirrels. The only other noteworthy trees in our little place are a few papaws. Asimina triloba, too shaded to bear. This fruit might be worthy of a little attention from the nut growers. The dictionary speaks of several other species of papaw.

Any of you who have outgrown the labor of caring for nut trees might find interest in mycology in which I found diversion and edibles for a while. Only beware the deadly Amanita and others of that ilk.

I cannot adequately express to you my heartfelt joy at the prosperity of our association. For one thing the great increase in the membership, for another the birth of three branch state associations, but above all the success in the production of nuts. In my time we had mostly, if not entirely, the promising production of specimen nuts only. We had nothing like the Jacobs Persian walnut with its imposing spread and its production of 200 pounds of nuts in one season; Mr. Kyhl's orchard with its many varieties of Persian walnuts; his success in grafting and his reporting of a tree which bears three or four bushels of heartnuts yearly; Mr. Best's 5,000 grafted pecan trees; Mr. Hirshi's chestnuts; the splendid results of the Persian walnut contests; and the almost spectacular increase in the number of nurseries selling grafted nut trees of many varieties. These facts, and many that I have not mentioned, make it certain that nut growing is now a firmly established and surely increasing industry. You may be sure that these facts give me great delight.

Some years ago while I was in possession of a mind as good as it had been at any time, I did a little grafting of nut trees in a commercial way for people at their country places, and I had the nerve to charge them fifty dollars a day. What's more I got paid and never got kicked, nor did I hear mutterings or see scowls. But then, you see, there was no other grafter, of the kind, around my part of the country. Almost a monopoly and, of course, a wicked one. But here my mind goes blank. I can't recall what luck I had with the grafting, nor can I recall the name of a single one for whom I did such work.

I strongly advise every one of you to have a good book in which you keep personal and geographic records of all your work with nut growing. All the details are vividly in your mind now, but when you get to be ninety you may find them, as I do, faded away and all washed up. Please go on with the good work.

Some more good friends have just taken me for a round trip to Litchfield where my little sister, who is 84, has just partly circumvented the squirrels and by going out very early in the morning to the chestnut tree has succeeded in getting a good big double handful of chestnuts, nice big ones.

She also called to my attention a good-sized Persian walnut which she says I once grafted on a black walnut and this year was quite well covered with nuts which she says the squirrels cut off while green, and she says they were helped by one of the black plumaged birds. Some time ago she gave me one of the nuts and I tried to husk it with my knife. But it was too immature. They would have matured this fall, I think but for the pests.

William C. Deming

Sweepstakes Award in Ohio Black Walnut Contest

L. WALTER SHERMAN, *Canfield, Ohio*

This I believe, is the third report to the Northern Nut Growers Association concerning the black walnut contest held in Ohio in 1946. The first report was given soon after the close of the contest. During the year following the contest (1947), I visited each of the ten prize winning trees, photographing them, and getting as complete a case history of each as was possible.

This, the third report, concerns mainly the process used to determine the winner of the $50.00 sweepstakes award given in 1951 for the best performance of a black walnut tree for a five-year period. The owners of the ten prize-winning trees in the 1946 contest were asked to report the amount of crop harvested each year as well as to send in samples of the nuts for a cracking test.

Complete data were recorded each year from the samples just as they had been for the 1946 contest. The average weight of nut, recovery of kernel at first cracking, total kernel content, and per cent of kernel content were recorded.

From these data tables and charts were compiled to make a visual comparison between the various nuts. Walnuts other than the prize winners were not excluded from this five-year competition and quite a few were submitted. However, only one of them, the "Chamberlin" was of special merit and it was given a place on these charts. No samples or crop records were received from the Davidson (sixth prize) and the Jackson (tenth prize) nuts, and so they are not shown on all the charts. One sample from the 1949 crop of Penn walnuts was lost to a pilfering squirrel, and the 1949 data used on the chart for the Penn walnut was therefore the average of all other samples of this variety. The weight of total crop harvested in 1949, however, is actual.

Table No. 1 gives the average weight in grams of the sample nuts. The Duke, (first prize) was the largest nut of all, in 1945, averaging just over 27 grams; but the Orth, in 1948, averaged almost a gram more. The Kuhn, which was the smallest of the eight nuts in 1946 and again in 1950, was the largest nut in 1949, and its size in 1949 was exceeded only four times by any of the other nuts during the contest. The nuts were large in size during the off year when only a small crop was produced and they were small when there was a heavy crop.

In table No. 2 the weight in grams of the kernel recovered on first crack, secured without the aid of nut pick, is recorded. In this comparison the Duke, because of large size, might be expected to be an easy winner and it was in 1946 and in 1950; but in 1948, though second in average weight of nut for that year, it was in fifth place in recovery of kernel at first cracking.

Table No. 3 records the average weight in grams of the kernels. Here the Duke, due largely to its size, is a consistent winner in all three years it produced nuts. However, in 1949, a small crop year for the Kuhn, the nuts of this variety were large and contained more kernel than the Duke did in 1948 or in 1950.

The per cent of kernel in the nuts as recorded in table No. 4 is interesting. The Burson, which was the smallest nut in 1947, had the highest per cent of kernel and also had the highest total kernel content of any sample in that year. Evidently the per cent of kernel is higher in well-filled nuts and this is largely determined by the weather and available food supply late in the season.

A comparison of the numerical score of the various nuts, figured out according to the T.V.A. score system, is given in Table No. 5. By this system, no variety had a consistent high score, but each varied greatly from year to year.

The nut characters studied so far in charts 1 to 5 inclusive have varied so much from year to year that any judgment based on these characters for any one year could not be relied upon.

What characteristic of a black walnut, then, can be used in evaluating it? In table No. 6 the percentage of the total kernel that is recovered at first cracking is given. Oliver and Penn show considerable consistency in that they remain above 91 per cent in all samples, but look at the Kuhn. It was perfect in 1950 but in 1948 only 65 per cent of the kernel was recoverable in the first cracking and Duke was nearly as bad, varying from 69 to 98 per cent recovery.

After careful study of these six charts, I am sure you will have to admit that any judgment of a black walnut variety based on these characters only is none too dependable.

These are the nut characters that we have been using in our contest! Some further method of evaluation is needed! Individual nut characters alone are not enough. A good farmer is concerned in quality of his produce but quantity is of more importance for financial success. The Elberta peach well illustrates this. There are many peaches of better quality, but the Elberta peach is a prolific producer and this is one reason more Elberta peaches are raised than any other variety. Quality without quantity means little.

With this in mind, the $50.00 sweepstakes prize was offered for the tree with the best five-year record. The judges interpreted this to mean the most pound of kernels produced that were recovered on first crack. Going back over the records, we find some trees have been much more productive than others.

At first it would seem unfair to compare the crop from trees of different size and age, but this time luck was with the judges. Take a look at Table No. 7 which gives the ages and sizes of the trees. There is not too much difference in size or age to make reasonable comparisons possible. However, it should be clearly understood that only trees of the same age growing in the same orchard and receiving the same care can be accurately compared. The trees we are dealing with were in different localities, with vast differences in soil conditions, air drainage, climate, etc.

Table No. 7 gives the total production for the five-year period for each tree, in bushels, the total amount of kernel as well as the amount of kernel recovered at first cracking. Only five trees had produced over four bushels of nuts each during the five year period.

The Oliver tree produced 1.8 bushels and 25 pounds more kernels than the Penn tree. The Kuhn tree, though producing four bushels less nuts than the Penn tree, did produce 4.1 pounds more kernels, with the same amount recovered on first cracking from the nuts of each tree—almost a photo finish for second place.

The sweepstakes award of $50.00 was therefore given to Mrs. Oliver Shaffer, of Lucasville, Ohio, who sent in the Oliver entry.

Referring to the case histories of these trees as written up in 1947, you will find that the Oliver, Kuhn, Penn, and Orth trees were reported on favorable sites, while the Duke and Burson were on very unfavorable ones so that the above results are only what might have been expected. The Orth tree, however, is in a favorable location and better production could have been expected of it.

Table 1. Size, as Weight of Unshelled Walnuts (Approximate).

===

Grams 1946 1947 1948 1949 1950 Average[24]
per nut

28 Orth
27 Duke
Duke
26 Penn
Oliver
Orth Duke Kuhn
25
Penn Orth Duke
Duke
Athens Penn
Williamson Penn Penn

23 Orth Williamson Oliver Oliver
Oliver Orth
Williamson Kuhn Duke
22 Oliver
Chamberlin
Burson Williamson
21 Oliver Penn
Athens Kuhn Burson Burson
Burson Burson, Athens Burson Kuhn
Athens
20 Athens
Chamberlin Williamson
19 Kuhn
18 Chamberlin
17
16
Kuhn

Judges for the contest were C. W. Ellenwood and O. D. Diller of the Ohio Experiment Station and L. Walter Sherman, then with the Department of Agriculture, Commonwealth of Pennsylvania.

FOOTNOTES:

[Footnote 24: Average of five years for Duke, Oliver, Burson and Kuhn; four years for Penn, which was not cracked in 1949, but interpolated in charts.

Note: To save time and the expense of redrawing and reproduction, these seven tables are printed instead of Mr. Sherman's graphic charts. With a ruler and pencil, lines can be drawn through the "D's of Duke", and so forth, to give an approximation of the original graphs.—Editor.]

Table 2. Kernel Recovery at First Crack, in Grams Per Nut (Approximate).

==

Grams 1946 1947 1948 1949 1950 Average[25]

7

Duke Orth Orth

6

Williamson Duke

Penn, Kuhn Duke, Orth Williamson Duke

Oliver Athens Kuhn

Burson, W'ms.

Athens Duke

5 Burson, Williamson

Ch'lin

Athens, Burson Orth, Oliver Penn, Burson

Penn Burson, Kuhn Kuhn, Oliver

Athens

Orth Oliver, Kuhn Penn Oliver, Penn

Ch'lin

Duke Bur., Wms., Ath.

Oliver Chamberlin

4

Kuhn

FOOTNOTES:

[Footnote 25: See note with Table 1.]

Attendance Register, Rockport, Ind., 1952

Ontario, Canada

O. Filman, Aldershot

District of Columbia

Howard Baker, Washington
Mr. & Mrs. E. L. Ford, Washington

Florida

Mrs. R. B. Pattie, St. Augustine

Georgia

Max B. Hardy, Leesburg
Mr. & Mrs. W. J. Wilson, Fort Valley

Illinois

Mr. & Mrs. R. B. Best, Eldred
C. R. Blyth, Urbana
S. C. Chandler, Carbondale

T. F. Clark, Peoria
A. S. Colby, Urbana
E. A. Curl, Urbana
Albert Dahlberg, Chicago
O. J. & Karl Eigsti, Normal
Mr. & Mrs. O. H. Fuller, Joliet
Mr. & Mrs. Louis Gerardi, Caseyville
J. C. McDaniel, Urbana
Mrs. R. E. Norris, Shawneetown
Mr. & Mrs. Royal Oakes, Bluffs
Elizabeth Sonnemann, Vandalia
Mr. & Mrs. W. F. Sonnemann, Vandalia
A. M. Whitford, Farina
Cullen Zethmayr, Westmount
Gordon Zethmayr, West Chicago

Indiana

Ralph Andrews & Son, John, Marion
Howard Bloomethol, Evansville
Ferd Bolton, Linton
L. E. Cooper, Rockport
Virginia M. Darning, Rockport
K. A. Dooley, Marion
Peter Glaser, Evansville
Jo Ann Hall, Rockport
A. W. Hamilton, Vincennes
Ray Kaufman, Peru
Charles Myer, Evansville
George Oberman, Evansville
Edward W. Pope, Marion

Carl Prell, South Bend
Adolph Risko, Monticello
L. E. Sawyer, Terre Haute
Ralph Schruber, New Albany
Barbara Sly, Rockport
Mr. & Mrs. Raymond Sly, Rockport
J. E. Talbott, Linton
Ford Wallick, Peru
Mr. & Mrs. W. B. Ward, West Lafayette
J. F. Wilkinson, Rockport

Iowa

A. B. Ferguson, Center Point
E. F. Huen, Eldora
Ira M. Kyhl, Sabula
Elizabeth Rohrbacher, Iowa City
Wm. Rohrbacher, Iowa City
D. C. Snyder, Center Point

Kentucky

Mr. & Mrs. Robert Alvis, Henderson
W. D. Armstrong, Princeton
W. W. Magill, Lexington
J. E. McClure, Owensboro

Maryland

John Flick, Riverdale
G. F. Gravatt, Beltsville

J. W. McKay & Family, College Park
Mr. & Mrs. Herbert Negus, Hyattsville

Michigan

Mr. & Mrs. Gilbert Becker, Climax
J. A. Becker, Climax
W. N. Beckert, Jackson
Ralph Emerson, Detroit
Frank J. Keplinger, Farwell
Edwin W. Lemke, Detroit
Mr. & Mrs. F. L. O'Rourke, East Lansing
L. L. Ricky, East Lansing

Missouri

H. W. Guengerich, Louisiana
R. E. Mangelsdorf, St. Louis

New York

Mr. & Mrs. S. Bernath, Poughkeepsie
David Caldwell & family, Syracuse
L. H. MacDaniels, Ithaca
Mr. and Mrs. George Salzer, Rochester
Rodman Salzer, Rochester
G. L. Slate, Geneva
Alfred Szego, Jackson Heights

Ohio

G. E. Craig, Dundas
F. L. Davell & family, Masillon

Mr. & Mrs. John Davidson, Xenia
John A. Gerstenmaier, Massillon
Edward A. Grad, Cincinnati
Frank M. Kintzel, Cincinnati
Shumzo Kodera, Columbus & Tokyo, Japan
Paul E. Machovina, Columbus
Christ Pataky, Jr., Mansfield
Sylvester Shessler, Genoa
Mr. & Mrs. R. E. Silvis, Massillon
Mr. & Mrs. John Underwood, Urbana
Martha Weber, Cincinnati

Pennsylvania

Mr. & Mrs. R. P. Allaman, Harrisburg
W. S. Clarke, Jr., State College
John Rick, Reading

Tennessee

Spencer Chase, Norris
H. O. Murphy, Chattanooga
Dr. & Mrs. Audrey Richard, Whiteville
Mr. & Mrs. W. J. Robinson, Jackson

Virginia

Bessie J. Gibbs, Linden
H. R. Gibbs, Linden
Miss Eloise Saddler, Fazewell
H. F. Stoke, Roanoke

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As of February 24, 1953

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** Honorary member
Sec. Contributing member
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+Hale, A. C., Fairview School, Camden
Schlan, Mrs. Agnes, Rt. 2, Mountainburg
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Wade, Clifton, Forest Ave., Fayetteville. Attorney
Wylie, W. D., Dept, of Entomology, U. of Ark., Fayetteville. Entomologist

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Brand, George, See Nebraska
+Buck, Ernest Homer, Three Arch Bay, 16 N. Portola, South Laguna
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